Neurocomputing 649 (2025) 130774

Contents lists available at ScienceDirect

Neurocomputing

journal homepage: www.elsevier.com/locate/neucom

Survey Paper

A review of few-shot image classification: Approaches, datasets and research trends

Jit Yan Lim [a], Kian Ming Lim [b,*], Chin Poo Lee [b], Yong Xuan Tan [a]

[a] School of Information Technology, Monash University Malaysia, Jalan Lagoon Selatan, Bandar Sunway, 47500, Selangor, Malaysia
[b] School of Computer Science, University of Nottingham Ningbo China, Taikang East Road, Yinzhou, Ningbo, Zhejiang, 315100, China

ARTICLE INFO

Communicated by G. Hu

Keywords:
Few-shot image classification
Few-shot learning
Meta-learning
Transfer learning
Computer vision
Survey

ABSTRACT

Over the past decade, deep learning has made significant advancements in image classification. However, these models struggle with data scarcity and distribution shifts, commonly referred to as the few-shot image classification (FSIC) problem. FSIC aims to recognize novel classes using only a limited number of labeled samples, posing challenges for conventional deep learning models that rely on large datasets for optimal performance. This paper provides a comprehensive review of FSIC methodologies, categorizing them into five main approaches: meta-learning, transfer learning, data augmentation, attribute-related, and vision-language foundation model adaptation. Meta-learning approaches are further classified into metric-based, model-based, and optimization-based methods, while transfer learning approaches are divided into hybrid and non-hybrid methods. Vision-language foundation model adaptation approaches are grouped into few-shot parameter tuning, dynamic or unsupervised tuning, and training-free adaptation methods. Beyond general FSIC, this paper also explores specialized FSIC methods in fine-grained classification, cross-domain classification, and class-incremental learning. Additionally, it reviews commonly used few-shot image datasets and compares the performance of representative methods through experimental results. Practical applications of FSIC across various domains are also discussed, highlighting its potential to address real-world challenges. Finally, the research trends of FSIC are identified, offering insights into the state-of-the-art FSIC methods and guiding future advancements in this field.

1. Introduction

Image classification is a fundamental task in computer vision with broad applications across various domains [1–4], aiming to accurately assign labels to given images. The success of deep learning approaches has significantly improved performance in image classification [5–8]. This is partly driven by advancements in computing technologies, such as high-end Graphics Processing Units (GPUs), and the availability of large benchmark datasets like ImageNet [9] and Meta-Dataset [10].

However, deep learning approaches face limitations in scenarios involving data shifting, where models struggle to classify unseen samples with distributions different from those in the training set. Additionally, these approaches typically require large amounts of training samples to build robust models, leading to high data labeling costs and complex data collection challenges, including concerns about privacy and safety. Moreover, deep learning models are often inadequate for low-data regimes, where acquiring labeled samples can be impossible, such as in the case of rare animal species. These challenges related to data shifting and scarcity are collectively referred to as the few-shot image classification (FSIC) problem [11].

In FSIC, the training and testing samples belong to different classes, commonly referred to as base and novel classes, respectively. The objective is to train a model using a limited number of samples from each base class, enabling it to effectively recognize unseen samples from novel classes. This paper categorizes existing FSIC approaches into five main groups: meta-learning, transfer learning, data augmentation, attribute-related, and vision-language foundation model adaptation. Among these, meta-learning approaches are further classified into metric-based, model-based, and optimization-based methods. Transfer learning approaches are divided into hybrid and non-hybrid methods, while vision-language foundation model adaptation approaches are categorized into few-shot parameter tuning, dynamic or unsupervised tuning, and training-free adaptation methods. Each FSIC method is classified based on its defining characteristics. While some approaches exhibit features that align with multiple categories, to ensure consis-

* Corresponding author.
E-mail addresses: lim.jityan@monash.edu (J.Y. Lim), kian-ming.lim@nottingham.edu.cn (K.M. Lim), chin-poo.lee@nottingham.edu.cn (C.P. Lee), tan.yongxuan@monash.edu (Y.X. Tan).

https://doi.org/10.1016/j.neucom.2025.130774
Received 26 September 2024; Received in revised form 4 May 2025; Accepted 14 June 2025
Available online 27 June 2025

tency, each method is assigned to a single category based on its primary characteristic.

At the end of each category, a summary table will be included that details various aspects of the reviewed methods, including publication year, backbone architecture, training mechanism, presence of a pre-training stage or pre-trained backbone, use of transductive learning, centroid representation, attention mechanisms, and the method's strengths and weaknesses. The backbone architecture denotes the feature extractor employed by each method. Most existing studies utilize Convolutional Neural Network (ConvNet) as shallow feature extractors, while deeper architectures such as Residual Network (ResNet), Wide Residual Network (WRN), and Vision Transformer (ViT) [12] are commonly adopted for enhanced representational capacity. Centroid representation refers to representing all support samples within each class via several methods such as mean vectors [13], summation [14], distributions [15], covariance matrices [16], or more.

Several surveys [17–19] have provided comprehensive overviews of few-shot learning (FSL), covering its methodologies and applications across diverse domains such as computer vision and natural language processing (NLP). Song et al. [17] provided an extensive review of FSL, classifying approaches by levels of prior knowledge and offering an analysis of applications, challenges, and future directions. Lu et al. [18] examined the evolution of FSL, categorizing methods by modeling principles, with an emphasis on meta-learning and emerging research topics. Wang et al. [19] summarized existing few-shot algorithms up to 2022 and explored their applications in computer vision, human–machine language interaction, and robotics. Compared to these surveys, this work provides a more detailed examination of FSIC and its related subfields.

Among the FSL surveys, several studies [20–24] specifically focus on reviewing FSIC. Liu et al. [20] provided an overview of general FSIC techniques up to 2022, covering algorithms, datasets, experimental results, and applications across various domains. Li et al. [21] conducted a comprehensive review focusing primarily on metric learning approaches for FSIC. He et al. [22] surveyed FSIC techniques from 2016 to 2023, primarily emphasizing meta-learning approaches. Zeng and Xiao [23] presented an overview of deep learning-based FSIC techniques, including discussions on datasets and performance evaluation. Ren et al. [24] conducted a review of existing FSL approaches, focused specifically on fine-grained image classification.

Compared to these existing surveys, this review provides an up-to-date analysis of general FSIC. Additionally, this work examines specialized approaches for few-shot fine-grained image classification (FSFGIC), cross-domain few-shot image classification (CDFSIC), and few-shot class-incremental learning (FSCIL). Furthermore, it includes a comprehensive summary of benchmark datasets used in existing FSIC studies across general, fine-grained, and cross-domain classification tasks. The review also discusses standard evaluation protocols adopted in the literature. Table 1 provides a comparative summary of previous surveys, highlighting their key contributions and limitations in relation to this work.

This paper makes the following key contributions:

1. Conducting an extensive review of recent literature and categorizing existing FSIC approaches into five primary groups: meta-learning, transfer learning, data augmentation, attribute-related, and vision-language foundation model adaptation.
2. Conducting a review on recent specialized FSIC approaches, including FSFGIC, CDFSIC, and FSCIL.
3. Providing a summary of the benchmark few-shot image datasets used in existing works across general, fine-grained, and cross domain image classification.
4. Discussing the standard evaluation protocols used in the literature.
5. Identifying a list of few-shot applications and research trends in FSIC.

The remainder of this paper is structured as follows: Section 2 provides a fundamental understanding of the problem definition and task structure in FSIC. Section 3 reviews existing FSIC methodologies, while Section 4 discusses specialized few-shot approaches. Section 5 presents an overview of few-shot image datasets relevant to general FSIC, FSFGIC, and CDFSIC. Section 6 outlines the evaluation protocols used for performance assessment in FSIC. In Section 7, real-world applications that can benefit from FSIC are explored. Section 8 highlights emerging research trends based on recent advancements in FSIC. Finally, Section 9 concludes the paper.

2. Fundamental

This section provides a fundamental understanding of FSIC problem, including its formal definition in Section 2.1 and the common training mechanisms and task structures employed in existing studies in Section 2.2.

2.1. Problem definition

FSIC is a machine learning problem where only a limited number of labeled images are available for learning. Let D represent a dataset with C number of classes:

$$D = \left\{ \left(\mathbf{x}_1, y_1\right), \left(\mathbf{x}_2, y_2\right), \dots, \left(\mathbf{x}_{|D|}, y_{|D|}\right) \right\}; y \in \{1, 2, \dots, C\} \tag{1}$$

where $(\mathbf{x}, y)$ denotes a labeled sample pair with an image $\mathbf{x}$ and its corresponding class label y. In conventional image classification, D is partitioned into the training set D_{train}, validation set D_{val}, and testing set D_{test}. Each set contains the same number of classes, with no overlapping images between them, as below:

$$D_{train} = \left\{ \left(\mathbf{x}_1, y_1\right), \dots, \left(\mathbf{x}_{|D_{train}|}, y_{|D_{train}|}\right) \right\}; y \in \{1, 2, \dots, C\} \tag{2}$$

$$D_{val} = \left\{ \left(\mathbf{x}_1, y_1\right), \dots, \left(\mathbf{x}_{|D_{val}|}, y_{|D_{val}|}\right) \right\}; y \in \{1, 2, \dots, C\} \tag{3}$$

$$D_{test} = \left\{ \left(\mathbf{x}_1, y_1\right), \dots, \left(\mathbf{x}_{|D_{test}|}, y_{|D_{test}|}\right) \right\}; y \in \{1, 2, \dots, C\} \tag{4}$$

where $|D| = |D_{train}| + |D_{val}| + |D_{test}|$.

FSIC and FSFGIC: unlike conventional classification, FSIC and FSFGIC split classes instead of images. D is divided into base classes C_{base}, validation classes C_{val}, and novel classes C_{novel}, ensuring no overlap among them as $C_{base} \cap C_{val} = \emptyset; C_{val} \cap C_{test} = \emptyset; C_{train} \cap C_{test} = \emptyset$ and $y \in \left\{C_{base} \cup C_{val} \cup C_{test}\right\}$. C_{base} is used for model training, C_{val} for validation, and C_{novel} for evaluation. The partitions of D_{train}, D_{val}, and D_{test} in FSIC and FSFGIC are formulated as follows:

$$\begin{aligned} D_{train} = & \left\{ \left(\mathbf{x}_1, y_1\right), \dots, \left(\mathbf{x}_{|D_{train}|}, y_{|D_{train}|}\right) \right\}; y \in C_{base} : C_{base} = \\ & \left\{1, \dots, |C_{base}|\right\} \end{aligned} \tag{5}$$

$$\begin{aligned} D_{val} = & \left\{ \left(\mathbf{x}_1, y_1\right), \dots, \left(\mathbf{x}_{|D_{val}|}, y_{|D_{val}|}\right) \right\}; y \in C_{val} : C_{val} = \\ & \left\{|C_{base}| + 1, \dots, |C_{base}| + |C_{val}|\right\} \end{aligned} \tag{6}$$

$$\begin{aligned} D_{test} = & \left\{ \left(\mathbf{x}_1, y_1\right), \dots, \left(\mathbf{x}_{|D_{test}|}, y_{|D_{test}|}\right) \right\}; y \in C_{test} : C_{test} = \\ & \left\{|C_{base}| + |C_{val}| + 1, \dots, C\right\} \end{aligned} \tag{7}$$

CDFSIC: the training D and evaluation D originate from different domains. The source domain, denoted as D_{source}, is used for training, while the target domain, D_{target}, is used for evaluation. Typically, the model is trained on C_{base} from D_{source} and evaluated on C_{novel} from D_{target}.

FSCIL: the classes in D are divided into $m+1$ sessions, with each class split into D_{train} and D_{test}. The training process is represented as a sequence of training sets $\left\{D^0_{train}, D^1_{train}, \dots, D^m_{train}\right\}$, while the evaluation is conducted on accumulated testing sets $\left\{D^0_{test}, D^1_{test}, \dots, D^m_{test}\right\}$. Each session is disjoint, ensuring no overlap in classes across sessions, i.e., $D^a \cap D^b = \emptyset$ where $a, b \in \{0, 1, \dots, m\}$.

Table 1
The summary of existing few-shot surveys.

Authors	Year	Scope	Key contributions	Limitations
Song et al. [17]	2023	FSL	Comprehensive review, categorized methods, discussed applications and challenges	Lack of specific focus on FSIC
Lu et al. [18]	2023	FSL	Examined FSL evolution, emphasized meta-learning and emerging topics	Limited analysis of FSIC datasets and benchmarks
Wang et al. [19]	2023	FSL	Investigated applications in vision, language, and robotics	Limited review on specialized FSIC approaches
Liu et al. [20]	2022	General FSIC	Reviewed general FSIC techniques, datasets, and applications	lacks the specific categorization and detailed analysis of methods
Li et al. [21]	2023	Specific FSIC	Specialized in metric learning approaches for FSIC	Lacked broader FSIC coverage
He et al. [22]	2023	General FSIC	Reviewed meta-learning-based FSIC approaches	Limited review on specialized FSIC approaches and application
Zeng and Xiao [23]	2024	General FSIC	Overview of deep learning-based FSIC techniques	Limited insights into specialized FSIC variants
Ren et al. [24]	2024	FSFGIC	Reviewed FSIC approaches specialized in fine-grained classification	Lacked FSIC application discussion

Fig. 1. The overall structure of the episodes (3-way 2-shot) vs. minibatches.

2.2. Training mechanism and task structure

Existing few-shot training mechanisms are categorized into two types: episode-based and minibatch-based approaches, as illustrated in Fig. 1. During training, the model learns from episodes or batches derived from D_{train}. In the evaluation phase, detailed in Section 6, the model is assessed on its ability to classify instances from D_{test}, based on the limited labeled samples encountered during training.

2.2.1. Episode-based

The episode-based training paradigm is widely adopted in FSIC. In this approach, a series of N-way K-shot episodes are sampled from D_{train}. Each episode consists of N randomly sampled classes from D_{train} with K labeled support samples and Q query samples per class. All support samples are denoted as D_s while all query samples are denoted as D_q. D_s is used for model learning while D_q is used for evaluation. In each episode, the model is tasked with predicting the y to which each query sample in D_q, based on the classes available in D_s. Both D_s and D_q consist of I and J sample pairs, constructed as follows:

$$D_s = \left\{ \left(\mathbf{x}_1, y_1\right), \left(\mathbf{x}_2, y_2\right), \dots, \left(\mathbf{x}_I, y_I\right) \right\}, \ I = N \times K \tag{8}$$

$$D_q = \left\{ \left(\mathbf{x}_1, y_1\right), \left(\mathbf{x}_2, y_2\right), \dots, \left(\mathbf{x}_J, y_J\right) \right\}, \ J = N \times Q \tag{9}$$

However, episode-based training poses significant challenges for conventional deep learning models. Due to the limited number of training samples, these models may struggle with poor generalization and overfitting when addressing N-way K-shot classification tasks. Furthermore, domain shifts exacerbate the generalization challenge, as the data distribution of C_{novel} may differ from that of C_{base}. Developing robust FSIC models that effectively generalize to novel classes remains a critical and open research problem.

2.2.2. Minibatch-based

Unlike episode-based training, the minibatch-based paradigm does not construct multiple episodes using D_s and D_q. Instead, D_{train} is divided into multiple fixed-size batches, without adhering to the N, K, and Q parameters. Each batch consists of labeled sample pairs from different classes. The construction of D_{train} in this paradigm is defined as follows:

$$D_{train} = \left\{ \left(\mathbf{x}_1, y_1\right), \left(\mathbf{x}_2, y_2\right), \dots, \left(\mathbf{x}_V, y_V\right) \right\}, \ V = \left\lceil \frac{\left|D_{train}\right|}{\text{batch size}} \right\rceil \tag{10}$$

where batch size denotes the maximum number of sample pairs available in each batch.

Minibatch-based training follows a conventional supervised learning approach, mitigating the limited data issue by exposing the model to all classes simultaneously, rather than a restricted subset of classes. This strategy enhances generalization performance. However, it does not explicitly consider relationships between tasks, which can pose challenges in cases where there is a substantial dissimilarity between base and novel class distributions, particularly in cross-domain scenarios.

3. Generic few-shot approaches

This section provides a comprehensive review of existing methodologies in general FSIC. The existing approaches can be broadly classified into five main categories: meta-learning, transfer learning, data augmentation, attribute-related, and vision-language foundation model adaptation, as illustrated in Fig. 2.

3.1. Meta-learning approaches

Meta-learning approaches aim to leverage prior knowledge to aid the model in learning new tasks, thereby enabling the model to develop the ability to learn how to learn and to acquire meta-level knowledge. Based on current research, meta-learning methods can be grouped into three main categories: metric-based, model-based, and optimization-based approaches. The overall concept of meta-learning is illustrated in Fig. 3.

3.1.1. Metric-based approaches

Metric-based meta-learning approaches aim to extract meta-knowledge by developing a feature space that can generalize across a variety of tasks. This feature space, typically represented by the weights of a neural network, forms the foundation for the model to acquire new knowledge related to novel tasks. The acquisition of this

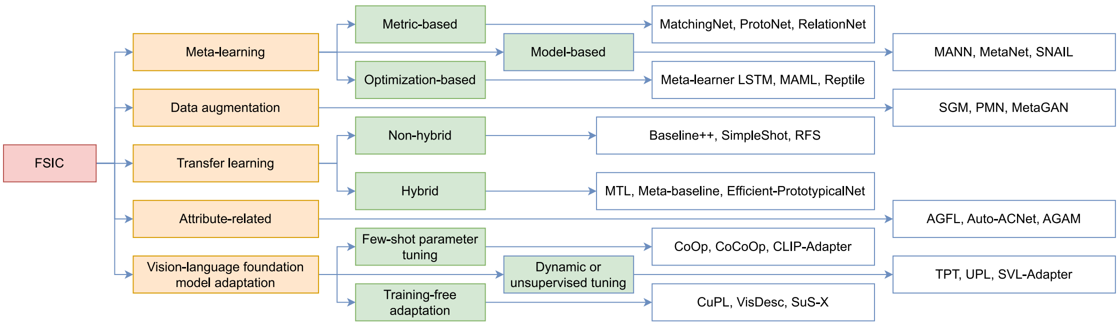


Fig. 2. The taxonomy of existing approaches for FSIC with three representative works for each category.

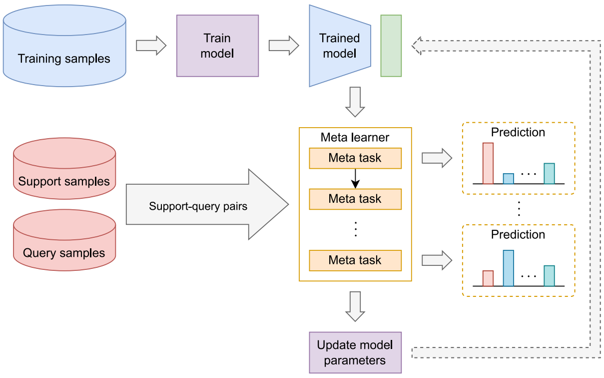


Fig. 3. The overall idea of meta-learning.

knowledge occurs by comparing novel tasks with samples from base tasks within the meta-learned feature space. The category of a novel sample is predicted by computing its similarity to each base sample; a higher similarity suggests that the novel sample belongs to the same category as the base sample. By leveraging prior knowledge embedded in the feature space, metric-based approaches facilitate FSL, enabling the rapid learning of new tasks.

To compute sample similarity, Siamese neural networks (SiameseNet) [25] examine the relationship between two samples during training. SiameseNet comprises two identical networks trained jointly to learn the relationship between inputs from both networks. Initially, the network is trained to predict whether the input samples belong to the same class. The similarity between samples is calculated using the L1 distance to learn discriminative features. Utilizing the prior knowledge from this initial stage, the model is then trained to predict the labels of unseen samples in the subsequent stage.

Unlike SiameseNet, which employs a distinct verification task to create the feature space, Matching Networks (MatchingNet) [26] learn the feature space across few-shot tasks through pairwise comparisons. During the process of outputting class label probabilities, an attention mechanism is used to compute the cosine similarity between the embedded query sample and the embedded support samples. Additionally, the proposed task-specific embeddings in MatchingNet achieve better performance than conventional embeddings.

Prototypical Networks (ProtoNet) [13] generate a mean vector (centroid) for each class to facilitate pairwise comparison with the query sample. This central point is derived from all support samples within the same class. In the pairwise comparison, the squared Euclidean distance between the embedded query sample and each mean vector is computed. The query sample is then classified based on the nearest mean vector. Since the pairwise comparisons only occur between mean vectors and the query sample, the computational cost of ProtoNet is lower than MatchingNet. Due to this centroid-based representation, ProtoNet achieves better performance than MatchingNet even without using task-specific embeddings. Inspired by its simplicity and efficiency, several extensions of ProtoNet have been proposed, offering improved performance, as shown in Fig. 4.

In contrast to earlier metric-based approaches that relied on predefined similarity metrics, Relation Networks (RelationNet) [27] introduced a trainable similarity metric. RelationNet consists of two main components: an embedding module and a relation module. The embedding module is responsible for embedding all samples into a feature space. As for the relation module, it is composed of a ConvNet to serve as the trainable metric. Similar to ProtoNet, RelationNet utilizes centroid representations, where a mean vector is generated for each class based on the embedded support samples. This mean vector is then concatenated with the embedded query sample, and the relation module predicts the class label of the query sample by evaluating its relationship with all the mean vectors. By employing a trainable metric, RelationNet demonstrates superior performance compared to the previously mentioned metric-based approaches. The overall distance comparison of RelationNet is depicted in Fig. 5.

To address more realistic FSL challenges, Ren et al. [28] extends [13] to incorporate unlabeled samples within a semi-supervised learning framework. In this approach, unlabeled data are sampled from random classes, which may or may not overlap with the classes in the support set. The model is trained using both labeled support samples and unlabeled samples, enhancing its ability to generalize in scenarios with limited labeled data. Similarly, Oreshkin et al. [29] expanded upon the work [13] by introducing the Task Dependent Adaptive Metric (TADAM). This approach integrates an additional task embedding network, enabling the model to adapt to a diverse range of tasks. Unlike the global representation shared across different tasks in ProtoNet, TADAM learns task-dependent representations. Furthermore, adaptive scaling factors are applied to adjust the similarity metric during training, leading to improved classification performance.

The idea of representing each class in the embedding space with a single mean vector from ProtoNet is vulnerable to noise and bias, particularly in scenarios with limited data. In view of this, several works [15,30] have been proposed to mitigate this issue. For instance, Zhang et al. [15] introduced a variational Bayesian framework that utilizes variational inference to estimate a distribution for each class. A later work by Allen et al. [30] proposed Infinite Mixture Prototypes (IMP), which represent a single class using a set of clusters. IMP demonstrates improved performance by interpolating between the prototypical representation and the nearest neighbor approach.

Liu et al. [31] identified two primary factors influencing the performance of ProtoNet. The first factor is the low-data regime, which can lead to the mean vector inadequately representing the class. To address this issue, the authors proposed a cosine similarity-based ProtoNet with a bias diminishing module (BD-CSPN), which incorporates pseudo-labeling through transductive inference. In this approach, the support set is expanded with predicted unlabeled samples from the query set

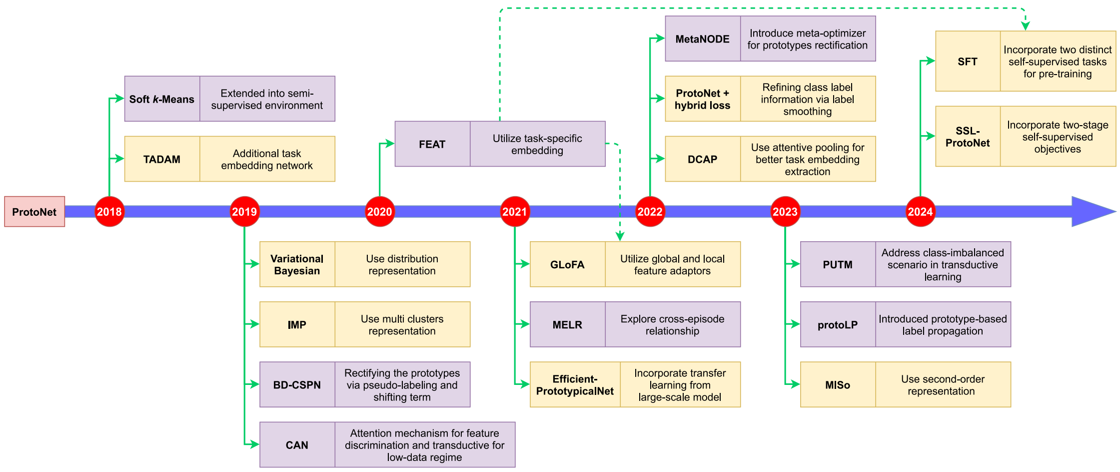


Fig. 4. The other extended works based on ProtoNet from 2018 to 2024. The dotted line represents the inheritance relationship between two works.

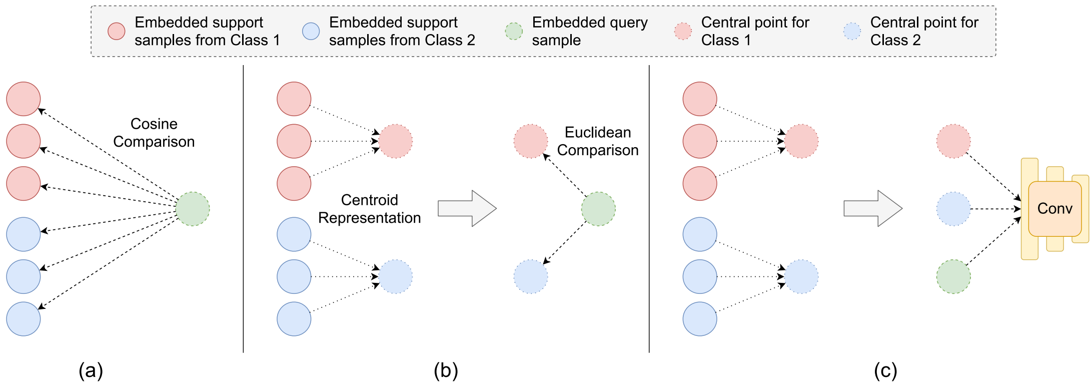


Fig. 5. The overall distance comparison of (a) MatchingNet, (b) ProtoNet, and (c) RelationNet.

that have high confidence scores. To account for potential misclassifications, the mean vector is computed using a weighted sum rather than simple averaging. The second factor identified is the discrepancy in distance between the mean vectors of the query set and the support set. To mitigate this issue, a shifting term was introduced to the query set, aligning its distribution relative to the support set. These strategies collectively offer promising solutions to enhance few-shot classification performance in low-data regimes.

Learning embedding representations based on base classes and applying them to novel classes is a task-agnostic approach that may not effectively discriminate between novel classes. To overcome this limitation, a new approach called Few-Shot Embedding Adaptation with Transformer (FEAT) [32] was introduced, which aims to obtain task-specific embeddings by transforming the task-agnostic embeddings through a set-to-set function. FEAT involves two training stages. In the first stage, the embedding model is pre-trained on base class samples using standard supervised learning to develop a generalized representation. In the second stage, FEAT employs a transformer as the set-to-set function, which is trained to transform the mean vectors derived from the embedding model into task-specific mean vectors. Additionally, the embedding model is fine-tuned during this stage.

Disregarding class diversity and obtaining task-specific information through a global transformation in FEAT motivated the development of the Global and Local Feature Adaptor (GLoFA) [33]. GLoFA addresses this issue by converting task-agnostic features into task-specific features using both global and local feature adaptors. The local adaptor facilitates the exploration of class-specific transformations, thereby enhancing the representation capability. Global masks are employed to capture the overall patterns of objects, while local masks focus on the detailed characteristics of objects. These adaptors enable GLoFA to adaptively measure the relationships between samples within tasks.

Transductive Propagation Network (TPN) [34] integrates transductive learning into the FSL framework to address challenges associated with low-data regimes. Like other metric-based approaches, TPN constructs a similarity graph by embedding the entire support and query sets. Labels are then propagated from the support set to the query set, followed by the application of transductive inference to enhance performance.

To achieve more discriminative features, Cross Attention Network (CAN) [35] integrates attention mechanisms into the meta-learner. This approach focuses on attending to important regions between the class feature map and the query feature map, thereby enhancing feature discriminability. Additionally, to address the challenges posed by low-data regimes, CAN employs a transductive technique that iteratively expands the support set with predicted query samples.

As the sample relationships are often neglected during the feature embedding stage, the deep K-tuplet network [36] aims to improve feature embedding by incorporating these relationships. To enhance generalization, the network updates its parameters using multiple negative samples within each training triplet, as opposed to the traditional approach that relies on a single negative sample.

RelationNet was initially designed to utilize ConvNet to learn the relations between support-query sample pairs. However, ConvNet's inherent local connectivity renders it sensitive to the spatial positioning of compared objects. To address this limitation, the Position-aware Relation Network (PARN) [37] was proposed, which employs a deformable feature extractor (DFE) to capture relevant features more effectively. Additionally, PARN mitigates the challenges posed by local connectivity through a dual correlation attention mechanism (DCA). The features obtained from self-correlation attention (SCA) and cross-correlation attention (CCA) are then combined to produce a more representative feature for classification.

Many metric-based approaches have conventionally focused on the relationship between the query sample and each support sample, often neglecting the relationships among the support samples themselves. To address this gap, He et al. [38] investigated the Memory-Augmented Relation Network (MRN), which enhances performance by leveraging these inter-support relationships. Unlike PARN that generates a more representative embedding from a single image, MRN improves representation by aggregating useful information from visually similar samples. Additionally, a relation module was introduced as a distance metric, and a memory slot for data augmentation was utilized to further enhance generalization.

To mitigate the background influence and huge intra-class variations, Zhang et al. [39] proposed DeepEMD to leverage discriminative local representations within a metric learning paradigm to achieve optimal image region matching. Instead of relying on global image features [13,40], the authors computed the structural distance between dense image representations using the Earth Mover's Distance (EMD), facilitating a more effective comparison of image regions. Furthermore, a cross-reference mechanism was incorporated to reduce the influence of background noise by assigning lower weights to unrelated regions, thereby enhancing the robustness and accuracy of the classification process.

Fei et al. [41] highlighted that the performance of episode-based few-shot approaches is highly dependent on the quality of few-shot sampling. To mitigate this issue, they proposed MELR, which exploits the relationships between episodes. By leveraging the relationship between two episodes containing the same group of classes during meta-training, MELR ensures that the model remains robust even under poor-sampling conditions during the evaluation phase.

In emphasizing the importance of learning a well-generalized embedding over developing a complex classifier, He et al. [42] introduced Dense Classification and Attentive Pooling (DCAP) to enhance task embedding extraction from pre-trained models. In the context of few-shot training, DCAP employs attentive pooling to aggregate feature maps for learning tasks, departing from the conventional use of features derived from a global average pooling layer. This strategic adjustment resulted in superior performance compared to other works that rely on global average pooling features during few-shot training.

Gao et al. [43] identified two critical issues within ProtoNet: unreliable label information and inflexible hyperparameters in the loss function. To address these challenges, the authors introduced a novel approach that incorporates label smoothing regularization to refine and enhance the reliability of class label information. Additionally, they integrated the distance matrix of the samples with the hyperparameters of the loss function, enabling greater adaptability to the specific requirements of the classification task. This strategic combination of label smoothing and flexible hyperparameter adjustments led to improved classification accuracy, effectively addressing the limitations identified in ProtoNet.

In another study, Zhang et al. [44] proposed a novel Multi-level Second-order (MlSo) network to tackle the challenges of FSL in both image classification and action recognition tasks. This innovative network leverages power-normalized second-order base learner streams alongside features that capture multiple levels of visual abstraction. Notably, it incorporates self-supervised discriminative mechanisms and utilizes second-order pooling to effectively manage convolutional feature maps of varying spatial sizes. By training in an end-to-end manner, MlSo demonstrates robust performance across various standard datasets.

To address the limitations of ConvNet in capturing global descriptions of samples and their tendency to overlook global task information, which weakens feature discrimination, Cui et al. [45] proposed the Dual Global-Aware method for label Propagation (DGAP). DGAP encodes dual global descriptions at both the sample and task levels. Unlike previous works [32,37] that utilize global descriptions, DGAP integrates a Global-Aware Sample Module (GSM) to enhance the feature representation capabilities of individual samples and a Global-Aware Task Module (GTM) to position task-specific features more effectively and discriminately within the feature space. The feature fusion module subsequently combines the features obtained from both the global sample and task perspectives, optimizing the label propagation module [34, 46] to achieve superior performance.

Existing approaches face limitations such as inaccurate prototype estimation and suboptimal graph construction in transductive learning. To address these issues, the protoLP [47] introduces a novel graph-based method that establishes relationships between prototypes and samples, moving beyond the conventional approach of treating prototypes merely as class centers. Additionally, current transductive methods [48–51] struggle with class-imbalanced scenarios, leading to inconsistent performance. To overcome these challenges, the Prototypes-oriented Unbiased Transfer Model (PUTM) [52] has been proposed specifically to tackle class-imbalanced transductive few-shot tasks. PUTM iteratively constructs transferable class-wise prototypes designed to capture unbiased statistics within class-imbalanced query samples. It utilizes task-adaptive forward and backward transport matrices with learnable marginal priors to ensure accurate estimation of class-wise marginal distributions.

Prior works [35,53,54] that focus on aligning semantically relevant regions encounter issues with similarity measures and intra-class variation. The proposed Class-aware Patch Embedding Adaptation (CPEA) [55] addresses these challenges by eliminating interference without aligning semantically relevant regions, thereby preventing supervision collapse. CPEA leverages self-supervised pre-training with Masked Image Modeling [56–59] to generate meaningful patch embeddings. By introducing a class-agnostic embedding into a transformer, CPEA transforms patch embeddings to be class-aware. This adaptation enhances relevance to classes and mitigates the scarcity of labeled images. Additionally, a proposed dense score matrix quantifies the similarity between class-relevant patch embeddings across images.

As the training process of ProtoNet neglects sample discrimination, Lim et al. [60] proposed SSL-ProtoNet to address this limitation. SSL-ProtoNet enhances model performance by incorporating sample discrimination via contrastive learning. This approach involves minimizing the distance between different representations of the same sample, thereby increasing its discriminability. The model is subsequently fine-tuned using a contrastive learning loss to ensure the generalization of its embeddings. Finally, self-distillation is employed to further refine the model's performance. This approach effectively addresses the initial limitation of ProtoNet by integrating sample-level discrimination, resulting in a more robust and generalized model.

To stabilize the fine-tuning process with a limited number of labeled samples, NegCosIC [61] employs a negative cosine similarity loss to preserve representation similarity during fine-tuning. Simultaneously, NegCosIC mitigates the risk of overfitting by introducing invariance and covariance losses as regularizers. To further enhance the robustness of the representations, DBDC-SSL [62] leverages the disparity between the product of marginals and the joint characteristic functions through Brownian Distance Covariance (BDC). Additionally, an auxiliary self-supervision pretext task is incorporated to further reduce the likelihood of overfitting. Finally, logistic regression is applied to predict the given few-shot tasks.

In contrast to FEAT, which emphasizes class-level discriminability, SFT [63] focuses on achieving instance-level diversity and discriminability to improve feature representation from the pre-training stage.

SFT utilizes two distinct self-supervised tasks to pre-train two separate models, thereby avoiding the challenges associated with training multiple tasks within a shared feature space. The trained models are then frozen and used as feature extractors to obtain features from the samples. These task-agnostic features are subsequently concatenated and transformed into task-specific features via a transformer, enhancing the model's ability to predict unseen tasks.

In summary, metric-based meta-learning approaches primarily focus on learning a feature space enriched with meta-level knowledge. These methods then employ pairwise similarity with previously seen samples to predict the classes of unseen samples. Generally, these approaches are simpler to implement and require less computational time during training. However, most metric-based meta-learning methods rely on a fixed embedding space and do not account for task-specific adjustments, making it challenging to classify novel tasks that differ significantly from the base tasks encountered during training. Additionally, the core component of these approaches — pairwise comparison — can lead to high computational costs as the task size increases. A summary of metric-based meta-learning approaches is provided in Table 2.

3.1.2. Model-based approaches

The objective of FSL is to acquire a classification strategy applicable across diverse tasks. In model-based meta-learning approaches, prior learning experiences are retained within a memory mechanism. These methods sequentially process support samples, storing the learning experience in memory as they proceed. The internal representation of tasks is statefully maintained based on the support sets, allowing the internal state of the model to adjust with each new input. This dynamic internal state captures relevant task-specific information, which is then utilized to predict new inputs. Since the prediction process relies on the dynamics of internal state, it is crucial to retain the information captured from previous inputs for accurate inference. Therefore, model-based approaches incorporate either an internal or external memory module to store learned knowledge.

Inspired by the Neural Turing Machine (NTM), Santoro et al. [64] proposed the Memory-Augmented Neural Network (MANN), which utilizes an external memory module instead of the internal memory found in Recurrent Neural Networks (RNN) or Long Short-Term Memory (LSTM). In this approach, encoded data about the classes of samples are stored in the external memory module and retrieved when new samples from the same class are encountered during prediction. The retrieval process was based on the cosine similarity with the input.

Subsequently, Meta Network (MetaNet) [65] was introduced to enable rapid generalization across different tasks. MetaNet consists of two subnetworks: the base-learner and the meta-learner. The base-learner encodes inputs into feature vectors for task learning, while the meta-learner receives loss gradients from the base-learner as meta-information. Unlike conventional neural networks that require time to compute weights through backpropagation, MetaNet employs the meta-learner to predict the parameters of the base-learner, referred to as "fast weights", facilitating faster task adaptation and improved generalization.

In contrast to approaches relying on external memory modules, the Simple Neural Attentive Meta-Learner (SNAIL) [66] introduces a specialized model architecture that serves as the memory module. Recognizing the limitations of RNN, such as constrained memory capacity and difficulty in pinpointing specific prior knowledge, SNAIL incorporates alternating causal temporal convolution layers and soft-attention layers. The temporal convolutions provide extensive memory access, while soft-attention enables precise knowledge retrieval. Similar to MANN, SNAIL processes tasks in a sequential manner. Different from RNN, SNAIL processes the entire sequence of the support set simultaneously during training. The input consists of a sequence of data-label pairs, including additional non-labeled data. Unlike MANN, which maps data at time t to labels at time $t-1$, SNAIL maps data to labels at the same time t, thereby accurately predicting labels for non-labeled data based on prior data-label pairs.

Alternatively, Cai et al. [67] proposed the Memory Matching Network (MM-Net), an approach based on MatchingNet with an integrated memory module. MM-Net stores embedded support samples in memory to generalize knowledge. During training, MM-Net employs a read controller to contextually embed support samples with the memory, while the contextual learner predicts the parameters of a ConvNet to embed the query sample. Similar to MatchingNet, a matching mechanism is then used to assign a label to the query sample by comparing it with support samples.

Several studies [68,69] have explored methods for dynamically generating classifier weights tailored to specific tasks. Among these approaches, Gidaris and Komodakis [70] introduced a model incorporating a lifelong memory structure designed to store feature representations of previously learned classes along with their corresponding parameters. To facilitate adaptation to novel tasks, the model employed a classification weight generator that leveraged an attention mechanism to produce task-specific weights. This enabled the classifier to effectively generalize to unseen samples by dynamically adjusting its parameters in response to new tasks.

In summary, model-based meta-learning approaches revolve around the concept of task internalization, where each encountered task is processed and represented within the hidden state of the model, i.e. memory module. The model then makes predictions on new tasks based on this hidden state, offering flexibility through its internal dynamics. However, as noted by Hospedales et al. [71], model-based approaches may not perform optimally when handling larger datasets. Additionally, Finn and Levine [72] highlighted that these approaches might struggle to generalize to tasks that are significantly different from previous knowledge. Consequently, researchers are shifting to the exploration of optimization-based and metric-based approaches. Table 3 summarizes the model-based meta-learning approaches.

3.1.3. Optimization-based approaches

In contrast to model-based approaches, optimization-based meta-learning employs a distinct strategy for FSL, focusing on explicit optimization for task learning. The core principle involves learning how to update the given model parameters to adapt to unseen tasks. This approach typically formulates FSL as a bi-level optimization problem, where a base-learner manages inner-level optimization, and a meta-learner handles outer-level optimization. The base-learner performs task-specific learning using an optimization strategy, such as gradient descent, while the feedback from this process informs the meta-learner's optimization across different tasks.

Andrychowicz et al. [73] incorporated the concept of meta-learning into the LSTM optimizer by proposing a novel update term that aligns the optimization process with a group of similar tasks. Their approach utilized two distinct networks to "learn to learn" an update policy across a set of tasks. The base-learner was responsible for managing specific tasks, whereas the meta-learner optimized the base-learner's parameters. Although their work did not apply the LSTM optimizer to a FSL setting, it laid the groundwork for subsequent developments in this area.

A later work by Ravi and Larochelle [74] extended the approach of [73] to the FSL scenario, explicitly modeling the optimization process within the meta-learner. In this context, the model performing the specific tasks was referred to as the learner, and the meta-learner efficiently updated the learner's parameters using a small portion of the support set, enabling the learner to quickly adapt to new tasks. The LSTM architecture was chosen due to its update process in the cell state, which is analogous to gradient-based optimization.

Given the complexity of the LSTM's gating mechanism, Model-Agnostic Meta-Learning (MAML) [75] was introduced as a simpler, gradient-based inner optimization algorithm designed for rapid task

Table 2
The summary of reviewed metric-based few-shot approaches. Under Backbone column: CN denotes ConvNet and RN denotes ResNet. Under Training column: E denotes episode and M denotes minibatch. P denotes pretraining. T denotes transductive learning, C denotes centroid representation, and A denotes attention mechanism.

Method	Year	Backbone	Training	P	T	C	A	Strength	Weakness
SiameseNet [25]	2015	CN	M	–	–	–	–	Effective for verification tasks	Struggles with complex multi-class tasks
MatchingNet [26]	2016	CN	E	–	–	–	✓	Improves adaptability via attention	Attention mechanisms increase computational overhead
ProtoNet [13]	2017	CN	E	–	–	✓	–	Simple and scalable implementation	Struggles with diverse intra-class distributions
RelationNet [27]	2018	CN	E	–	–	✓	–	Use a learnable metric for complex feature adaptation	Learnable metric introduces additional resource overhead
Soft k-Means [28]	2018	CN	E	–	✓	✓	–	Improves ProtoNet with unlabeled data	Sensitive to unlabeled data quality
TADAM [29]	2018	RN	E	–	–	✓	–	Enables dynamic task adaptation	Complex architecture with hyperparameter sensitive
TPN [34]	2019	CN, RN	E	–	✓	–	–	Leverages unlabeled query data during inference to exploit task-specific structure	Graph construction and label propagation scale poorly with large test sets
PARN [37]	2019	CN	E	–	–	–	✓	Improve robustness by adapting to object spatial variations	Computational overhead and domain-sensitive
Variational Bayesian [15]	2019	CN, RN	E	✓	–	✓	–	More reliable with probabilistic distributions	Variational inference introduce more computational overhead
IMP [30]	2019	CN	E	–	–	✓	–	Infers number of cluster per class dynamically to handles complex distributions	Computational complexity and sensitive to initialization
CAN [35]	2019	RN	E	–	✓	✓	✓	Dynamic feature adaptation	High compute cost with complex training
BD-CSPN [31]	2020	CN, RN, WRN	M	–	✓	✓	–	Robust prototypes with bias mitigation	Dependence on query set quality
Deep K-Tuplet network [36]	2020	RN	E	–	–	–	–	Semi-hard mining strategy stabilizes training and speeds up convergence	Computational overhead for large K
FEAT [32]	2020	CN, RN, WRN	M+E	✓	✓	✓	✓	Exploit task-specific features	High complexity and tuning sensitivity
MRN [38]	2020	CN	E	–	✓	✓	–	Context-ware embedding with flexible metric	Memory overhead during storing and retrieving memory
DeepEMD [39]	2020	RN	M+E	✓	–	✓	–	Focus on discriminative local features while minimizing background noise	Requires computationally intensive optimization
GLoFA [33]	2021	RN	M+E	✓	–	✓	–	Captures both broad patterns and fine details	Additional complexity with dual feature adaptors
MELR [41]	2021	RN	M+E	✓	✓	✓	✓	Against poorly-sampled data via episode relationships	Sampling two episode with same classes increase computational overhead
DCAP [42]	2022	CN, RN	M+E	✓	–	✓	✓	Improves embeddings via dense classification and attentive pooling	Dense local matching increases method complexity
ProtoNet + hybrid loss [43]	2022	CN, RN	E	–	–	✓	–	Dynamically adjusting loss function hyperparameters	Introduces complexity to the loss function
DGAP [45]	2022	CN, RN	M+E	✓	✓	–	✓	Enhances feature discrimination by encoding global sample and task descriptions	Introduces additional computational complexity
protoLP [47]	2023	RN, WRN	M+E	✓	✓	✓	–	Accurate prototype refinement	Computationally intensive
PUTM [52]	2023	RN, WRN	M+E	✓	✓	✓	✓	Effectively handles class-imbalanced data	Transport matrices and iterative optimization increase computational cost
MlSo [44]	2023	CN, RN	E	–	–	✓	✓	Improves discriminative power by capturing complex feature relationships	Second-order pooling increases memory and runtime costs

(*continued on next page*)

adaptation. The meta-learner was aimed to explicitly optimize the base-learner for task adaptation in a single step. It utilized the validation loss from the base-learner to complete the gradient descent optimization, thereby deriving a well-generalized initial parameter set for the base-learner in future tasks. Through this process, the base-learner acquired task-specific knowledge, while the meta-learner gained meta-level knowledge. After a few gradient updates during training, the model demonstrated smooth performance on specific tasks.

Table 2 (*continued*).

CPEA [55]	2023	ViT	M+E	✓	–	–	✓	Improves patch embeddings by making them class-relevant	Additional complexity in the embedding adaptation process
SSL-ProtoNet [60]	2024	CN	M+E	✓	–	✓	–	Enhances sample discrimination via self-supervised learning	Requires multi-stage training process
NegCosIC [61]	2024	RN, WRN	M+E	✓	–	–	–	Stabilizes the fine-tuning process and regularizes variance	Added regularization terms increase the training complexity
DBDC-SSL [62]	2024	RN	M+E	✓	–	–	–	Captures higher-order dependencies via BDC	BDC computation is resource-intensive
SFT [63]	2024	RN	M+E	✓	–	✓	✓	Improves feature diversity and discriminability	Complex multi-stage training pipeline

Table 3
The summary of reviewed model-based few-shot approaches. Under Backbone column: CN denotes ConvNet, RN denotes ResNet. Under Training column: E denotes episode and M denotes minibatch. A denotes attention mechanism.

Method	Year	Backbone	Training	A	Strength	Weakness
MANN [64]	2016	–	E	–	Excel in memory-intensive tasks	Memory bottlenecks and computational overhead
MetaNet [65]	2017	CN	E	✓	Offer dynamic and rapid adaptation	Training complexity and resource demands
SNAIL [66]	2018	CN, RN	E	✓	Avoids hand-designed components	Lack of interpretability
MM-Net [67]	2018	CN	E	–	Leverages memory for rapid adaptation	Memory and computational overhead
Dynamic-Net[a] [70]	2018	CN, RN	M+E	✓	Mitigates catastrophic forgetting	Dependency on base class training

[a] Involved pretrain and centroid.

The generalization capability of MAML allows it to work with any model utilizing gradient descent. However, the computation of second-order gradients in MAML led to increased time and memory consumption. To address this, Finn et al. [75] also proposed First-Order MAML (FOMAML), which eliminates second-order gradients to reduce computational costs. Despite the simplification, the performance of FOMAML was comparable to MAML, indicating that first-order gradients were sufficient for effective initialization updates. This was particularly advantageous for FSL, which requires multiple gradient steps during training. Although FOMAML improved performance on tasks requiring more gradient steps, it still struggled to retain most gradient information compared to MAML. Additionally, both MAML and FOMAML were heavily dependent on the model architecture, prone to overfitting, and required extensive computation time to optimize hyperparameters. To address these challenges, several variants of MAML have been proposed, offering various improvements, as illustrated in Fig. 6.

Li et al. [76] introduced Meta-SGD, a simple yet effective approach for FSL. Like MAML, Meta-SGD learns initial parameters but additionally incorporates the initialization of learning rates within the meta-learner. Moreover, Meta-SGD updates the base-learner's direction, similar to prior works [73,74], leading to simpler implementation, faster learning, and improved generalization. Notably, Meta-SGD adapts to new tasks with just a single inner-step update.

In an effort to reduce complexity, Nichol et al. [77] proposed Reptile, a simplified optimization algorithm for FSL. Like FOMAML, Reptile is easy to implement, uses gradient descent in meta-optimization, and is model-agnostic. However, Reptile diverges from FOMAML by eliminating the train–test split paradigm, instead utilizing task sampling and multiple gradient descent steps during task training. After training, Reptile adjusts the initial model weights towards the trained weights, embedding an appropriate inductive bias across related tasks.

Subsequently, several studies [78–80] integrated probabilistic models into MAML to enhance performance. For instance, this study [78] introduced Laplace Approximation for Meta-Adaptation (LLAMA), a framework that employs MAML as its foundational probabilistic model, using Laplace approximation to compute uncertainty. Unlike MAML, LLAMA produces multiple solutions with varying uncertainties by using a Bayesian approach rather than a single solution. Similar to LLAMA, Finn et al. [79] proposed PLATIPUS based on the probabilistic interpretation. Unlike LLAMA that computed probabilistic distribution for the task-specific parameters, PLATIPUS prioritized global initialization parameters akin to MAML. They focus on the probabilistic distribution of these parameters to address the challenges posed by ambiguous FSL tasks. In another related work [80], they proposed Bayesian MAML (BMAML), a modified MAML utilizing a probabilistic model. BMAML introduced multiple solutions within a more complex Bayesian framework and employed Stein Variational Gradient Descent (SVGD) for the inner-loop update. Additionally, BMAML incorporated a Coined Chaser loss for meta-level optimization, enhancing both stability and performance.

Lee and Choi [81] introduced Mask Transformation Networks (MT-net), which employ a meta-learner to learn the activation space of all layers, forming a learned distance metric for the base-learner. In this method, each task is assigned its own specific knowledge rather than sharing knowledge across all tasks, allowing MT-net to achieve better results with fewer training steps compared to MAML.

Vuorio et al. [82] proposed Multi-Modal Model-Agnostic Meta-Learner (MUMOMAML), a variant of MAML designed to address tasks with high diversity arising from a multimodal task distribution. MUMOMAML comprises a model-based meta-learner and a gradient-based meta-learner. The model-based meta-learner identifies the task's modes based on a few samples, subsequently modulating the gradient-based meta-learner's parameters to quickly adapt to the specific task within the detected mode. The gradient-based meta-learner, similar to that in MAML, learns a good parameter initialization through a few update steps for faster task adaptation.

Antoniou et al. [83] introduced MAML++ to address several issues inherent in MAML. First, MAML++ reduces training time by applying first-order gradients during the initial 50 epochs. Second, it uses a separate set of biases for each layer to learn the feature distribution specifically for each step. Third, MAML++ employs a different learning rate in the inner-loop step. Finally, a meta-learner is utilized during training to prevent overfitting and improve generalization across tasks.

To reduce memory consumption, this study [84] proposed iMAML, which approximates higher-order derivatives with lower memory consumption instead of eliminating them. iMAML focuses solely on the

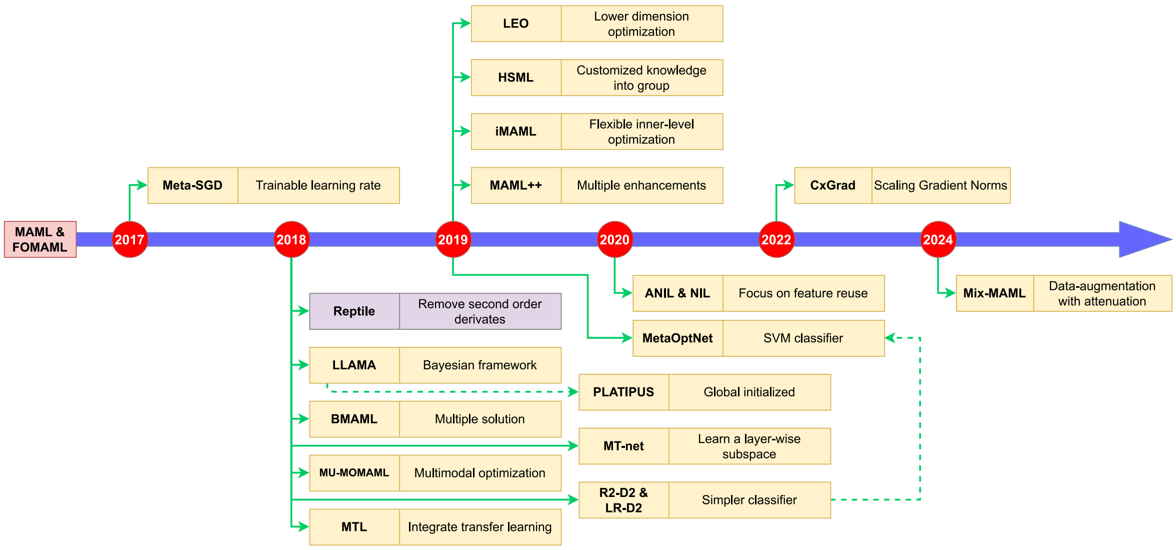


Fig. 6. The relationship between MAML and the other variants from 2017 to 2024. The dotted line represents the inheritance relationship between two works.

solution to the inner-level optimization and requires explicit regularization to prevent overfitting, particularly when using a greater number of inner gradient update steps. With its reduced memory consumption, iMAML is suitable for use with non-differentiable inner optimizers.

To address the poor generalization often encountered when MAML utilizes limited samples to compute gradient information in high-dimensional spaces, Latent Embedding Optimization (LEO) [85] was developed. LEO improved generalization by learning optimal initial parameters for the base-learner in a lower-dimensional embedding space. Samples were encoded into feature vectors using RelationNet [27], and these features were then used to compute task-specific weights in a low-dimensional space through a decoder.

In contrast to previous approaches, Bertinetto et al. [86] introduced two novel methods: Ridge Regression Differentiable Discriminator (R2-D2) and Logistic Regression Differentiable Discriminator (LR-D2). These methods employed simple base-learners to enable faster learning speeds within a closed-form solution while still exploring task-specific information. Similarly, MetaOptNet [87] replaced MAML's base-learner classifier with a linear SVM, demonstrating that a linear classifier could achieve better generalization than a nearest-neighbor classifier.

Recognizing the limitations of global knowledge sharing in MAML for handling task uncertainty, Yao et al. [88] proposed Hierarchically Structured Meta-Learning (HSML). This framework tailored knowledge to distinct groups of similar tasks, consolidating generalized knowledge within each task group to generate initial parameters. In a continual learning environment, new tasks were categorized based on existing groups and updating the group structure as needed. This approach allowed the model to expand the clustering structure when new tasks did not fit into the current task groups.

Raghu et al. [89] investigated MAML from the perspective of feature reuse rather than rapid adaptation, proposing Almost No Inner Loop (ANIL). ANIL avoided inner loop optimization by freezing all layers except the final ones during both training and testing phases, thus highlighting the feature reuse capability. They further introduced No Inner Loop (NIL), which entirely removed the inner loop network during testing, relying on learned features from the outer loop to perform classification using cosine similarity.

One of the limitations in MAML is focusing on generic task features over backbone layers, which led to the introduction of Contextual Gradient Scaling (CxGrad) [90]. CxGrad involved scaling the gradient norms of the model's backbone in a task-specific manner, enhancing the acquisition of task-specific knowledge. This approach significantly improved MAML's performance in both in-domain and cross-domain few-shot classification scenarios.

To preserve the simplicity and model-agnostic nature of MAML while addressing generalization challenges, Jia et al. [91] introduced Mix-MAML. This approach incorporates a novel data augmentation technique with an attenuation strategy to better explore task relationships, rather than applying a uniform global initialization across all tasks. Additionally, the study examines the use of diverse backbone architectures and varying resolutions to enhance performance, moving beyond the shallow backbones that typically limit model efficacy.

In summary, optimization-based meta-learning approaches have proven highly effective in achieving superior performance across a wide range of tasks, primarily by leveraging the base-learner's ability to learn task-specific information. The meta-learner then incorporates feedback from the base-learner to generate generalized knowledge, enabling the base-learner to adapt quickly to new tasks. However, as noted by this work [71], these approaches can be computationally expensive, as the base-learner must optimize for each task encountered. Furthermore, this study [89] found that the success of optimization-based approaches hinges more on robust feature extraction than on rapid adaptation in the base-learner. Table 4 summarizes the discussed optimization-based meta-learning approaches.

3.2. Transfer learning approaches

In addition to meta-learning approaches, recent studies [40,92–94] have demonstrated that pre-training an embedding model before applying it to FSL significantly enhances performance. This approach leverages transfer learning to provide well-generalized features that are suitable for few-shot tasks, as illustrated in Fig. 7. Transfer learning approaches for FSIC can be broadly categorized into hybrid and non-hybrid approaches.

3.2.1. Hybrid approaches

Several studies [93,95] have integrated transfer learning with meta-learning to enhance generalization performance. These hybrid approaches leverage the ability of transfer learning to extract useful representations from pre-trained models while utilizing meta-learning to improve classification in low-data scenarios. By combining these methodologies, researchers aim to develop more robust models capable of adapting to novel tasks with minimal supervision.

Table 4
The summary of reviewed optimization-based few-shot approaches. Under Backbone column: CN denotes ConvNet and RN denotes ResNet. Under Training column: E denotes episode and M denotes minibatch.

Method	Year	Backbone	Training	Strength	Weakness
Meta-Learner LSTM [74]	2017	CN	E	Flexible learned optimizer	Complex LSTM's gating mechanism and limited scalability
MAML [75]	2017	CN	E	Simple model-agnostic method with rapid task adaptation	High memory and time consumption
Meta-SGD [76]	2017	CN	E	Higher capacity with faster adaptation than MAML	Higher memory and computation cost than MAML
Reptile [77][c]	2018	CN	E	Efficient first-order approximation	Omitted curvature information
LLAMA [78]	2018	CN	E	Enables improvement via Bayesian techniques	Complex implementation
PLATIPUS [79]	2018	CN	E	Handles uncertainty in FSL tasks	High computational cost
BMAML [80]	2018	CN	E	Robust to via meta-level overfitting	Scalability issues with high-dimensional models
MT-net [81]	2018	CN	E	Enables layerwise adaptation with stable training	Architecture complexity
MU-MOMAML [82][b]	2018	CN	E	Handles multimodal tasks with faster adaptation	Complex modulation network
R2-D2 [86]	2018	CN	E	Avoids iterative fine-tuning for rapid adaptation	Limited scalability
MAML++ [83]	2019	CN	E	Improves stability and efficiency of MAML	Second-order gradients limiting scalability
iMAML [84]	2019	CN	E	Approximating higher-order derivatives with lower memory consumption	Computationally intensive and sensitive to curvature
LEO [85][a]	2019	CN, WRN	M+E	High-dimensional scalability and aware task uncertainty	Complex architecture and latent space limitations
HSML [88]	2019	CN	E	Handles task heterogeneity with continual learning	Computationally complex and cluster-sensitive
MetaOptNet [87]	2019	CN, RN	E	Learning high-quality embeddings without excessive computational overhead	Convexity dependency and feature-sensitive
ANIL [89]	2020	CN	E	Computational efficient than MAML via reusable features	Limited to scenarios with applicable feature reuse
CxGrad [90]	2022	CN	E	Scales backbone gradients to learn task-specific features	Requires tuning of scaling factors for different domains
Mix-MAML [91]	2024	CN, RN	E	Robust to overfitting in deeper architectures	High computational overhead

[a] Pre-training.
[b] Attention mechanism.
[c] Transductive learning.

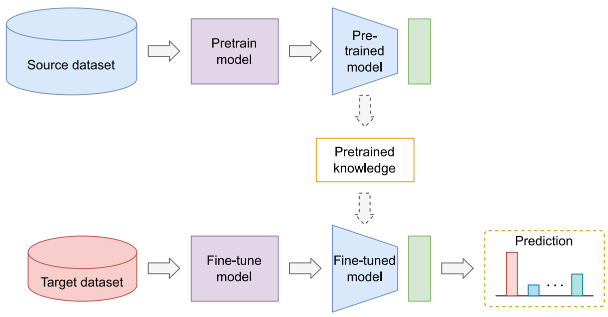


Fig. 7. The overall concept of transfer learning.

Chen et al. [93] combined the strengths of pre-training and meta-learning in their Meta-Baseline approach. They pre-trained an embedding model as a feature extractor and used it to compute average features for each class. These features were then used to calculate cosine nearest-centroid metrics between the embedded support set and query set. Lim et al. [95] explored the use of transfer learning with a pre-trained vision foundation model, which served as a feature extractor to obtain features for each input. These features were subsequently fed into a metric-based few-shot model for the FSL process. To further enhance the model's performance, sequential self-knowledge distillation was employed, allowing the model to distill its own knowledge, thus improving performance and mitigating overfitting in low-data regimes.

Huang et al. [96] introduced the Poisson Transfer Network (PTN) for semi-supervised FSL, which operates in three stages: pre-training, unsupervised transfer, and label inference. During the pre-training stage, an embedding representation was learned from a large set of base class samples [40,94]. To address domain shifts between base and novel classes, PTN utilized contrastive learning to transfer knowledge from pre-trained embeddings of base classes to novel classes. This method minimized intra-class variation while maximizing inter-class separability. In the label inference stage, PTN incorporated Poisson learning [97] within a graph-based semi-supervised FSL framework, effectively reducing cross-class bias.

Zhang et al. [98] proposed MetaNODE, which employed a meta-optimizer to address the prototype bias issue identified in earlier work [93]. Prototype bias occurs when the computed mean vector deviates from the true mean due to a limited number of support samples. MetaNODE comprises two phases: pre-training and meta-training. The pre-training phase follows the prior work [99] to ensure generalizable feature learning. During the meta-training stage, the meta-optimizer refines the prototype mean vector before computing cosine similarity.

Prior works [100,101] introduced Meta-Transfer Learning (MTL), integrating transfer learning with meta-learning to improve task adaptation. MTL adjusts the weights of a pre-trained deep neural network using scaling and shifting functions, enabling efficient adaptation to new tasks. The Hard Task (HT) meta-batch strategy was implemented to enhance model adaptability by identifying failure classes within each task and re-combining them to create more challenging learning scenarios. This approach makes MTL architecture-agnostic, allowing broad applicability in various network designs.

3.2.2. Non-hybrid approaches

Alternatively, several studies have demonstrated that a well-generalized pre-trained embedding model alone can achieve strong performance on few-shot tasks, even without meta-learning. Chen et al. [40] showed that pre-training an embedding model on a large number of base class samples using standard cross-entropy loss yielded performance comparable to conventional FSIC methods. After pre-training, model weights were frozen, and a new classifier was trained on novel samples. The authors highlighted the importance of minimizing intra-class variation to optimize performance, particularly when using shallow embedding models.

Wang et al. [92] proposed SimpleShot, a method that applied feature transformations (e.g., mean-subtraction and L2-normalization) to a nearest-neighbor classifier. The model was initially trained through conventional supervised learning, and the embedded features were used for FSIC with additional feature transformations. Similarly, Tian et al. [94] demonstrated that a simple, well-learned embedding model could outperform complex FSIC algorithms. Their approach involved training a multivariate logistic regression classifier on the features extracted from the frozen embedding model, followed by refinement through self-distillation.

To enhance feature embeddings for few-shot tasks, several studies [102–104] incorporated additional distortion-based self-supervised tasks [105] into the training framework. Rajasegaran et al. [102] proposed the self-supervised knowledge distillation (SKD) that preserves intra-class diversity while capturing inter-class knowledge. SKD comprises two stages: Generation-zero and Generation-one. In Generation-zero, an embedding model was pre-trained similarly to [94], but with added self-supervised learning to enhance feature representations. Unlike [99], which applied self-supervised learning at the embedding layer, SKD constrained the output manifold by predicting the rotation label on the outcomes from the output layer. In Generation-one, the model adopted a teacher-student framework for knowledge distillation, ensuring robust manifold learning.

Conversely, prior works [103,104] employed one-stage training without explicit knowledge distillation as in [94,102]. Singh and Mazumder [103] introduced a novel distortion-based self-supervised task that modified the rotation degrees of image regions, forming a complex classification task extending [105]. Lim et al. [104] proposed Self-supervised Contrastive Learning (SCL), a model designed to enhance generalization through sample discriminability and diversity. SCL incorporated contrastive-based self-supervision, which minimized distances between augmented and original samples, thereby improving feature discrimination. Additionally, rotation-based self-supervision enabled the model to recognize rotation degrees, enhancing sample diversity. The evaluation followed the methodology of [94], employing multivariate logistic regression to assess performance based on extracted features. The general architecture of RFS [94], SKD [102], and SCL [104] is shown in Fig. 8.

In summary, non-hybrid approaches bypass episode-based training and instead merge all episodes into a single task, utilizing conventional supervised learning. This strategy improves class-level representation but may neglect episode-level relationships, potentially reducing effectiveness in distinguishing novel or closely related classes. Conversely, hybrid approaches employ multi-stage training to explore episode-level relationships, facilitating better novel class recognition. However, these methods require complex mechanism design and incur higher computational costs. Table 5 summarizes the reviewed transfer learning approaches.

3.3. Data augmentation approaches

Rather than designing complex architectures for meta-learning or leveraging pre-trained embeddings for generalization, data augmentation approaches offer an alternative pathway. These methods directly address the low-data regime by generating additional training samples based on limited available data, enhancing sample diversity and mitigating overfitting during FSL, as shown in Fig. 9.

To tackle the challenges of limited data, Hariharan and Girshick [106] introduced a data augmentation approach that synthesizes new feature vectors for novel classes. By increasing intra-class variation, their method mitigates overfitting, ensuring robust feature extraction through a loss function that penalizes differences between base and novel class classifiers. Instead of synthesizing diverse and realistic samples, Wang et al. [107] proposed a trainable hallucinator that generates useful classification samples by training it jointly with a meta-learner in an end-to-end manner. Their proposed approach also integrates strength of both MatchingNet [26] and ProtoNet [13] to enhance representation learning.

Inspired by the success of Generative Adversarial Networks (GANs) [108], Mehrotra and Dukkipati [109] developed a GAN-based regularizer for a novel metric-based model. Unlike fixed-distance metric approaches [13,25,26], their model employs a trainable distance measure using a residual network architecture, Skip Residual Pairwise Network (SRPN), to compute expressive pairwise similarity objectives. Zhang et al. [110] extended this concept with MetaGAN, a task-conditioned generator that sharpens decision boundaries in most of the FSIC methods by training a classifier to distinguish real from generated samples in an adversarial manner. Instead of conventional GANs, Li et al. [111] introduced a conditional Wasserstein GAN (cWGAN) with two novel regularizers, ensuring the generated features are both diverse and discriminative while maintaining training stability.

To further enhance model performance, Chen et al. [112] proposed a trainable image deformation sub-network within a meta-learning framework. Unlike methods that synthesize new features from scratch, their model deforms existing samples via a learned transformation, augmenting the support set for improved embedding learning. Inspired by the success of the saliency network, Zhang et al. [113] leveraged a pre-trained saliency network [114] to segment foreground and background components of images, merging them to create composite samples for training a similarity network.

To address the feature deterioration and unreliable prediction in the metric-based meta-learning approaches, Tang et al. [115] introduced BlockMix, a data augmentation technique designed to improve generalization in metric-based meta-learning. Their method randomly selects and mixes sub-blocks from different samples and assigns them mixed labels, while a meta-regularization task predicts the mixed sub-blocks to enhance prototype accuracy in novel classes. Osahor and Nasrabadi [116] addressed overfitting with a low displacement rank (LDR) regularization strategy that incorporates multiple augmentation techniques. Their method, based on a doubly block-Toeplitz (DBT) matrix structure, improves intra-class feature embeddings while reducing memory and computational requirements.

Rather than relying on external data, Zhang et al. [117] proposed a novel augmentation approach, self-mixup (SM), which mixes different

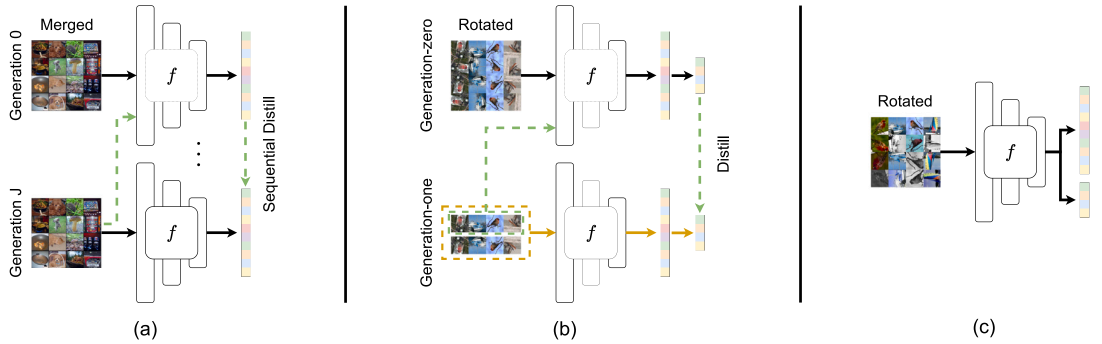


Fig. 8. The overall concept of (a) RFS [94], (b) SKD [102], and (c) SCL [104]. They used logistic regression as the classifier during the evaluation stage.

Table 5
The summary of reviewed transfer learning few-shot approaches. # denotes the hybrid category (H) and non-hybrid category (NH). Under Backbone column: CN denotes ConvNet, RN denotes ResNet, DenseNet denotes DN, MobileNet denotes MN, and EfficientNet denotes EN. Under Training column: E denotes episode and M denotes minibatch. P denotes pretraining. T denotes transductive learning, and C denotes centroid representation.

#	Method	Year	Backbone	Training	P	T	C	Strength	Weakness
H	MTL [100]	2019	RN	M+E	✓	–	–	Fast adaptation with better training stability	Requires careful task-specific tuning
	PTN [96]	2020	WRN	M	✓	✓	✓	More stable label propagation	High complexity and unlabeled data dependency
	Meta-baseline [93]	2021	RN	M+E	✓	–	✓	Simple pipeline with scalability	Performance relies heavily on pre-training quality
	Efficient-PrototypicalNet [95]	2021	EN	M+E	✓	–	✓	Offer better generalization with simple structure	Performance relies heavily on pre-training embeddings
	MetaNODE[a] [98]	2022	RN	M+E	✓	✓	✓	Model prototype dynamics more accurately	Resource-intensive and hyperparameter sensitivity
NH	Baseline++ [40]	2019	CN, RN	M	✓	–	–	Simple pipeline with scalability	Performance relies heavily on pre-training quality
	SimpleShot [92]	2019	CN, RN, WRN, DN, MN	M	✓	–	✓	Avoids meta-training overhead with minimal preprocessing	Performance relies heavily on pre-training embeddings
	RFS [94]	2020	CN, RN	M	✓	–	–	Avoids complex meta-learning design	Requires multi-stage training
	SKD [102]	2021	RN	M	✓	–	–	Improved generalization with multitask pre-training	Requires balancing of entropy objectives in two-stage pipeline
	DCRL [103]	2022	RN	M	✓	–	–	Complex pretext tasks enhanced embedding generalization	Performance relies heavily on pre-training quality
	SCL [104]	2023	RN	M	✓	–	–	Rich feature learning through multiple pretext tasks	High computational cost and hyperparameter sensitivity

[a] Attention mechanism.

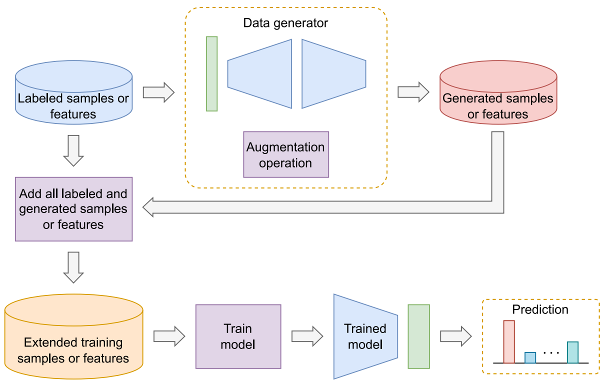


Fig. 9. The overall concept of data augmentation.

augmented variations of the same sample at the feature level. To further refine feature extraction in metric-based FSL, they introduced Calibration-Adaptive Downsampling (CADS), a method that minimizes information loss during feature extraction.

Autoencoder-based approaches have also been explored for data augmentation. Schwartz et al. [118] introduced Delta-encoder, which synthesizes additional samples using a pre-trained feature extractor within an autoencoder framework. Chen et al. [119] extended the idea of autoencoder with a dual TriNet architecture, where the encoder maps features into a semantic space, and the decoder reconstructs them into the visual domain for dataset expansion. Xian et al. [120] combined Variational Autoencoder (VAE) and GAN techniques to generate discriminative and interpretable visual features under zero-shot and few-shot environments. Their approach utilizes a non-conditional discriminator to infer feature distributions from unlabeled novel samples, allowing the model to operate in both inductive and transductive settings.

In summary, data augmentation approaches aim to generate additional training samples to alleviate overfitting in low-data scenarios.

Table 6
The summary of reviewed data augmentation few-shot approaches. Under Backbone column: CN denotes ConvNet and RN denotes ResNet. Under Training column: E denotes episode and M denotes minibatch. P denotes pretraining. T denotes transductive learning, and C denotes centroid representation.

Method	Year	Backbone	Training	P	T	C	Strength	Weakness
SGM [106]	2017	RN	M	✓	–	–	Combined feature regularization and hallucination	Handcrafted feature transformations may not generalize well
SRPN [109]	2017	WRN	M	–	–	–	Learns an adaptive similarity metric	Suffers from unstable GAN training
PMN[a] [107]	2018	RN	M+E	✓	–	✓	Works seamlessly with different meta-learners	Hallucinations may lack diversity, and performance depends on pre-trained features
Delta-encoder [118]	2018	VGG, RN	M	✓	–	–	Learns non-linear deformations between same-class pairs	Struggles with cross-class generalization
MetaGAN [110]	2018	CN	E	–	–	–	Sharpens decision boundaries using adversarial fake data	Relies on generator quality for meaningful fake data
Dual TriNet [119]	2019	RN	M	–	–	–	Smartly augmenting features using semantic relationships	Relies on pre-trained semantic data and adds computational overhead
f-VAEGAN-D2 [120]	2019	RN	M	✓	✓	–	Generates high-quality features even with no labeled data	Heavily relies on good semantic embeddings with complex computation
IDeMe-Net [112]	2019	RN	E	–	–	✓	Leveraging smart deformations that retain critical semantic information	High complexity and reliance on good gallery images
SalNet [113]	2019	CN	E	✓	–	–	Cost-effective hallucination via saliency maps instead of GANs	Performance depends on saliency map accuracy
AFHN [111]	2020	RN	M+E	✓	–	✓	Generate diverse and discriminative features	Complex adversarial training with computationally expensive and unstable
BlockMix [115]	2020	RN	E	–	✓	✓	Forcing smarter feature learning and more reliable predictions	Improvement shrinks when more data are available
Ortho-shot [116]	2022	CN	E	–	–	–	Reduce computational complexity via DBT matrices	DBT-based regularization requires careful matrix structuring and tuning
Self-Mixup [117]	2024	RN	M	–	–	✓	Improved data utilization and feature extraction without external data	Increased computational complexity and dependence on augmentation quality

[a] Attention mechanism.

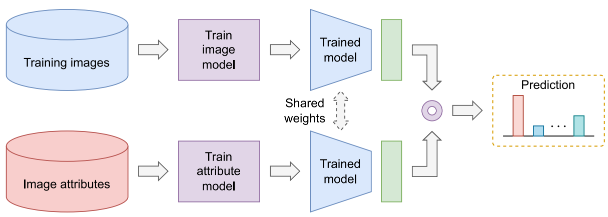


Fig. 10. The overall concept of attribute-related approaches.

However, their effectiveness is inherently constrained by the quality of the synthesized samples, limiting their overall impact on generalization. Table 6 presents a summary of the discussed data augmentation approaches.

3.4. Attribute-related approaches

Attribute-related approaches aim to establish connections between samples from different classes by leveraging attribute annotations to obtain additional semantic information. This additional information guides the model learning process by providing a visual link between base and novel classes, thereby enhancing generalization. Unlike other FSIC methods, approaches within this category typically require at least two types of inputs: images and attribute annotations, as depicted in Fig. 10.

Zhu et al. [121] investigated the role of attribute learning in FSIC and introduced Attribute-Guided Feature Learning (AGFL). AGFL utilizes attribute information to establish connections between base and novel classes by recognizing both attributes and categories in a multitask learning environment. This approach enables AGFL to leverage prior knowledge by identifying shared attributes across novel classes. Furthermore, AGFL can be integrated with existing metric-based meta-learning approaches [13,26]. Similarly, Auto-ACNet [122] employed a neural architecture search (NAS) technique, specifically Differentiable Architecture Search (DARTS), to automatically determine the optimal network structure for improving FSIC performance.

Several existing works [123,124] have focused on using auxiliary semantics to enhance only support set representations while ignoring query samples. To address this limitation, Huang et al. [125] proposed the Attributes-Guided Attention Module (AGAM), which incorporates query sample information. AGAM employs two parallel branches: an attributes-guided branch and a self-guided branch, integrated with spatial- and channel-wise attention modules. The attributes-guided branch extracts discriminative features from support sets based on both visual and attribute information, while the self-guided branch refines purely visual representations for both support and query samples. To ensure meaningful feature extraction of the self-guided branch in the absence of attribute annotations, AGAM aligns the attention weights of both branches through an attention alignment mechanism, thereby mitigating mismatches between support and query samples with the same label.

ADRGN [126] further advanced attribute-based learning by developing a transformation network to create specific attribute representations, highlighting the relationship between attributes and visual patterns. To capture visual diversity across different samples, ADRGN employs inter-sample routing, while inter-attribute routing models relationships between attributes to determine their significance in distinguishing classes. Additionally, knowledge distillation is used to enhance attribute prediction and capture fine-grained sample relations, facilitating improved graph-based inference for information propagation.

Prototype-related methods [127,128] have also been explored within attribute-related approaches. Xu et al. [127] extended [129] into FSIC and proposed the Attribute Prototype Network (APN). APN aims to enhance the locality of image representations by leveraging semantic visual attributes while reducing background noise through the identification of informative regions. APN utilizes attribute prototypes to generate similarity maps that highlight relevant image regions for specific attributes. In contrast, Lyu and Wang [128] introduced the Compositional Prototypical Network (CPN), which learns component prototypes based on attribute annotations and fuses them with visual prototypes obtained from ProtoNet [13]. CPN employs a learnable weight generator to adaptively combine these prototypes, improving FSIC performance by leveraging both visual and attribute-based information. Like previous works [123–125], CPN also utilizes category-level attribute score vectors to enhance classification.

AADNet [130] incorporates attribute distribution similarity alongside class similarity to enhance FSIC performance. AADNet pre-trained a feature extractor to learn fundamental attributes from the training set, which are subsequently refined by an adaptive task attribute-enhancing module. This module selectively enhances relevant attributes using the LLE algorithm. AADNet then computes the similarity between the attribute distributions of the support class and query sample, integrating it with class similarity via a fusion module to generate final classifications. This approach enables AADNet to leverage both attribute and class-level information to improve FSIC performance.

Building on AGAM [125], Wang et al. [131] proposed the Attribute-Guided Attention Network (AGAN), which utilizes self-attention and cross-attention mechanisms to refine learned representations. AGAN analyzes feature correlations within and across samples to improve representation learning. Additionally, an attribute prediction module for query samples is introduced to reduce the discrepancy between visual and attribute representations, leading to more accurate prototype generation and improved classification.

In summary, attribute-related approaches enhance model generalization in low-data regimes by incorporating auxiliary attribute information, thereby establishing a stronger connection between base and novel classes beyond visual features alone. However, these approaches are constrained to scenarios where attribute annotations are available, and designing mechanisms that fully exploit this information remains a significant challenge. Table 7 provides a summary of the reviewed attribute-related approaches.

3.5. Vision-language foundation model adaptation approaches

Recent studies have increasingly leveraged the prior knowledge embedded in large vision-language models (VLMs) [132–135], given their success in various tasks. One notable example is CLIP [132], which was trained on large-scale image–text pairs and demonstrated remarkable capability in capturing both visual and semantic representations of open-world concepts. This makes it a strong foundational model for diverse downstream vision tasks, including general image classification [9,136], fine-grained image classification [137–141], and action recognition [142]. These tasks benefit from the learned representations of CLIP in a zero-shot manner. Typically, vision-language foundation model adaptation approaches are trained with a limited number of samples (e.g., 1, 2, 4, 8, or 16) per base class and evaluated across all novel class samples, distinguishing them from conventional few-shot approaches that employ fixed N-way K-shot settings in both training and evaluation. The overall concept of CLIP is shown in Fig. 11.

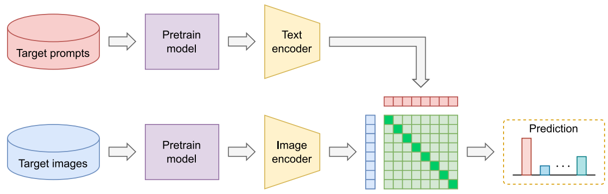


Fig. 11. The overall concept of CLIP.

Initially, CLIP was trained to predict whether an image and a textual description were semantically aligned within a target dataset. In zero-shot classification, CLIP utilized all class names as potential text pairings and predicted the most semantically aligned class based on the given image. This process involved converting class labels into textual prompts via a prompt template, as shown in Table 8. Label prediction for new images was achieved by computing similarity scores between image embeddings and text prompts, with the highest-scoring prompt determining the predicted label. Motivated by CLIP's capabilities, numerous studies have explored its few-shot adaptation, as illustrated in Fig. 12. These approaches can be broadly categorized into few-shot parameter tuning approaches, dynamic or unsupervised tuning approaches, and training-free adaptation approaches.

3.5.1. Few-shot parameter tuning approaches

These methods require a small amount of labeled data from the downstream task to tune a limited set of parameters before the model is deployed for inference. The core VLM weights remain frozen. This category primarily encompasses techniques like prompt learning [143–147], where learnable vectors representing textual context are optimized, and adapter tuning [148–151], where lightweight neural modules are inserted and trained within the VLM architecture. These approaches aim to balance effective adaptation with parameter efficiency, leveraging direct supervision from the few available labeled examples.

CoOp [152] extended this concept to automate prompt design for CLIP, replacing handcrafted prompts with learnable continuous vectors that capture context from input samples while keeping CLIP's parameters frozen. Two variations of CoOp exist: a unified context, where all classes share the same context vectors, and a class-specific context, where each class has unique context vectors suited for fine-grained datasets. However, CoOp exhibited overfitting to base classes, limiting its generalizability to novel classes. To address this, their extended work, CoCoOp [153], introduced a dynamic prompt mechanism conditioned on input images. This approach employed a lightweight neural network to generate context-aware vectors, thereby improving robustness against class shifts.

Conventional prompt learning approaches [152–154] relied on a single textual prompt per class, which often failed to capture intra-class variability. To address this, Prompt Learning with Optimal Transport (PLOT) [155] introduced multiple prompts to represent diverse visual features (e.g., habitat, texture, or color for a bird image). Optimal Transport [156] was leveraged to align textual prompts with different visual components, enhancing fine-grained cross-modal matching.

ProDA [157] further improved prompt representation by learning a distribution of prompts instead of relying on a single one. This approach helped mitigate domain shifts, a limitation of previous methods such as CoOp. Similarly, SoftCPT [158] introduced a meta-network to generate task-specific prompts rather than using a global prompt across all tasks. By employing multitask learning, SoftCPT facilitated better knowledge transfer among related tasks.

To address overfitting issues in soft-prompt approaches [152,153], Language-Aware Soft Prompting (LASP) [159] optimized learnable

Table 7
The summary of reviewed attribute-related few-shot approaches. Under Backbone column: CN denotes ConvNet, RN denotes ResNet, and D denotes DARTS. Under Training column: E denotes episode and M denotes minibatch. P denotes pretraining. T denotes transductive learning, C denotes centroid representation, and A denotes attention mechanism.

Method	Year	Backbone	Training	P	T	C	A	Strength	Weakness
AGFL [121]	2020	CN, RN	E	–	–	–	–	Works with diverse FSL frameworks and no need for data augmentation	Performance drops with shallow backbones and attributes may be noisy
Auto-ACNet [122]	2021	D[a]	M	–	–	–	✓	Automates architecture design	Complex implementation with high search cost
AGAM [125]	2021	CN, RN	E	–	–	–	✓	Simple integration into existing models	Requires annotated attributes and limited to metric-based methods
ADRGN [126]	2022	CN, RN	E	–	✓	–	✓	Dynamically identifying task-specific attribute importance	higher computational complexity due to dynamic routing and graph-based inference
APN [127]	2022	RN	M/E[b]	✓[b]	–	–	✓	Enhance image representations with localized attribute features	Struggles with abstract or global attributes
CPN [128]	2023	CN, RN	M+E	✓	–	✓	–	Utilize component prototypes to improve few-shot classification	Relies heavily on predefined attribute annotations
AADNet [130]	2024	CN, RN	M+E	✓	–	✓	–	Additional modules do not increase the model's complexity	Relies on the capability of the pre-trained backbone
AGAN [131]	2024	CN, RN	E	–	–	✓	✓	Enhance visual features using multiple attention modules	Higher computational costs

[a] The architecture is automatically search by DARTS.
[b] Depend on the selected FSL method; pretrain not applicable to N-way K-shot task.

Table 8
The text prompts for some datasets. The {LABEL} will be replaced by the ground-truth text label for each class.

Datasets	Prompts
Aircraft	*"a photo of a* {LABEL}*, a type of aircraft.", "a photo of the* {LABEL}*, a type of aircraft."*
EuroSAT	*"a satellite photo of a* {LABEL}*.", "a centered satellite photo of a* {LABEL}*.", "a centered satellite photo of the* {LABEL}*."*
Flower102	*"a photo of a* {LABEL}*, a type of flower."*
Food101	*"a photo of* {LABEL}*, a type of food."*
OxfordPets	*"a photo of a* {LABEL}*, a type of pet."*

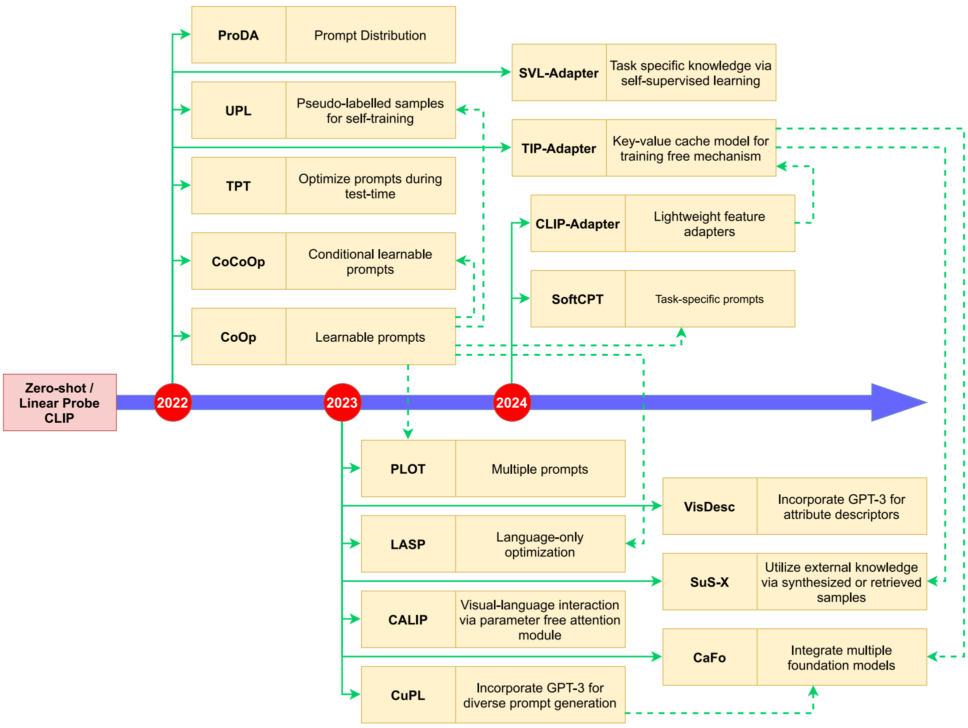


Fig. 12. The relationship between CLIP and the other variants from 2022 to 2024. The dotted line represents the inheritance relationship between two works.

prompt vectors solely within the text encoder, reducing dependency on visual features. Additionally, LASP fine-tuned layer normalization in the visual encoder to correct sample distribution shifts.

The adapter concept has been extended to VLMs as an alternative to prompt learning for FSIC. CLIP-Adapter [160] incorporated lightweight feature adapters at the fully connected layer of CLIP's backbone, reducing overfitting and improving parameter efficiency. By employing residual connections, CLIP-Adapter effectively fused fine-tuned and pre-trained knowledge.

3.5.2. *Dynamic or unsupervised tuning approaches*

These methods also tune parameters, but they avoid using labeled few-shot data, instead relying on unlabeled target data during training or dynamically optimizing parameters for each sample at test time.

While CoOp and CoCoOp fine-tune the model using labeled samples from downstream tasks, Test-Time Prompt Tuning (TPT) [161] optimized prompts dynamically during inference using a single test sample. TPT augmented the input image using AugMix [162] and enforced consistency across augmented variants to refine the prompt. By filtering out low-confidence augmentations, TPT improved adaptation by preventing misleading predictions.

To eliminate dependency on labeled target samples [152,153], Unsupervised Prompt Learning (UPL) [154] utilized pseudo-labels generated by CLIP to optimize prompts, enhancing scalability. Unlike CoOp, which required annotations, UPL selected only the most confident pseudo-labeled samples per class, mitigating biases inherent in CLIP's predictions.

Previous studies [152,157,160,163] have demonstrated poor performance on samples with significant visual discrepancies from those used during CLIP's pre-training, such as medical or remote sensing images. To address this issue, SVL-Adapter [164] integrated self-supervised learning with CLIP adaptation to incorporate both general knowledge from CLIP and task-specific knowledge from self-supervised learning. An additional self-supervised encoder learned representations from unlabeled target samples, which were combined with CLIP's outputs for improved classification.

3.5.3. *Training-free adaptation approaches*

This category covers methods that adapt the model without any gradient-based training on the target task, using inference-time mechanisms, pre-built caches, or external models (like LLMs) to enhance performance.

Unlike [152,160], TIP-Adapter [163] introduced a training-free approach by using a key–value cache model to store visual features extracted from CLIP and corresponding labels from the dataset. Predictions for novel samples were obtained by aggregating similarities with stored keys, reducing computational overhead. CALIP [165] further enhanced cross-modal interactions by incorporating a parameter-free attention module that refined both textual and visual representations, improving zero-shot and few-shot performance.

Customized Prompts via Language Models (CuPL) [166] employed GPT-3 [167] to generate diverse and descriptive class-specific textual prompts, as conventional CLIP prompts often fail to capture distinctive visual features or class nuances. Instead of generic prompts such as "a photo of a tree frog", CuPL generated more detailed descriptions, e.g., "A tree frog is a small amphibian with large eyes and sticky toe pads". Handcrafted prompt templates used to guide GPT-3 are shown in Table 9. Since GPT-3-generated prompts may lack descriptiveness or reliability, Menon and Vondrick [168] refined this approach by prompting GPT-3 to generate feature- or attribute-based descriptions rather than full-text prompts to facilitate more interpretable classification decisions.

Rather than refining textual semantics via external knowledge [166,168], Udandarao et al. [169] enhance visual representations by synthesizing additional training samples using text-to-image models like Stable Diffusion [170] and retrieving samples from LAION-5B [171]. By dynamically constructing support sets, the method achieved superior generalization under zero-shot conditions.

CaFo [172] cascaded knowledge from multiple foundation models, integrating GPT-3 [167], DALL-E [173], CLIP [132], and DINO [174] in a single pipeline. By leveraging linguistic, generative, and contrastive learning knowledge, CaFo demonstrated robust performance in FSIC. While CaFo does perform minimal training on its cache model keys, its dominant reliance on frozen external components placed it conceptually closer to this category despite not being strictly training-free.

In summary, vision-language foundation model adaptation approaches focus on adapting VLMs like CLIP to the FSIC task. These approaches offer significant strengths, such as improved generalization to novel classes, enhanced cross-modal alignment, and reduced reliance on extensive labeled data. Additionally, leveraging learnable prompts, various adapter structures, and external knowledge enhances their adaptability to few-shot tasks. However, challenges persist in optimizing the integration of these techniques, particularly in balancing model efficiency and performance while minimizing overfitting to base classes. Table 10 provides a summary of the vision-language foundation model adaptation approaches discussed in this section.

In conclusion, FSIC approaches can be broadly categorized into meta-learning, transfer learning, data augmentation, attribute-related, and vision-language foundation model adaptation methods, each offering distinct advantages and limitations, as summarized in Table 11.

4. Specialized few-shot approaches

This section provides a comprehensive review of specialized few-shot approaches, including FSFGIC, CDFSIC, and FSCIL. While FSFGIC and CDFSIC share similar training and evaluation mechanisms with general FSIC, the key distinction in CDFSIC lies in the use of training and evaluation samples from different domains ($D_{source} \neq D_{target}$). For consistency, the taxonomy established in Section 3 is maintained in this section to classify the approaches. Given the conceptual similarity between general FSIC, FSFGIC, and CDFSIC, the discussion will primarily focus on these two domains. FSCIL, which presents unique challenges related to incremental learning, will be briefly introduced following the review of FSFGIC and CDFSIC.

4.1. *Few-shot fine-grained image classification*

This section presents a review of recent studies in FSFGIC. Unlike general FSIC, FSFGIC requires models to capture subtle visual cues to differentiate objects belonging to similar categories with only a few available samples. Consequently, FSFGIC benchmark datasets (Section 5.2) are characterized by high intra-class variability and low inter-class distinctiveness.

While existing FSIC approaches have achieved notable success in classifying generic patterns (e.g., Omniglot dataset) and general objects (e.g., ImageNet-based and CIFAR100-based datasets), they often fail to capture subtle image cues, limiting their effectiveness in FSFGIC. To address this gap, Wei et al. [175] introduced the first dedicated FSFGIC approach, incorporating a bilinear feature learning module to extract fine-grained features and a piecewise classifier mapping module to refine decision boundaries. The bilinear features were derived from a fine-tuned AlexNet, while the sub-classifier mapping approach improved representation learning with parameter efficiency.

Zhu et al. [176] proposed MattML, a multi-attention meta-learning approach that leverages convolutional block attention modules (CBAMs) to identify discriminative regions. A RNN task learner further refined the classifier by generating task-specific attention weights, enabling the model to adapt dynamically to new tasks. Similarly, Li et al. [177] highlighted the limitations of image-level feature-based measures in existing FSIC approaches [13,25–27] and introduced the Deep Nearest Neighbor Neural Network (DN4), which employs a local descriptor-based image-to-class metric inspired by Naive-Bayes Nearest Neighbor (NBNN) [178]. Another work [16] from the same authors

Table 9
The GPT-3 prompts for some datasets. The {LABEL} will be replaced by the ground-truth text label for each class.

Datasets	GPT-3 Prompts
Aircraft	*"Describe a* {LABEL} *aircraft"*
EuroSAT	*"Describe an aerial satellite view of* {LABEL}*", "How does a satellite photo of a* {LABEL} *look like", "Visually describe a centered satellite view of a* {LABEL}*"*
Flower102	*"What does a* {LABEL} *flower look like", "Describe the appearance of a* {LABEL}*", "A caption of an image of* {LABEL}*", "Visually describe a* {LABEL}*, a type of flower"*
Food101	*"Describe what a* {LABEL} *looks like", "Visually describe a* {LABEL}*", "How can you tell that the food in this photo is a* {LABEL}*?"*
OxfordPets	*"Describe what a* {LABEL} *pet looks like", "Visually describe a* {LABEL}*, a type of pet"*

Table 10
The summary of reviewed vision-language foundation model adaptation few-shot approaches. # denotes few-shot parameter tuning methods (T), dynamic or unsupervised tuning methods (U), and training-free adaptation methods (F). VE denotes the vision encoder and RN denotes ResNet. R denotes required training. L denotes involved labeled data. T denotes required target data distribution.

#	Method	Year	VE	R	L	T	Strength	Weakness
–	Zero-shot CLIP [132]	2021	RN, ViT	–	–	–	Excels at zero-shot image classification by leveraging natural language supervision	Struggles with complex or abstract tasks and requires massive computational resources
	Linear Probe CLIP [132]	2021	RN, ViT	✓	✓	✓		
T	CoOp [152]	2022	RN, ViT	✓	✓	✓	Learning optimal context tokens directly from data	Struggles with interpretability
	CoCoOp [153]	2022	ViT	✓	✓	✓	Improves generalization to unseen classes by generating image-specific prompts	Additional flexibility that trades off specialization for generalization
	PLOT [155]	2023	RN	✓	✓	✓	Use of optimal transport to align visual and textual features	Increased computational cost due to the optimal transport calculations
	ProDA [157]	2022	RN	✓	✓	✓	Effectively capture the variance in visual representations and improve generalization to unseen examples	Complexity involved in learning and handling the distribution of prompts
	SoftCPT [158]	2024	RN, ViT	✓	✓	✓	Enabling effective knowledge sharing across related tasks	Improvement on generalized datasets with less related tasks is less significant
	LASP [159]	2023	ViT	✓	✓	✓	Use of a text-to-text loss to guide soft prompt learning	Lies in its reliance on hand-crafted textual prompts
	CLIP-Adapter [160]	2024	RN	✓	✓	✓	Improves performance by fine-tuning lightweight feature adapters with residual connections	Requires few-shot training data
U	TPT [161]	2022	RN, ViT	✓	–	–	Adapt prompts for VLMs using only a single test sample	Reliance on the consistency of predictions across augmented views of the test sample
	UPL [154]	2022	RN, ViT	✓	–	✓	adapt VLMs for downstream image recognition without requiring any labeled data	Reliance on pseudo-labels generated by the pre-trained VLM
	SVL-Adapter [164]	2022	RN	✓	–	✓	Effectively combines pretraining with self-supervised representation learning	Relies on the underlying self-supervised learning method and the quality of the pre-trained VLM
F	TIP-Adapter [163]	2022	RN, ViT	–	✓	✓	Training-free unseen task adaptation	Not fully utilize the potential of the available few-shot data
	CALIP [165]	2023	RN, ViT	–	–	–	Enhances zero-shot performance with a parameter-free cross-modal attention mechanism	Depends on CLIP's pre-trained alignment quality
	CuPL [166]	2023	ViT	–	–	–	Generating rich, discriminative prompts using LLMs	Depends on the quality and specificity of generated text
	VisDesc [168]	2023	ViT	–	–	–	Enhances the interpretability of VLMs by grounding classification decisions in descriptive features	Relies on the descriptors generated by the LLM
	SuS-X [169]	2023	RN, ViT	–	–	–	Avoiding the need for labeled images or fine-tuning during task adaptation	Reliance on the pre-trained VLM's existing knowledge
	CaFo [172]	2023	RN	✓	–	–	Leverage diverse knowledge via multiple pre-trained models	Relies heavily on the quality of synthetic images from DALL-E

in [177] proposed a covariance-based metric that utilizes second-order statistics to represent class distributions instead of a single mean vector [13]. Although DN4 able to summarize local similarities between support and query samples, it struggled to capture structural information within local patches, leading to inflexibility across varying scales. To overcome this, Yu et al. [179] introduced the Local Spatial Alignment Network (LSANet), which aligns local spatial regions between support and query samples using a traversal scanning approach,

Table 11
The advantages and disadvantages of existing few-shot categories under general image classification.

Categories	Advantages	Disadvantages
Metric-based Meta-learning	Simple implementation	Lack of task-specific adjustment
Model-based Meta-learning	Flexible internal dynamics	Poor performance on large datasets
Optimization-based Meta-learning	Fast adaptation	High resource consumption
Hybrid Transfer Learning	Explore task-specific relationship	Complex design with high resource consumption
Non-hybrid Transfer Learning	Simple training paradigm	Poor performance on distant classes
Data augmentation	Overfitting prevention	Highly relied on the quality of generated samples
Attribute-related	Semantic-level generalization	Relied on attribute annotations
Few-shot Parameter Tuning	Direct optimization toward downstream tasks	Require labeled samples
Dynamic or Unsupervised Tuning	Reduce dependency on data labeling	Sensitive to the quality of unlabeled samples
Training-free Adaptation	Highly efficient in low resource settings	Dependent on the quality of external resources

aggregating patch-level similarities for improved classification accuracy and mitigated background interference.

One major challenge in FSFGIC is spatial misalignment, where differences in object posture (e.g., standing vs. diving) introduce significant intra-class variation. While prior works [16,175,176] paid limited attention to this issue, several studies [14,180,181] sought to address it. Huang et al. [14] introduced the Low-Rank Pairwise Alignment Bilinear Network (LRPABN), which aligns support-query features via a shallow multilayer perceptron (MLP) and extracts second-order features through low-rank bilinear pooling, adapted from self-bilinear pooling [175]. This method enhances subtle difference detection while maintaining computational efficiency. In their subsequent study, Huang et al. [180] proposed the Target-Oriented Alignment Network (TOAN), which improves intra-class alignment via a Target-Oriented Matching Mechanism (TOMM) and enhances inter-class discrimination using Group Pairwise Bilinear Pooling (GPBP). Similarly, Wu et al. [181] introduced the Object-Aware Long-Short-Range Spatial Alignment Network (OLSA), which refines spatial alignment through three modules: a Foreground Object Enhancement (FOE) module to suppress background interference, a Long-Range Semantic Correspondence (LSC) module for global feature alignment, and a Short-Range Spatial Manipulation (SSM) module to refine local feature alignment.

Beyond spatial alignment, Li et al. [182] argued that a single similarity measure [40,177] is insufficient for FSFGIC and proposed the Bi-Similarity Network (BSNet), which utilized two distinct similarity computation by combining cosine similarity with existing metric-based approaches [13,26,27,177]. This method enhances feature compactness and allows models to focus on discriminative image regions while maintaining model complexity. As prior studies overlooked the relationship between support-query samples, Zhang et al. [183] proposed HelixFormer, a Transformer-based approach that leverages cross-image semantic relations for improved feature comparison. By computing consistent cross-image semantic relation maps (CSRMs) between support and query samples, HelixFormer refines local feature discrimination to better capture subtle object differences.

Feature reconstruction has also been explored in FSFGIC. Inspired by prior work [39,53], Wertheimer et al. [184] introduced Feature Map Reconstruction Networks (FRN), which reconstruct query feature maps from support features. Unlike earlier reconstruction-based models that relied on iterative processes [39] or computationally expensive attention mechanisms [35,53], FRN employs a closed-form ridge regression for greater efficiency. Expanding on this, Wu et al. [185] proposed the Bi-Directional Feature Reconstruction Network (Bi-FRN), which not only reconstructs query features from support features but also reconstructs support features from query features to reduce intra-class variation. In their subsequent study, Wu et al. [186] introduced a snapshot ensemble strategy, integrating cyclical cosine annealing learning rate schedules with episode training. This method enables models to converge to multiple local minima, improving generalization without increasing computational costs. To further enhance discriminative feature extraction, Li et al. [187] proposed the Local Content-Enriched Cross-Reconstruction Network (LCCRN) on top of FRN, integrating a Local Content-Enriched Module (LCEM) to extract fine-grained local features and a Cross-Reconstruction Module (CRM) to reconstruct query features using both local and global representations.

Lee et al. [188] noted that existing feature alignment methods [32,35,37,184] performed suboptimally on FSFGIC due to insufficient feature discrimination. To address this, they proposed the Task Discrepancy Maximization (TDM) framework, which enhanced class-wise discriminative localization through a Support Attention Module (SAM) and a Query Attention Module (QAM). These components assigned class-specific and object-relevant attention weights, producing task-adaptive feature maps that improved FSFGIC.

One main challenge in FSFGIC was the background interference. Tang et al. [189] tackled this issue using an attention-guided refinement strategy, refining multi-scale pyramid features to isolate discriminative regions while mitigating background noise. Meanwhile, Zha et al. [190] proposed BSFA, a novel framework combining a Background Activation Suppression (BAS) module to enhance foreground objects, a Foreground Object Alignment (FOA) module to realign support features, and a Local-to-Local (L2L) similarity metric to compute spatial similarity.

Instead of relying solely on high-level features, recent works [191,192] explored both mid- and high-level features for more detailed fine-grained representations. Ma et al. [191] proposed C2-Net to address sample-level noise and feature mismatch in FSFGIC. The Cross-Layer Feature Refinement (CLFR) module integrates and refines multi-layer feature maps for comprehensive representation learning, while the Cross-Sample Feature Adjustment (CSFA) module aligns support-query features via channel activation and spatial position adjustments. In contrast, Ma et al. [192] introduced the Bi-Directional Task-Guided Network (BTG-Net), which employs a Semantic-Guided Noise Filtering (SGNF) module to refine mid-level features and a General Knowledge Prompt Modeling (GKPM) module to enhance generalization across novel tasks.

Inspired by [113,193–196], saliency-based approaches have also been explored to improve FSFGIC. Liu et al. [196] proposed the Robust Saliency-Aware Distillation (RSaD) model, which refines object-specific information through mutual learning between raw and saliency-enhanced image branches. The Saliency-Aware Guidance (SaG) strategy enforces intrinsic sub-category relationships using a distillation technique, while the Representation Highlight & Summarize (RHS) module produces generalized contextual embeddings with minimal computational cost. To address background interference, Wang et al. [193] introduced the Foreground Object Transformation (FOT) method, which employs a pre-trained saliency detection model as [113], to segment foreground and background regions. This technique removes background noise and applies posture transformations to base samples, generating augmented novel class instances for training. FOT can be integrated with various FSFGIC models to enhance fine-grained classification performance.

In summary, the reviewed FSFGIC approaches primarily aim to capture fine-grained discriminative features from limited samples while effectively classifying novel categories. Initially, research focused on

Table 12
The summary of reviewed fine-grained-specialized few-shot approaches. # denotes optimization-based methods (O), metric-based methods (M), and data augmentation-based methods (A). Under Backbone column: CN denotes ConvNet and RN denotes ResNet. Under Training column: E denotes episode and M denotes minibatch. P denotes pretraining, C denotes centroid representation, and A denotes attention mechanism.

#	Method	Year	Backbone	Training	P	C	A	Strength	Weakness
O	PCM [175]	2019	AlexNet	M+E	✓	✓	✓	Piecewise mappings reduces model complexity	Relies heavily on high-dimensional bilinear features
	MattML [176]	2020	CN	E	–	✓	✓	Uses attention to capture discriminative local parts	Multiple attention mechanisms increase method complexity
M	DN4 [177]	2019	CN	E	–	✓	–	Uses local features for better image comparison	Relies on the quality of local descriptors
	CovaMNet [16]	2019	CN	E	–	✓	–	Uses covariance to capture second-order statistics	Incorporating covariance likely adds model complexity
	LRPABN [14]	2020	CN	E	–	✓	–	Uses low-rank bilinear pooling and feature alignment	Introduces additional complexity and cost
	BSNet [182]	2020	CN	E	–	✓	✓[a]	Two similarity measures for more discriminative features	Scalability with deeper backbone
	TOAN [180]	2021	CN, RN	E	–	✓	✓	Address high intra-class variance	Complex network architecture
	OLSA [181]	2021	CN, RN	M+E	✓	✓	✓	Enhances discriminative object details	Computationally intensive
	FRN [184]	2021	CN, RN	M+E	✓	✓	–	Preserves spatial details without overfitting to pose	Heavily depends on the quality of feature maps
	TDM [188]	2022	–	E	–	✓	✓	Dynamically highlight discriminative channels	Specialized in fine-grained details instead of coarse-grained
	HelixFormer [183]	2022	CN, RN	M+E	✓	–	✓	Uses Transformers to focus on discriminative local features	Increases complexity and computational cost
	AGPF [189]	2022	CN, RN	E	–	✓	✓	Capturing fine-grained details and reducing background noise	Inconsistent object localization for varying sizes
	LSANet [179]	2022	CN, RN	E	–	✓	✓	Aligns local spatial patches between samples	Higher computational complexity
	LCCRN [187]	2023	CN, RN	E	–	✓	–	Uses local features and cross-reconstruction	Additional modules which may add complexity
	BSFA [190]	2023	RN	E	–	✓	–	Suppressing background and aligning foregrounds	Complexity in training and optimization
	Bi-FRN [185]	2023	CN, RN	E	–	✓	✓	Increases inter-class and reduces intra-class variations	Complex network architecture and training
	Snapshot [186]	2024	CN, RN	E	–	✓	✓	Improves accuracy without extra training costs.	Requires saving multiple models
	C2-Net [191]	2024	CN, RN	E	–	✓	✓	Refines features across layers and aligns features across samples	Involves complex cross-layer and cross-sample modules
	BTG-Net [192]	2024	CN, RN	E	–	✓	✓	Utilizes mid-level features and explicit cross-task knowledge	Increases network complexity
	RSaD [196]	2024	CN, RN	M+E	✓	✓	✓	Uses saliency-aware guidance to focus on key object regions	Underperform in scenarios with crucial high-dimensional local descriptors
A	FOT[b] [193]	2021	CN, RN	M	✓	–	–	Removes background noise and transforms object postures	Relies on accurate foreground extraction and transformation

[a] Only applicable for using MatchingNet as the classifier.
[b] Only applicable to the transductive base method.

localizing discriminative image regions [175,176]. Subsequently, attention shifted toward addressing spatial misalignment through feature alignment [14,180,181] and feature reconstruction [184–187]. However, alignment processes are susceptible to background interference, leading to further exploration of foreground–background relationships to enhance robustness [189,190,192,196]. Table 12 provides a summary of the reviewed FSFGIC approaches.

4.2. Cross-domain few-shot image classification

CDFSIC differs from general FSIC and FSFGIC by addressing the challenge of categorizing unseen samples with significant domain gaps from the seen samples. This means that the base and novel classes originate from distinct domains, making it particularly relevant when collecting sufficient samples from the same domain is impractical. For instance, the source domain may contain natural images, while the target domain consists of medical or satellite images. The objective is to develop models capable of adapting to new tasks across different domains using only a few samples, which is essential for real-world applications.

Existing works have extensively explored CDFSIC for general object images [32,36,40,42,65,88,90,184,186] and fine-grained object images [181,183]. Prior metric-based meta-learning approaches struggle with domain adaptation due to overfitting to the data distribution in source domain. To mitigate this, Tseng et al. [197] introduced Feature-wise Transformation (FWT) layers that modify intermediate feature activations through affine transformations, learning domain insensitive features and enhancing generalization.

Despite the domain disparity inherent in the CDFSIC setup employed by prior studies, the base and novel classes frequently exhibit significant visual similarities. For example, prior works [40,65] used Omniglot as the source domain and MNIST or EMNIST [198] as the target domain, all of which are symbolic images. Other works employed natural images [40,42,184,186] or bird-specific images [181, 183] for both domains. The high similarity between domains raised concerns about generalizability, prompting Guo et al. [199] to evaluate meta-learning methods on more challenging CDFSIC tasks, including plant disease, satellite, dermoscopic, and X-ray images. Their study revealed that simple fine-tuning outperformed previous meta-learning approaches.

STARTUP [200] addressed CDFSIC by adapting learned feature representations to vastly different novel classes through self-training and knowledge distillation. It consisted of two stages: representation learning, where a teacher model generated pseudo-labels for novel samples, and evaluation, where a linear classifier was trained on the frozen backbone. Layer-wise Relevance Propagation (LRP) [201] improved meta-learning models by assigning relevance scores to feature maps, guiding models toward learning domain-relevant representations for better generalization.

Adversarial Task Augmentation (ATA) [202,203] tackled domain shift by augmenting training tasks with challenging virtual tasks that enhance adaptation. Noise-enhanced Supervised Autoencoder (NSAE) [204] utilized supervised autoencoders for feature reconstruction, incorporating noise-augmented data and fine-tuning to enhance domain generalization. Meta-FDMixup [205] leveraged limited labeled novel samples and performed meta-training with mixed-domain episodes. A subsequent improvement, [206], introduced Mixup-P and ConL modules to reinforce instance-level generalization through contrastive learning.

Unlike ATA [202] that performed task-level augmentation, Adversarial Feature Augmentation (AFA) [207] performed feature-level augmentation by simulating feature distribution mismatches across domains via adversarial training. It introduced a feature augmentation module to maximize domain discrepancies while training a domain discriminator, promoting domain-invariant feature learning. The P>M>F pipeline [208] combined pre-training, meta-learning, and fine-tuning, using self-supervised learning to build a strong foundation model, followed by ProtoNet-based meta-training and optional fine-tuning for domain adaptation.

Ranking Distance Calibration (RDC) [209] approached CDFSIC as an image retrieval problem, calibrating distances between samples via a re-ranking process based on k-nearest neighbors and Jaccard distance. StyleAdv [210] improved robustness by introducing adversarial style perturbations through Style-FGSM, enabling models to recognize samples under diverse visual styles. Local–global Distillation Prototypical Network (LDP-net) [211] enhanced ProtoNet [13] with a dual-branch structure and knowledge distillation, enforcing semantic consistency between global images and local patches to improve cross-domain generalization. In contrast to previous studies [199,202,204,209], LDP-net did not require fine-tuning on the target domain.

FLoR [212] improved knowledge transfer and fine-tuning by flattening loss landscapes in the representation space, thereby mitigating sharp minima that hinder adaptation. AttnTemp [213] refined ViT performance in CDFSIC by adjusting attention mechanisms to enhance target-domain generalization. ADAPTER [214] leveraged self-supervised learning with DINO [174] to learn domain-invariant features from unlabeled base and target domain samples, enhancing reliability through label smoothing.

In summary, CDFSIC approaches focus on transferring or adapting knowledge from a source domain to a target domain with significant domain gaps. Various techniques, including augmentation [204–207,210] and fine-tuning [199,202,209,210,212], have been employed for domain adaptation. Despite progress, CDFSIC remains challenging, necessitating robust methods to mitigate domain shifts. Table 13 summarizes the reviewed CDFSIC approaches.

4.3. Few-shot class-incremental learning

FSCIL [215] introduces a distinct training and evaluation paradigm compared to previous learning domains. Unlike conventional approaches that assume access to all training classes simultaneously, FSCIL necessitates the sequential learning of novel classes from a limited number of samples. This paradigm aligns with real-world scenarios where models must continuously acquire new knowledge while retaining previously learned information.

FSCIL training typically commences with a base session, followed by multiple incremental sessions. During the base session, the model is trained on a large set of samples and classes to establish a robust feature representation. This stage predominantly employs the standard cross-entropy loss [215–219]. Recent studies have further enhanced feature learning by integrating self-supervised learning techniques during the base session [220,221].

Each incremental session involves training the model on a small number of labeled samples, typically formulated as an N-way K-shot task, where new classes are introduced without any overlap with previously encountered classes. The model can only access training samples from the current session. To adapt to new classes while minimizing catastrophic forgetting, various approaches have been proposed. Some methods update the model backbone using neural gas network adjustments [215], triplet loss [222], or partial parameter selection [223].

Conversely, another set of approaches mitigates catastrophic forgetting by freezing the backbone during incremental sessions [219,224–226]. Building on these frozen-backbone strategies, subsequent works have tackled the stability-plasticity dilemma by exploring flatten minima [226], leveraging episodic training [224,225,227], addressing forward compatibility [228], employing ensemble learning [222], incorporating explicit memory [229], fine-tuning the projection layer [230], implementing transformational adaptation strategies [231], and utilizing selective fine-tuning with prompts [232].

Throughout all sessions, model evaluation encompasses both previously learned and newly encountered classes, as depicted in Fig. 13. This ensures that the model retains its ability to recognize earlier classes while learning new ones. The principal challenges in FSCIL include catastrophic forgetting, where previously acquired knowledge deteriorates, and the stability-plasticity dilemma, wherein a frozen backbone hinders the learning of new tasks. To address these challenges, researchers have employed techniques such as knowledge distillation [217,218,222,232], neural gas networks [215], meta-learning [221,229,231], self-supervised learning [219–221], attention mechanisms [218,222,225,229], and prototype-based methods [222–225, 228–232].

By employing these strategies, FSCIL research continues to progress toward developing models capable of effective and efficient lifelong learning in dynamic environments.

5. Few-shot image datasets

This section provides an overview of the most commonly utilized benchmark datasets for FSIC. The datasets are categorized into three groups: general classification, fine-grained classification, and cross-domain classification.

5.1. General classification

Datasets for general FSIC typically contain coarse-grained or generic categories. The most widely used datasets in this category include Omniglot [233], *mini*ImageNet [26], *tiered*ImageNet [28], CIFAR-FS [86], and FC100 [29]. Some samples from these datasets are illustrated in Fig. 14.

Table 13
The summary of reviewed cross-domain-specialized few-shot approaches. # denotes metric-based methods (M), non-hybrid transfer learning methods (NH), hybrid transfer learning methods (H), and data augmentation-based methods (A). Under Backbone column: CN denotes ConvNet and RN denotes ResNet. Under Training column: E denotes episode and M denotes minibatch. P denotes pretraining, T denotes transductive learning, C denotes centroid representation, and A denotes attention mechanism.

#	Method	Year	Backbone	Training	P	T	C	A	Strength	Weakness
M	FWT [197]	2020	RN	M+E	✓	–	–	–	Simulates diverse feature distributions	Requires careful tuning for hyperparameters optimization
	LRP [201]	2021	RN	E	–	✓	–	–	Incorporates explanation scores during the training process	Reliance on explanation methods introduces computational overhead
	LDP-net [211]	2023	RN	M+E	✓	✓	✓	–	Enforces local–global semantic consistency through knowledge distillation	May struggle with extreme domain shifts
NH	Transductive fine-tuning [199]	2020	RN	M	✓	✓	–	–	Leverages test data statistics to improve performance	Assumes access to all test data upfront
	STARTUP [200]	2021	RN	M	✓	–	–	–	Bridges large domain gaps by using unlabeled target data	Requires unlabeled data from the target domain
	RDC [209]	2022	RN	M	✓	✓	✓	✓	Boost performance without requiring extra labeled data	Computationally intensive
H	P>M>F [208]	2022	RN, ViT	M+E	✓	–	✓	✓[a]	Simpler than complex meta-learning approaches	Limited to domains where pre-trained models are available
	ATA [202,203]	2021	RN	M+E	✓	✓[c]	–	–	Generates challenging virtual tasks to enhance model robustness	Requires careful tuning of hyperparameters
	FLoR [212]	2024	RN, ViT	M+E	✓	✓	✓[b]	✓[a]	Enhance transferability and fine-tuning by flattening long-range loss landscapes	Requires additional computation and careful tuning of flatness-promoting techniques
	AttnTemp [213]	2024	ViT	M+E	✓	✓	✓[b]	✓	Enhances the transferability of ViT	Reliance on temperature adjustments limit flexibility
	ADAPTER [214]	2024	CCT	M+E	✓	✓	–	✓	Learns transferable features between different domains	Require significant computational resources
A	NSAE [204]	2021	CN, RN	M+E	✓	✓	✓	–	Learns broader feature variations through joint reconstruction and classification	Require additional computational resources
	Meta-FDMixup [205]	2021	RN	M+E	✓	–	–	–	Utilize both source and target data and extract domain-invariant features	Reliance on having a few labeled target examples
	GMeta-FDMixup [206]	2021	RN	M+E	✓	–	–	–	Learns more transferable features	Reliance on having a few labeled target examples
	AFA [207]	2022	RN	M+E	✓	✓[c]	–	–	Simulates diverse feature distributions	Requires careful balancing of adversarial training
	StyleAdv [210]	2023	RN, ViT	E	✓[a]	–	✓[a]	✓[a]	Robust to diverse visual styles through adversarial training	Additional computational resources for adversarial training process

[a] Only applicable to ViT network.
[b] Only applicable to ProtoNet classifier.
[c] Only applicable to the transductive base method.

- **Omniglot:** This dataset [233] comprises 1623 unique handwritten characters from 50 different alphabets. Each character contains 20 samples, each produced by a different individual. To expand the dataset, additional samples were generated through image rotations of 90°, 180°, and 270°. Santoro et al. [64] constructed a training set comprising 1200 character classes and 3600 augmented classes, leading to a total of 4800 classes. The remaining character classes, including their augmented versions, are allocated to the testing set.
- ***mini*ImageNet:** This dataset [26] is a smaller subset of ImageNet [9]. It consists of 60,000 RGB images distributed across 100 classes, each containing 600 images. As the original train–test split proposed by [26] is unavailable, Ravi and Larochelle [74] introduced a revised split, dividing the dataset into 64 training classes, 16 validation classes, and 20 testing classes.
- ***tiered*ImageNet:** This dataset [28] is a larger subset of ImageNet [9] compared to *mini*ImageNet. It contains 779,165 images spanning 608 classes. The dataset is split into 351 training classes, 97 validation classes, and 160 testing classes, as outlined by [28].
- **CIFAR100 few-shots (CIFAR-FS):** This dataset [86] is derived from the CIFAR100 dataset [234]. It contains 60,000 images with a resolution of 32 × 32 pixels, spread across 100 classes. The CIFAR-FS dataset follows a similar split methodology to *mini*ImageNet, with 64 classes for training, 16 for validation, and 20 for testing.
- **Fewshot-CIFAR100 (FC100):** This dataset [29] is another adaptation of CIFAR100, employing a split methodology similar to *tiered*ImageNet. It consists of 100 classes, each with 600 images, and is partitioned into 60 training classes, 20 validation classes, and 20 testing classes.

5.2. Fine-grained classification

FSFGIC datasets typically feature highly similar categories with fine-grained distinctions. Notable datasets in this category include CUB-2010 [235], CUB-2011 [236], Stanford Dogs [237], Stanford Cars [139], Aircraft [137], NABirds [238], meta-iNat [239], and tiered meta-iNat [239]. Some samples are shown in Fig. 15.

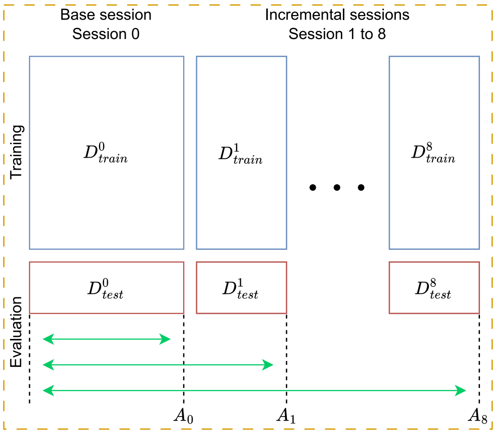


Fig. 13. The overall concept of FSCIL with an 8-step incremental task. D consists of 100 classes, where D^0 includes classes 1 to 60, and new classes are introduced incrementally from D^1 (classes 61–65) to D^8 (classes 96–100). In each session, D_{train} and D_{test} represent the training and testing sets, respectively. The accumulated accuracy A_i measures the model's performance from session 0 to session i, reflecting its ability to retain and learn new classes over time.

Fig. 14. Some samples from (a) Omniglot, (b) *mini*ImageNet, (c) *tiered*ImageNet, and (d) CIFAR-FS and FC100 datasets.

- **Caltech-UCSD Birds-200-2010 (CUB-2010):** This dataset [235] consists of 6033 images across 200 bird species. Based on the split criteria established in [177], the dataset is divided into 130 training classes, 20 validation classes, and 50 testing classes.
- **Caltech-UCSD Birds-200-2011 (CUB-2011):** An extended version of CUB-2010 [235], this dataset [236] comprises 11,788 images across 200 bird categories. It can be utilized in two classification settings: raw images [40] or preprocessed images with human-annotated bounding boxes [32]. Following the split criteria in [240], it is divided into 100 training classes, 50 validation classes, and 50 testing classes.
- **Stanford Dogs:** This dataset [237] is derived from the ImageNet dataset [9], which contains 20,580 images covering 120 dog breeds. According to [177], it is split into 70 training classes, 20 validation classes, and 30 testing classes, which are commonly used in existing research.

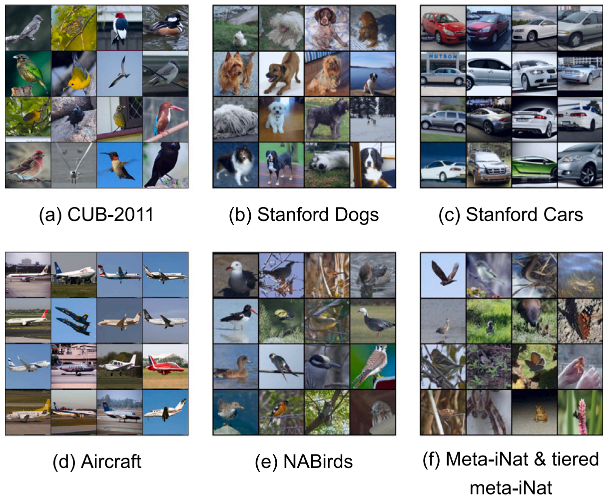


Fig. 15. Some samples from (a) CUB-2011, (b) Stanford Dogs, (c) Stanford Cars, (d) Aircraft, (e) NABirds, and (f) meta-iNat and tiered meta-iNat datasets.

- **Stanford Cars:** This dataset [139] consists of 16,185 images of 196 car types. As per the common practice in [177], it is split into 130 training classes, 17 validation classes, and 49 testing classes. For CDFSIC, Tseng et al. [197] utilized only 8144 samples from the training split in [139] as one of their evaluation datasets. The dataset is divided into 98 training, 49 validation, and 49 testing classes.
- **Fine-Grained Visual Classification of Aircraft (Aircraft):** This dataset [137] contains 10,000 aircraft images across 100 models. It provides hierarchical annotations at four levels (model, variant, family, and manufacturer) along with bounding box information. Following [184], the dataset is split into 50 training classes, 25 validation classes, and 25 testing classes, randomly.
- **North America Birds (NABirds):** This dataset [238] consists of 48,562 bird images from 555 North American species, including part annotations and bounding box data. According to [14], the dataset is split into 350 training classes, 66 validation classes, and 139 testing classes.
- **Meta-iNat:** This dataset [239] is derived from the iNaturalist Classification and Detection Dataset (iNat2017) [241], which contains 1135 animal species, with each class having around 50 to 1000 images. This dataset is originally designed for localization tasks [239], it has since been adapted for FSFGIC in [184]. Following [239], the dataset is split into training and testing sets in a 4:1 ratio.
- **Tiered meta-iNat:** This dataset [239] retains the same number of classes as meta-iNat but applies a super-category-based train–test split, resulting in 781 training classes and 354 testing classes. Compared to meta-iNat, tiered meta-iNat introduces a larger domain gap between training and testing classes.

5.3. Cross-domain classification

Several benchmark datasets in CDFSIC, including *mini*ImageNet [26], CUB-2011 [236], and Stanford Cars [139], have been discussed in previous sections. This section focuses on additional datasets such as Places [242], Plantae [241], CropDiseases [243], EuroSAT [244], ISIC [245,246], ChestX [247], and Meta-Dataset [10]. Examples are presented in Figs. 16 and 17.

- **Places365-Standard (Places):** This dataset [242] contains over 10 million images from 365 scene categories. As part of the CDFSIC benchmark in [197], a subset of 73,000 images was created by sampling 200 images per class from the original training set. The split

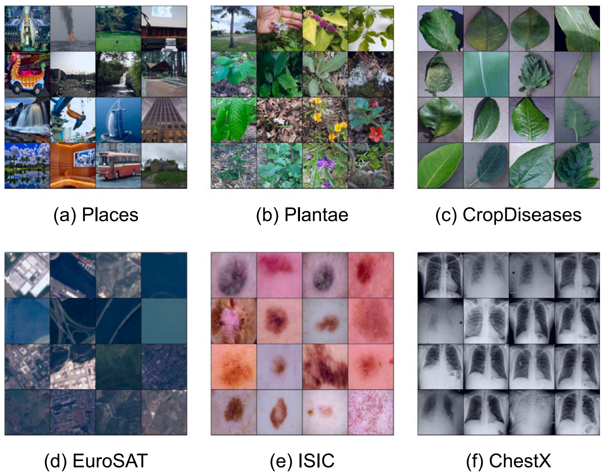


Fig. 16. Some samples from (a) Places, (b) Plantae, (c) CropDiseases, (d) EuroSAT, (e) ISIC, and (f) ChestX datasets.

includes 183 training classes, 91 validation classes, and 91 testing classes.

- **Plantae:** This dataset [241] is derived from iNat2017 [241], which consists of over 2000 plant species. A subset of 200 classes with more than 40k images was selected for CDFSIC [197], with splits of 100 training classes, 50 validation classes, and 50 testing classes.
- **CropDiseases:** This dataset [243] is derived from PlantVillage [248], which focuses on agricultural disease recognition. It consists of plant leaf images with various plant diseases collected under controlled conditions. Same as [242], only the original training samples were used by Guo et al. [199] in their CDFSIC benchmark. Under Fu et al. [206], 25 classes were used for training and 13 for testing, with no validation set.
- **EuroSAT:** This dataset [244] contains 27,000 Sentinel-2 satellite images categorized into 10 classes, originally designed for land use and land cover classification. Guo et al. [199] used the whole dataset in their CDFSIC benchmark. Based on Fu et al. [206], it was divided into 6 training classes and 4 testing classes, without a validation set.
- **ISIC2018 (ISIC):** This dataset [245,246] consists of 13,000 dermoscopic images classified into 7 categories, originally created for the International Skin Imaging Collaboration (ISIC) challenge in 2018 for human skin lesions classification. The full training set in the original dataset was used in the CDFSIC [199]. According to Fu et al. [206], the dataset is further divided into 4 training and 3 testing classes.
- **NIH Chest X-rays (ChestX):** This dataset [247] includes 112,120 frontal-view chest X-ray images from 30,805 unique patients. The images are labeled with 14 common thoracic pathologies. For CDFSIC [199], 7 classes with 25,848 images were used and Fu et al. [206] further divided them into 4 training and 3 testing classes.
- **Meta-Dataset:** This dataset [10] combines 10 diverse datasets: ImageNet (ILSVRC-2012) [9], Omniglot [233], Aircraft [137], CUB-2011 [236], Describable Textures (DTD) [249], Quick Draw [250], Fungi [251], VGG-Flower [140], Traffic Signs [252], and MSCOCO [253].

A summary of these datasets is presented in Table 14.

5.4. Challenges in dataset utilization

This section outlines key challenges in the utilization of few-shot image datasets. Three primary issues are identified: inconsistencies in train–test split ratios, variations in class allocations across studies, and the unavailability of official dataset sources.

Inconsistencies in Train–Test Split Ratios: a major challenge is the lack of standardized train–test split ratios across different studies. For example, the widely used CUB-2011 dataset typically follows a 100/50/50 split. However, several studies [44,176,185,186,193] have adopted the 130/20/50 split, originally proposed by Li et al. [177] for CUB-2010. Other variations include 140/20/40 [121] and 120/-/80 [122]. Similarly, Li et al. [182] used a 2:1:1 split for both Stanford Dogs and Stanford Cars, differing from most studies. For the Aircraft dataset, prior works [176,189] have used a 60/15/25 split instead of the more common 50/25/25 ratio.

Variations in Class Allocations: beyond differences in split ratios, inconsistencies also arise in class allocations. For instance, FEAT [32] and Baseline++ [40] both employ a 100/50/50 split for CUB-2011, yet their class assignments for training, validation, and testing differ. Similarly, DN4 [177] and Bi-FRN [185] apply a 130/17/49 split for Stanford Cars, but with differing class distributions across partitions. These discrepancies complicate direct comparisons of model performance and necessitate careful consideration when interpreting results.

Dataset Availability Issues: another challenge is the discontinuation of official dataset sources. For example, the original websites for Plantae [241] and Stanford Cars [139] are no longer accessible. While alternative sources, such as GitHub repositories or dataset reconstructions (e.g., Plantae from iNat2017 or *mini*ImageNet from ImageNet), allow researchers to obtain the data, these methods introduce the risk of inconsistencies in sample distribution due to random sampling process. This variation can lead to discrepancies in experimental results, making it difficult to ensure reproducibility and fair comparisons with prior studies.

Given these challenges, researchers should exercise caution when utilizing few-shot image datasets, ensuring that dataset splits and sources align as closely as possible with prior works for accurate performance evaluation.

6. Evaluation protocol

The evaluation protocol is a standardized procedure used to assess the performance of a model. Typically, a set of N-way K-shot episodes is sampled from novel classes to measure the model's ability in classifying previously unseen classes. Two common evaluation settings are the 5-way 1-shot episode and the 5-way 5-shot episode. The former assesses performance with only one support sample per class, while the latter evaluates performance with five support samples per class.

To ensure reliable and consistent results, a number of episodes are randomly sampled from the novel classes, and the average classification accuracy, along with a 95% confidence interval, is computed. The average classification accuracy with the confidence interval is computed as:

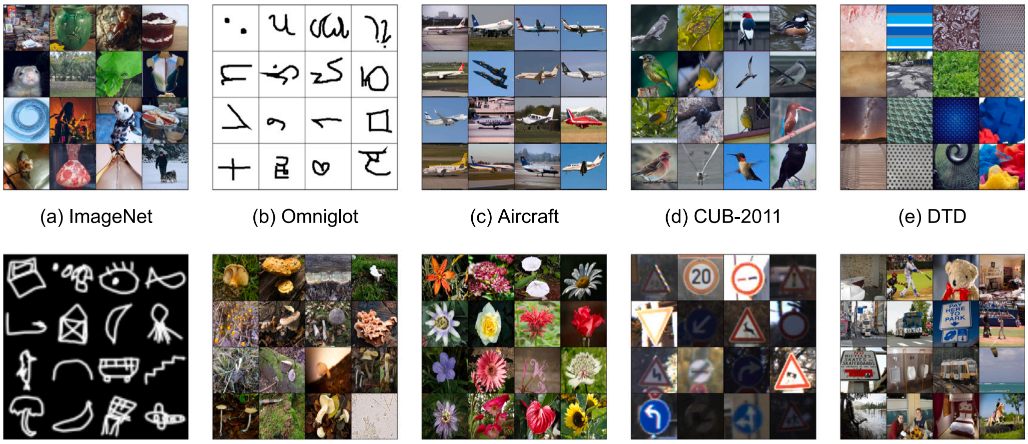
$$\mathrm{AC} = \frac{\sum_T \text{classification accuracy}}{T}\% \pm 1.96 \times \frac{\sigma}{\sqrt{T}} \tag{11}$$

where AC represents the average classification accuracy, σ denotes the standard deviation of the task accuracies, and T indicates the number of novel episodes.

Based on the evaluation protocol mentioned above, the performance of existing methods for general FSIC across various benchmark datasets is summarized in Tables 15, 16, 17, and 18.

For FSFGIC, the comparative performance of state-of-the-art approaches on multiple fine-grained benchmark datasets is reported in Tables 19, 20, 21, and 22. The reported performance follows the common split for each of the fine-grained datasets.

For CDFSIC, performance evaluation is conducted using three established benchmarks [10,197,199]. The benchmarks [197,199] train models on *mini*ImageNet and evaluate them on four different datasets.



(a) ImageNet (b) Omniglot (c) Aircraft (d) CUB-2011 (e) DTD

(f) Quick Draw (g) Fungi (h) VGG-Flower (i) Traffic Signs (j) MSCOCO

Fig. 17. Some samples inside the Meta-Dataset.

Table 14
The summary of few-shot image datasets. The class split here is shown in train/validation/test format, and the reported class split is based on the widely used ratio. *The Plantae dataset is sampled directly from the iNat2017 dataset based on the code provided by [197] due to the original Plantae dataset no longer available.

Dataset	No. Samples	Class split	Description and split criteria (if available)
Omniglot [233]	32,000	1200/-/423	A dataset containing black and white character images written by 20 different individuals.
*mini*ImageNet [26]	60,000	64/16/20	A dataset constructed from a smaller subset of ImageNet [9] spanning hundreds of object categories.
*tiered*ImageNet [28]	779,165	351/97/160	A dataset constructed from a larger subset of ImageNet [9] and split based on higher-level category to create a more challenging and realistic environment.
CIFAR-FS [86]	60,000	64/16/20	A low-resolution dataset derived from CIFAR100 [234] with the same criteria as *mini*ImageNet.
FC100 [29]	60,000	60/20/20	A low-resolution dataset derived from CIFAR100 [234] and split based on superclass to prevent information overlap and make it more challenging, similar to *tiered*ImageNet.
CUB-2010 [235]	6033	130/20/50	An image dataset containing various bird categories, specifically designed for fine-grained classification tasks. The dataset is split randomly.
CUB-2011 [236]	11,788	100/50/50	An extended version of the CUB-2010 dataset featuring nearly twice the number of samples per class, along with newly added part location annotations. The dataset is split randomly.
Stanford Dogs [237]	20,580	70/20/30	A dog dataset constructed from ImageNet [9] for fine-grained classification purposes.
Stanford Cars [139]	16,185	130/17/49	A dataset comprising different car models for fine-grained image classification.
Aircraft [137]	10,000	50/25/25	An airplane dataset that consists of 100 aircraft models with four-level hierarchical notations: model, variant, family, and manufacturer for fine-grained classification. The dataset is split randomly.
NABirds [238]	48,562	350/66/139	A bird dataset contains different North American bird species for fine-grained classification.
Meta-iNat [239]	243,986	908/-/227	Constructed from iNat2017 [241], an animal species dataset that is class imbalanced and heavy-tailed with both coarse-grained and fine-grained classes. The dataset is split randomly.
Tiered meta-iNat [239]	243,986	781/-/354	Constructed from iNat2017 [241] with a large domain gap in between base and novel classes. The dataset is split based on superclass.
Places [242]	73,000	183/91/91	A scene dataset that consists of a wide range of environments encountered in the world. The dataset is split randomly.
Plantae [241]	47,242*	100/50/50	A plant dataset derived from iNat2017 [241] for plantae recognition. The dataset is split randomly.
CropDiseases [243]	43,456	25/-/13	A plant leaf dataset that consists of diseased and healthy leaves for agricultural disease recognition.
EuroSAT [244]	27,000	6/-/4	A landscape satellite image dataset that is designed for satellite imagery recognition.
ISIC [245,246]	10,015	4/-/3	A dataset contains the dermoscopy images of human skin lesions.
ChestX [247]	25,848	4/-/3	A dataset with human chest X-ray images for medical diagnosis.
Meta-Dataset [10]	–	–	A combination of 10 datasets demonstrating a more realistic and diverse scenario.

The benchmark introduced by Tseng et al. [197] assesses model performance on CUB-2011 [236], Stanford Cars [139], Places [242], and Plantae [241]. Meanwhile, the benchmark proposed by Guo et al. [199] presents a more challenging evaluation scenario by testing models on CropDiseases [243], EuroSAT [244], ISIC [245,246], and ChestX [247].

Meta-Dataset [10] is the largest-scale benchmark CDFSIC. It features two experimental setups: (1) training on a single dataset and (2) training on multiple datasets. In both setups, models are subsequently evaluated on all available datasets. Unlike conventional FSIC benchmarks, Meta-Dataset does not use fixed values for the N and K in few-shot episodes. Instead, the values of N and K vary across datasets within Meta-Dataset. As a result, its performance evaluations are not reported using the standard 5-way K-shot setup. The benchmark results are summarized in Tables 23 and 24.

Table 15
The N-way K-shot accuracy of reviewed few-shot approaches on the Omniglot dataset.

Category	Method	Backbone	T	Omniglot [233]			
				20-way 1-shot	20-way 5-shot	5-way 1-shot	5-way 5-shot
Metric	SiameseNet [25]	ConvNet	–	88.1	97.0	97.3	98.4
	MatchingNet [26]	ConvNet	–	93.8	98.5	98.1	98.9
	ProtoNet [13]	ConvNet	1000	96.0	98.9	98.8	99.7
	RelationNet [27]	ConvNet	1000	97.6 ± 0.2	99.1 ± 0.1	99.6 ± 0.2	99.8 ± 0.1
	PARN [37]	ConvNet	1000	98.55 ± 0.18	99.48 ± 0.05	99.91 ± 0.08	99.93 ± 0.03
	Variational Bayesian [15]	ConvNet	1000	97.95 ± 0.17	99.24 ± 0.10	99.67 ± 0.13	99.83 ± 0.06
	IMP [30]	ConvNet	500	95.0 ± 0.1	98.6 ± 0.1	98.4 ± 0.3	99.5 ± 0.1
	MlSo [44]	ConvNet	1000	98.7	99.7	99.9	99.9
	BD-CSPN[a] [31]	ConvNet	600	–	–	99.62	99.76
	BD-CSPN[a] [31]	WRN	600	–	–	99.08	99.85
Model	MANN [64]	–	–	–	–	82.8	94.9
	MetaNet [65]	ConvNet	400	97.0	–	98.95	–
	SNAIL [66]	ConvNet	–	97.64 ± 0.30	99.36 ± 0.18	99.07 ± 0.16	99.78 ± 0.09
	MM-Net [67]	ConvNet	500	97.16 ± 0.10	98.93 ± 0.05	99.28 ± 0.08	99.77 ± 0.04
Optimization	MAML [75]	ConvNet	–	95.8 ± 0.3	98.9 ± 0.2	98.7 ± 0.4	99.9 ± 0.1
	Meta-SGD [76]	ConvNet	–	95.93 ± 0.38	98.97 ± 0.19	99.53 ± 0.26	99.93 ± 0.09
	Reptile[a] [77]	ConvNet	–	89.43 ± 0.14	97.12 ± 0.32	97.68 ± 0.04	99.48 ± 0.06
	MT-net [81]	ConvNet	600	96.2 ± 0.4	–	99.5 ± 0.3	–
	MU-MOMAML [82]	ConvNet	–	97.2	99.4	99.7	99.9
	MAML++ [83]	ConvNet	600	97.65 ± 0.05	99.33 ± 0.03	99.47	99.93
	iMAML [84]	ConvNet	–	96.18 ± 0.36	99.14 ± 0.10	99.50 ± 0.26	99.74 ± 0.11
	R2-D2 [86]	ConvNet	10,000	96.24 ± 0.05	99.20 ± 0.02	98.91 ± 0.05	99.74 ± 0.02
Transfer	Baseline++ [40]	ConvNet	600	–	–	95.41 ± 0.39	99.38 ± 0.10
	Efficient-PrototypicalNet [95]	EfficientNet	1000	–	–	94.95	98.79
Augmentation	MetaGAN+RelationNet [110]	ConvNet	–	97.64 ± 0.17	99.21 ± 0.1	99.67 ± 0.18	99.86 ± 0.11
	SRPN [109]	WRN	100 × 100	94.8	–	–	-

[a] Transductive result.

Prior research [13,32,75,87,94] has utilized various numbers of T, leading to inconsistent performance evaluations due to different numbers of episodes yielding varying results. It has been observed that a low number of novel episodes can result in high variance in performance [85]. To address this, the authors of [32] increased the number of novel episodes in the evaluation stage to obtain more reliable results. An experiment was conducted to examine performance variations with different values of T, and the results are presented in Table 25.

In this study, ProtoNet and RFS were re-implemented with a ConvNet backbone, while the remaining methods utilized pre-trained models from their official sources. All models were evaluated using varying numbers of novel episodes, denoted as $t = \{600, 2000, 10000\}$, during the evaluation stage. As shown in Table 25, the results varied across all methods and datasets depending on the number of novel episodes. Notably, the highest difference between the minimum and maximum results exceeded 1%, as evidenced by SCL on the 1-shot *tiered*ImageNet dataset (minimum: 69.93% for 600 episodes, maximum: 70.99% for 10,000 episodes). To ensure consistent comparisons, future studies should consider using a uniform number of novel episodes for performance evaluation.

7. Few-shot application

FSIC has significant practical applications across various fields, including medical diagnosis, fault diagnosis, and smart agriculture. These domains often encounter challenges such as limited sample sizes, scarce annotations, and rare categories, making it difficult to apply conventional deep learning models that typically require large-scale labeled datasets. FSL effectively addresses these challenges, enabling the deployment of high-performance models in diverse applications. This section provides an overview of the specific applications of FSIC across medical diagnosis, fault diagnosis, and smart agriculture.

7.1. Medical diagnosis

In medical diagnosis, accurate disease identification and treatment heavily rely on medical imaging. FSL facilitates the rapid identification of various medical conditions, even when only a limited number of samples are available for rare diseases. Several studies have demonstrated the efficacy of FSL in disease detection. Cai et al. [254] proposed an attention-based ProtoNet for brain tumor classification under a few-shot scenario, achieving superior performance compared to traditional transfer learning methods. Additionally, during the COVID-19 pandemic, research efforts focused on leveraging FSL for rapid and accurate disease diagnosis. Due to limited data availability, studies [255,256] emphasized the importance of mitigating model bias caused by sensitivity parameter selection in generative approaches and weak data distribution. Chen et al. [255] employed contrastive learning to train an encoder capable of extracting expressive feature representations from lung images, subsequently integrating ProtoNet for COVID-19 classification using chest CT images. Similarly, Jadon [256] combined transfer learning with a Siamese network to enhance classification performance in low-data scenarios.

Beyond individual disease classification, Dai et al. [257] introduced PFEMed, a framework applicable to various public disease datasets. PFEMed employs a dual-encoder structure to extract both general and specific features from medical images, further refining them using a prior-guided VAE module for enhanced classification under a transductive learning paradigm. Additionally, Ouahab and Ahmed [258] developed ProtoMed, a ProtoNet variant incorporating an auxiliary contrastive learning branch to improve feature discrimination and enhance learning from limited medical images. This method effectively captures intra-class and inter-class relationships, resulting in a lightweight and efficient model suitable for low-prevalence disease detection. Furthermore, Krenzer et al. [259] proposed an automated polyp classification system utilizing deep metric learning for NICE classification, demonstrating the viability of FSL in data-scarce medical environments.

7.2. Fault diagnosis

FSL has also been applied to industrial quality control, where detecting defective products is crucial for maintaining product standards. Several studies have explored the use of few-shot models for identifying defects with minimal training samples, reducing reliance on manual

Table 16
The 5-way K-shot accuracy of reviewed metric-based and model-based few-shot approaches on ImageNet-related datasets.

Category	Method	Backbone	T	*mini*ImageNet [26]		*tiered*ImageNet [28]	
				1-shot	5-shot	1-shot	5-shot
Metric	MatchingNet [26]	ConvNet	–	46.6	60.0	–	–
	ProtoNet [13]	ConvNet	600	49.42 ± 0.78	68.20 ± 0.66	–	–
	RelationNet [27]	ConvNet	600	50.44 ± 0.82	65.32 ± 0.7	–	–
	FEAT[a] [32]	ConvNet	10,000	57.04 ± 0.20	72.89 ± 0.16	–	–
	TPN[a] [34]	ConvNet	600	53.75 ± 0.86	69.43 ± 0.67	57.53 ± 0.96	72.85 ± 0.74
	PARN [37]	ConvNet	600	55.22 ± 0.84	71.55 ± 0.66	–	–
	Variational Bayesian [15]	ConvNet	600	57.15 ± 0.31	71.54 ± 0.23	–	–
	CovaMNet [16]	ConvNet	600	51.19 ± 0.76	67.65 ± 0.63	–	–
	DN4 [177]	ConvNet	600 × 5	51.24 ± 0.74	71.02 ± 0.64	–	–
	MELR [41]	ConvNet	2000	55.35 ± 0.43	72.27 ± 0.35	56.38 ± 0.48	73.22 ± 0.41
	LSANet [179]	ConvNet	1000	56.16	72.85	58.35	75.62
	MRN[a] [38]	ConvNet	1000	57.83 ± 0.69	71.13 ± 0.50	62.65 ± 0.84	74.20 ± 0.64
	IMP [30]	ConvNet	500	49.6 ± 0.8	68.1 ± 0.8	–	–
	DCAP [42]	ConvNet	1000	53.68 ± 0.63	68.62 ± 0.60	55.63 ± 0.73	70.78 ± 0.60
	ProtoNet + hybrid loss [43]	ConvNet	600	52.26 ± 0.52	70.33 ± 0.69	–	–
	DGAP[a] [45]	ConvNet	600	54.65 ± 0.62	69.22 ± 0.47	59.28 ± 0.68	75.04 ± 0.54
	MlSo [44]	ConvNet	600	58.03 ± 0.80	73.06 ± 0.72	61.97 ± 0.91	78.83 ± 0.77
	LCCRN [187]	ConvNet	10,000	53.93 ± 0.20	70.41 ± 0.16	–	–
	SSL-ProtoNet [60]	ConvNet	2000	52.58 ± 0.45	70.87 ± 0.36	55.14 ± 0.49	74.23 ± 0.40
	TADAM [29]	ResNet	500 × 10	58.5 ± 0.3	76.7 ± 0.3	–	–
	FEAT [32]	ResNet	10,000	66.78	82.05	70.80 ± 0.23	84.79 ± 0.16
	GLoFA [33]	ResNet	600	66.12 ± 0.42	81.37 ± 0.33	69.75 ± 0.33	83.58 ± 0.42
	TPN[a] [34]	ResNet	600	59.46	75.65	–	–
	Deep K-Tuplet Network [36]	ResNet	600	58.30 ± 0.84	72.37 ± 0.63	–	–
	Variational Bayesian [15]	ResNet	600	61.23 ± 0.26	77.69 ± 0.17	–	–
	CAN[a] [35]	ResNet	2000	67.19 ± 0.55	80.64 ± 0.35	73.21 ± 0.58	84.93 ± 0.38
	DeepEMD [39]	ResNet	600	65.91 ± 0.82	82.41 ± 0.56	71.16 ± 0.87	86.03 ± 0.58
	GNN+FWT [197]	ResNet	1000	66.32 ± 0.80	81.98 ± 0.55	–	–
	FRN [184]	ResNet	10,000	66.45 ± 0.19	82.83 ± 0.13	71.16 ± 0.22	86.01 ± 0.15
	MELR [41]	ResNet	2000	67.40 ± 0.43	83.40 ± 0.28	72.14 ± 0.51	87.01 ± 0.35
	LSANet [179]	ResNet	1000	66.78	83.45	72.78	86.42
	DCAP [42]	ResNet	1000	65.20 ± 0.67	80.93 ± 0.53	70.15 ± 0.74	85.33 ± 0.55
	ProtoNet + hybrid loss [43]	ResNet	600	59.44 ± 0.67	74.93 ± 0.51	59.64 ± 0.82	75.44 ± 0.71
	DGAP[a] [45]	ResNet	600	61.35 ± 0.62	78.85 ± 0.46	70.10 ± 0.67	84.99 ± 0.46
	MlSo [44]	ResNet	600	66.79 ± 0.86	83.41 ± 0.67	–	–
	LCCRN [187]	ResNet	10,000	62.24 ± 0.20	79.19 ± 0.14	–	–
	DBDC-SSL [62]	ResNet	2000	68.64 ± 0.43	86.02 ± 0.28	73.88 ± 0.48	89.03 ± 0.29
	SFT [63]	ResNet	2000	68.90 ± 0.44	83.81 ± 0.29	71.84 ± 0.51	85.69 ± 0.35
	Bi-FRN+Snapshot [186]	ResNet	10,000	63.76 ± 0.20	80.33 ± 0.15	–	–
	BD-CSPN[a] [31]	WRN	600	70.31 ± 0.93	81.89 ± 0.60	78.74 ± 0.95	86.92 ± 0.63
	protoLP[a] [47]	WRN	10,000	83.07 ± 0.25	89.04 ± 0.13	89.04 ± 0.23	92.80 ± 0.13
	PUTM[a] [52]	WRN	3000	73.8	85.7	81.1	90.4
	NegCosIC [61]	WRN	500	69.41 ± 0.52	85.98 ± 0.40	–	–
	CPEA [55]	ViT	1000	71.97 ± 0.65	87.06 ± 0.38	76.93 ± 0.70	90.12 ± 0.45
Model	MetaNet [65]	ConvNet	400	49.21 ± 0.96	–	–	–
	MM-Net [67]	ConvNet	500	53.37 ± 0.48	66.97 ± 0.35	–	–
	Dynamic-Net [70]	ConvNet	600	56.20 ± 0.86	73.00 ± 0.64	–	–
	Dynamic-Net [70]	ResNet	600	55.45 ± 0.89	70.13 ± 0.68	–	–
	SNAIL [66]	ResNet	–	55.71 ± 0.99	68.88 ± 0.92	–	-

[a] Transductive result.

inspection and lowering operational costs. Xu and Ma [260] enhanced ProtoNet [13] by integrating multiple attention modules to improve defect classification accuracy in automotive components. Additionally, Tong et al. [261] developed a modified deep Siamese network for metal surface defect detection under low-data conditions, demonstrating superior efficiency compared to traditional manual inspection methods.

For cross-domain few-shot fault diagnosis, Tang et al. [262] introduced NLRN, which utilizes a lightweight encoder module for efficient feature extraction and a semi-supervised calibration method to mitigate domain shift in limited labeled datasets. Similarly, Hu et al. [263] proposed JTFMN, a hybrid approach combining metric-based meta-learning and transfer learning to align distribution differences between source and target domains. Furthermore, Hu et al. [264] incorporated a graph convolutional network to learn feature relationships and applied domain-adversarial training to minimize domain shift errors. The method computes sample similarity using a scaled distance metric, enabling effective learning with limited labeled data. For bearing fault diagnosis, Shao et al. [265] introduced a task-supervised ANIL model, enhancing feature reuse and task adaptability by optimizing the inner loop structure with a residual network and a supervised task-adaptive loss function. Moreover, Zhang et al. [266] proposed a self-supervised approach integrating intra-domain and cross-domain prototypical contrast to improve class discrimination and reduce domain discrepancies in few-shot fault diagnosis.

7.3. Smart agriculture

Nevertheless, FSL can be utilized in smart agriculture, particularly in scenarios where obtaining large-scale labeled datasets is challenging or expensive. A key application area is plant disease recognition, where FSL enables rapid and accurate disease identification, reducing reliance on expert visual inspection. Several studies have demonstrated the efficacy of metric-based few-shot approaches in detecting cotton leaf spots [267] and various plant diseases [268]. Wang et al. [269] proposed a multi-modal collaborative representation learning framework that integrates both image and textual data to mitigate the impact of complex backgrounds in vegetable disease identification.

Table 17
The 5-way K-shot accuracy of remaining few-shot approaches on ImageNet-related datasets.

Category	Method	Backbone	T	*mini*ImageNet [26]		*tiered*ImageNet [28]	
				1-shot	5-shot	1-shot	5-shot
Optimization	Meta-Learner LSTM [74]	ConvNet	–	43.44 ± 0.77	60.6 ± 0.71	–	–
	MAML [75]	ConvNet	–	48.7 ± 1.84	63.11 ± 0.92	–	–
	FOMAML [75]	ConvNet	–	48.07 ± 1.75	63.15 ± 0.91	–	–
	Meta-SGD [76]	ConvNet	–	50.47 ± 1.87	64.03 ± 0.94	–	–
	Reptile[a] [77]	ConvNet	–	49.97 ± 0.32	65.99 ± 0.58	–	–
	LLAMA [78]	ConvNet	600	49.40 ± 1.83	–	–	–
	PLATIPUS [79]	ConvNet	100	50.13 ± 1.86	–	–	–
	BMAML [80]	ConvNet	–	50.01 ± 1.86	–	–	–
	MT-net [81]	ConvNet	4000	51.7 ± 1.84	–	–	–
	MAML++ [83]	ConvNet	600	52.15 ± 0.26	68.32 ± 0.44	–	–
	iMAML [84]	ConvNet	–	49.30 ± 1.88	–	–	–
	R2-D2 [86]	ConvNet	10,000	51.8 ± 0.2	68.4 ± 0.2	–	–
	LR-D2 [86]	ConvNet	10,000	51.9 ± 0.2	68.7 ± 0.2	–	–
	HSML [88]	ConvNet	1000	50.38 ± 1.85	–	–	–
	ANIL [89]	ConvNet	2000	46.7 ± 0.4	61.5 ± 0.5	–	–
	NIL [89]	ConvNet	2000	48.0 ± 0.7	62.2 ± 0.5	–	–
	CxGrad [90]	ConvNet	600 × 3	51.80 ± 0.46	69.82 ± 0.42	55.55 ± 0.46	73.55 ± 0.41
	Mix-MAML [91]	ConvNet	–	53.17 ± 0.49	70.18 ± 0.46	–	–
	MetaOptNet [87]	ResNet	1000	62.64 ± 0.61	78.63 ± 0.46	65.99 ± 0.72	81.56 ± 0.53
	CxGrad [90]	ResNet	600 × 3	60.19 ± 0.45	75.17 ± 0.40	65.47 ± 0.44	82.52 ± 0.35
	Mix-MAML [91]	ResNet	–	59.38 ± 0.49	76.93 ± 0.42	–	–
	LEO [85]	WRN	10,000	61.76 ± 0.08	77.59 ± 0.12	66.33 ± 0.05	81.44 ± 0.09
Transfer	Baseline++ [40]	ConvNet	600	48.24 ± 0.75	66.43 ± 0.63	–	–
	MTL [100]	ResNet	600	64.3 ± 1.7	80.9 ± 0.8	72.0 ± 1.8	85.1 ± 0.8
	PTN[a] [96]	ResNet	600	82.66 ± 0.97	88.43 ± 0.67	84.70 ± 1.14	89.14 ± 0.71
	STARTUP [200]	ResNet	600	54.20 ± 0.81	76.48 ± 0.63	55.33 ± 0.85	77.78 ± 0.67
	Classifier-baseline [93]	ResNet	2000	58.91 ± 0.23	77.76 ± 0.17	68.07 ± 0.26	83.74 ± 0.18
	Meta-baseline [93]	ResNet	2000	63.17 ± 0.23	79.26 ± 0.17	68.62 ± 0.27	83.29 ± 0.18
	RFS [94]	ResNet	600	64.82 ± 0.60	82.14 ± 0.43	71.52 ± 0.69	86.03 ± 0.49
	SKD-GEN0 [102]	ResNet	600	65.93 ± 0.81	83.15 ± 0.54	71.69 ± 0.91	86.66 ± 0.60
	SKD-GEN1 [102]	ResNet	600	67.04 ± 0.85	83.54 ± 0.54	72.03 ± 0.91	86.50 ± 0.58
	MetaNODE[a] [98]	ResNet	600	77.92 ± 0.99	85.13 ± 0.62	83.46 ± 0.92	88.46 ± 0.57
	DCRL [103]	ResNet	1000	66.25 ± 0.61	83.29 ± 0.39	72.88 ± 0.70	87.52 ± 0.45
	SCL [104]	ResNet	2000	66.79 ± 0.43	83.39 ± 0.29	70.58 ± 0.50	85.39 ± 0.35
	SimpleShot [92]	DenseNet	10,000	64.29 ± 0.20	81.50 ± 0.14	71.32 ± 0.22	86.66 ± 0.15
	Efficient-PrototypicalNet [95]	EfficientNet	2000	76.16 ± 0.40	91.90 ± 0.19	44.84 ± 0.45	71.49 ± 0.39
Augmentation	MetaGAN+RelationNet [110]	ConvNet	–	52.71 ± 0.64	68.63 ± 0.67	–	–
	FOT+Baseline++ [193]	ConvNet	–	51.46 ± 0.76	68.36 ± 0.45	–	–
	Ortho-shot [116]	ConvNet	–	67.39 ± 0.34	83.44 ± 0.24	–	–
	Dual TriNet [119]	ResNet	600	58.12 ± 1.37	76.92 ± 0.69	–	–
	IDeMe-Net [112]	ResNet	–	59.14 ± 0.86	74.63 ± 0.74	–	–
	AFHN [111]	ResNet	600	62.38 ± 0.72	78.16 ± 0.56	–	–
	BlockMix[a] [115]	ResNet	–	69.79	80.19	–	–
	Self-Mixup [117]	ResNet	10,000	66.56 ± 0.19	82.74 ± 0.13	72.04 ± 0.22	86.47 ± 0.15
	SRPN [109]	WRN	100 × 100	55.2	69.6	–	–
Attribute	ADRGN[a] [126]	ConvNet	10,000	69.91 ± 0.43	84.21 ± 0.56	71.32 ± 0.52	87.48 ± 0.52
	AGFL+MatchingNet [121]	ResNet	600	53.51 ± 0.59	66.19 ± 0.55	–	–
	ADRGN[a] [126]	ResNet	10,000	72.14 ± 0.42	86.25 ± 0.56	74.64 ± 0.52	88.76 ± 0.58
	AADNet [130]	ResNet	600	67.87 ± 0.68	82.77 ± 0.50	73.22 ± 0.52	86.02 ± 0.44

[a] Transductive result.

Additionally, Zhong et al. [270] introduced conditional adversarial autoencoders for the classification of Citrus aurantium L. diseases. Rezaei et al. [271] developed PMF+FA, a few-shot plant disease detection model that employs a three-step pipeline enhanced with a feature attention module for improved classification performance in complex field environments.

Beyond disease recognition, FSL has been utilized in pest identification and crop detection. Automated pest identification plays a critical role in developing robotic pest control systems, reducing dependence on manual and chemical interventions. Previous studies have demonstrated the feasibility of FSL for cotton pest identification [272] and soybean heartworm detection [273]. In crop detection, FSL has been employed to locate crop seeds and cultivated soil areas, facilitating automated farmland monitoring. This technology has been integrated into Unmanned Aerial Vehicles (UAVs) to detect crop seed locations [274] and in two-dimensional scene analysis for guiding autonomous tractors [275]. Additionally, Li et al. [276] proposed a Siamese domain transfer network for detecting corn residues, enhancing precision in agricultural automation.

7.4. Other applications

Beyond the aforementioned domains, FSL has been explored in autonomous vehicle systems, which rely heavily on computer vision for accurate object detection. Few-shot approaches can facilitate the rapid and reliable classification of pedestrians, vehicles, and obstacles, thereby improving the safety and reliability of autonomous transportation. Wang et al. [277] introduced a two-stage anomaly detection framework for identifying defects in power line insulators using UAV-captured aerial images, reducing reliance on human inspection. Additionally, Patsiouras et al. [278] incorporated [70] into UAV-based

Table 18
The 5-way K-shot accuracy of reviewed few-shot approaches on CIFAR-100-related datasets.

Category	Method	Backbone	T	CIFAR-FS [86]		FC100 [29]	
				1-shot	5-shot	1-shot	5-shot
Metric	LCCRN [187]	ConvNet	10,000	–	–	35.38 ± 0.16	48.76 ± 0.17
	SSL-ProtoNet [60]	ConvNet	2000	60.41 ± 0.52	76.52 ± 0.38	–	–
	TADAM [29]	ResNet	500 × 10	–	–	40.1 ± 0.4	56.1 ± 0.4
	DeepEMD [39]	ResNet	600	–	–	46.47 ± 0.78	63.22 ± 0.71
	ProtoNet + hybrid loss [43]	ResNet	600	–	–	41.41 ± 0.53	55.26 ± 0.51
	LCCRN [187]	ResNet	10,000	–	–	38.71 ± 0.17	52.24 ± 0.17
	DBDC-SSL [62]	ResNet	2000	75.60 ± 0.44	88.49 ± 0.31	–	–
	SFT [63]	ResNet	2000	75.71 ± 0.46	87.93 ± 0.33	–	–
	protoLP[a] [47]	WRN	10,000	87.69 ± 0.23	90.82 ± 0.15	–	–
	PUTM[a] [52]	WRN	3000	81.4	88.6	–	–
	NegCosIC [61]	WRN	500	73.98 ± 0.65	87.33 ± 0.48	–	–
	CPEA [55]	ViT	1000	77.82 ± 0.66	88.98 ± 0.45	47.24 ± 0.58	65.02 ± 0.60
Optimization	R2-D2 [86]	ConvNet	10,000	65.4 ± 0.2	79.4 ± 0.2	–	–
	LR-D2 [86]	ConvNet	10,000	65.3 ± 0.2	78.3 ± 0.2	–	–
	Mix-MAML [91]	ConvNet	–	61.58 ± 0.48	77.46 ± 0.41	–	–
	MetaOptNet [87]	ResNet	1000	72.0 ± 0.7	84.2 ± 0.5	41.1 ± 0.6	55.5 ± 0.6
	Mix-MAML [91]	ResNet	–	68.90 ± 0.46	83.62 ± 0.37	–	–
Transfer	Efficient-PrototypicalNet [95]	EfficientNet	2000	77.41 ± 0.44	90.02 ± 0.27	45.73 ± 0.41	63.05 ± 0.38
	SimpleShot [92]	ResNet	10,000	–	–	40.13 ± 0.18	53.63 ± 0.18
	MTL [100]	ResNet	600	–	–	46.0 ± 1.7	61.3 ± 0.8
	RFS [94]	ResNet	600	73.9 ± 0.8	86.9 ± 0.5	44.6 ± 0.7	60.99 ± 0.6
	SKD-GEN0 [102]	ResNet	600	74.5 ± 0.9	88.0 ± 0.6	46.4 ± 0.8	63.3 ± 0.7
	SKD-GEN1 [102]	ResNet	600	76.9 ± 0.9	88.9 ± 0.6	47.3 ± 0.8	63.8 ± 0.7
	DCRL [103]	ResNet	1000	77.18 ± 0.38	89.36 ± 0.25	46.85 ± 0.35	63.56 ± 0.33
	SCL [104]	ResNet	2000	75.45 ± 0.46	88.10 ± 0.32	46.14 ± 0.43	62.59 ± 0.42
Augmentation	Ortho-shot [116]	ConvNet	–	76.41 ± 0.25	87.68 ± 0.24	–	–
	Dual TriNet [119]	ResNet	600	63.41 ± 0.64	78.43 ± 0.62	–	–
	AFHN [111]	ResNet	600	68.32 ± 0.93	81.45 ± 0.87	–	–
	Self-Mixup [117]	ResNet	10,000	73.23 ± 0.21	87.67 ± 0.14	–	–

[a] Transductive result.

sports cinematography, where FSL is essential for recognizing specific athletes within broader classification categories.

FSL has also been applied to human–object interaction (HOI) recognition, where it addresses the challenges posed by long-tail category distributions and instance imbalances when recognizing the relationship between human actions and surrounding objects. Ji et al. [279] developed SAPNet, a metric-based approach that leverages semantic information to guide attention and classify HOI categories. Similarly, Ji et al. [280] proposed SADG-Net, a dynamic generation model that utilizes task-specific representations to enhance action and object classification accuracy.

Another critical domain where FSL has been applied is remote sensing scene classification (RSSC). Few-shot RSSC aims to classify remote sensing images with only a limited number of labeled samples, which is essential for applications such as land cover analysis, urban planning, and disaster monitoring. Xu et al. [281] introduced an attention-based model to enhance feature extraction and applied a dictionary-based contrastive loss to improve classification accuracy. Li et al. [282] employed global-local contrastive learning to address high inter-class similarities in remote sensing images, while Hou et al. [283] developed an unsupervised approach leveraging random augmentation sampling and channel-driven metric learning to enhance feature diversity and adaptability.

8. Research trends

Recent advancements in computer vision have demonstrated the effectiveness of Transformers [12,284] across various tasks. Motivated by their success, researchers have explored leveraging Transformer architectures to enhance FSL. However, the data-intensive nature of Transformers, combined with their lack of inductive bias, poses significant challenges in few-shot scenarios where labeled samples are scarce. To mitigate these limitations, several studies [32,53–55,63,285] have proposed innovative approaches to establish stable training paradigms for Transformers, thereby improving their applicability to FSIC tasks.

Another emerging trend in FSIC involves leveraging prior knowledge for cross-domain adaptation. Given the limited availability of labeled data, prior knowledge can facilitate model generalization across domains. Recent research [286] has explored techniques such as graph-based methods to capture inter-class relationships, while domain adaptation strategies have been employed to transfer knowledge from a source domain to a target domain with potentially different data distributions. Various studies [287–297] have examined domain adaptation approaches under challenging benchmarks [197,199]. Furthermore, the introduction of Meta-Dataset [10] has enabled the evaluation of models under more realistic conditions by incorporating class imbalances and varying sample sizes within tasks. Studies [298–301] have leveraged this benchmark to assess model performance beyond traditional fixed N-way K-shot settings.

The adoption of foundation models has also gained traction in FSIC. Initially, single-modality pre-trained models such as EfficientNet [8] and ViT [12] were utilized for few-shot tasks [95,208]. Subsequently, multi-modal foundation models, such as CLIP [132], were introduced to handle few-shot classification under both zero-shot and low-shot settings [302–307]. Recent studies highlight the effectiveness of prompt learning in adapting multi-modal foundation models for few-shot tasks. Beyond prompt learning, multi-task fine-tuning [308] has been explored as a potential strategy to enhance model performance. Additionally, parameter-efficient fine-tuning [309] and domain-specific adaptations [310–312] are active research directions, aiming to reduce computational overhead while improving domain-specific generalization.

Addressing the inherent challenges of FSIC in low-data regimes, self-supervised learning has emerged as a promising approach. Self-supervised learning techniques enable models to derive meaningful representations from limited labeled data. Previous studies [44,99,102,104,313,314] have incorporated self-supervised tasks such as rotation prediction, patch prediction, and contrastive learning to enhance representation learning. Different methodologies [54,55,60,315] integrate self-supervised learning either by pre-training the model in a fully

Table 19
The 5-way K-shot accuracy of reviewed few-shot approaches on CUB-related datasets. Under version column, 2010 denotes CUB-2010 [235] while 2011 denotes CUB-2011 [236].

Category	Method	Backbone	T	Version	Raw Input		Preprocessed Input	
					1-shot	5-shot	1-shot	5-shot
Metric	CovaMNet [16]	ConvNet	600	2010	52.42 ± 0.76	63.76 ± 0.64	–	–
	DN4 [177]	ConvNet	600 × 5	2010	53.15 ± 0.84	81.90 ± 0.60	–	–
	BD-CSPN[a] [31]	ConvNet	600	2011	75.10	87.25	–	–
	FEAT [32]	ConvNet	10,000	2011	–	–	68.87 ± 0.22	82.90 ± 0.15
	BSNet (R&C) [182]	ConvNet	600	2011	65.89 ± 1.00	80.99 ± 0.63	–	–
	OLSA [181]	ConvNet	2000	2011	–	–	73.07 ± 0.46	86.24 ± 0.29
	FRN [184]	ConvNet	10,000	2011	–	–	73.48	88.43
	FRN+TDM [188]	ConvNet	10,000	2011	–	–	74.39	88.89
	HelixFormer [183]	ConvNet	2000	2011	–	–	79.34 ± 0.45	91.01 ± 0.24
	AGPF [189]	ConvNet	600	2011	74.03 ± 0.90	86.54 ± 0.50	–	–
	LCCRN [187]	ConvNet	10,000	2011	–	–	76.22 ± 0.21	89.39 ± 0.13
	BSFA [190]	ConvNet	2000	2011	65.48 ± 0.51	76.02 ± 0.41	–	–
	C2-Net [191]	ConvNet	2000	2011	78.66 ± 0.46	89.43 ± 0.28	–	–
	BTG-Net [192]	ConvNet	2000	2011	79.19	89.71	–	–
	RSaD [196]	ConvNet	600	2011	71.15 ± 0.92	84.03 ± 0.62	–	–
	DeepEMD [39]	ResNet	600	2011	–	–	75.65 ± 0.83	88.69 ± 0.50
	BSNet (R&C) [182]	ResNet	600	2011	69.61 ± 0.92	83.24 ± 0.60	–	–
	OLSA [181]	ResNet	2000	2011	–	–	77.77 ± 0.44	89.87 ± 0.24
	FRN [184]	ResNet	10,000	2011	83.55 ± 0.19	92.92 ± 0.10	83.16	92.59
	MELR [41]	ResNet	2000	2011	–	–	70.26 ± 0.50	85.01 ± 0.32
	FRN+TDM [188]	ResNet	10,000	2011	84.36 ± 0.19	93.37 ± 0.10	83.36	92.80
	HelixFormer [183]	ResNet	2000	2011	–	–	81.66 ± 0.30	91.83 ± 0.17
	AGPF [189]	ResNet	600	2011	78.73 ± 0.84	89.77 ± 0.47	–	–
	LCCRN [187]	ResNet	10,000	2011	–	–	82.97 ± 0.19	93.63 ± 0.10
	BSFA [190]	ResNet	2000	2011	82.27 ± 0.46	90.76 ± 0.26	86.00 ± 0.41	92.53 ± 0.23
	DBDC-SSL [62]	ResNet	2000	2011	84.67 ± 0.39	94.76 ± 0.16	–	–
	BTG-Net [192]	ResNet	2000	2011	86.44	94.20	–	–
	RSaD [196]	ResNet	600	2011	82.45 ± 0.79	92.02 ± 0.44	–	–
	BD-CSPN[a] [31]	WRN	600	2011	87.45	91.74	–	–
	protoLP[a] [47]	WRN	10,000	2011	91.69 ± 0.18	94.18 ± 0.09	–	–
	NegCosIC [61]	WRN	500	2011	85.68 ± 0.78	94.66 ± 0.41	–	–
	PUTM[a] [52]	WRN	3000	2011	86.9	92.2	–	–
Transfer	Baseline++ [40]	ConvNet	600	2011	60.53 ± 0.83	79.34 ± 0.61	–	–
	MetaNODE[a] [98]	ResNet	600	2011	90.94 ± 0.62	93.18 ± 0.38	–	–
Augmentation	Dual TriNet [119]	ResNet	600	2011	69.61 ± 0.46	84.10 ± 0.35	–	–
	AFHN [111]	ResNet	600	2011	70.53 ± 1.01	83.95 ± 0.63	–	–
	BlockMix[a] [115]	ResNet	–	2011	83.82	90.22	–	–
	Self-Mixup [117]	ResNet	10,000	2011	76.47 ± 0.21	89.45 ± 0.12	–	–
Attribute	AGAM+ProtoNet [125]	ConvNet	10,000	2011	75.87 ± 0.29	81.66 ± 0.25	–	–
	ADRGN[a] [126]	ConvNet	10,000	2011	78.72 ± 0.48	90.52 ± 0.42	–	–
	AGAN [131]	ConvNet	2000	2011	77.47 ± 0.40	82.44 ± 0.31	–	–
	AGAM+ProtoNet [125]	ResNet	10,000	2011	79.58 ± 0.25	87.17 ± 0.23	–	–
	ADRGN[a] [126]	ResNet	10,000	2011	82.32 ± 0.51	92.97 ± 0.35	–	–
	APN+DPGN [127]	ResNet	–	2011	77.4 ± 0.44	92.2 ± 0.24	–	–
	CPN [128]	ResNet	5000	2011	87.29 ± 0.20	92.54 ± 0.14	–	–
	AADNet [130]	ResNet	600	2011	82.34 ± 0.61	92.11 ± 0.53	–	–
	AGAN [131]	ResNet	2000	2011	79.95 ± 0.41	89.34 ± 0.26	–	-

[a] Transductive result.

self-supervised setting before fine-tuning it on few-shot tasks or by concurrently training it on both self-supervised and few-shot tasks. Despite the variations in approach, these studies share the common objective of improving model generalization and performance in low-data settings.

In summary, ongoing research trends in FSIC include Transformer-based methodologies, cross-domain adaptation, foundation model adaptation, and self-supervised learning. These approaches collectively aim to address the fundamental challenges associated with limited labeled data, ultimately enhancing model robustness and generalization in few-shot scenarios. The integration of multiple strategies, such as combining self-supervised learning with foundation models [164,172,208,214,308], continues to be an exciting area of research, pushing the boundaries of FSIC towards more practical and robust solutions.

9. Conclusion

In conclusion, this paper presents a comprehensive review of existing approaches in FSIC, categorizing them into five primary groups: meta-learning, transfer learning, data augmentation, attribute-related, and vision-language foundation model adaptation. Meta-learning approaches are further classified into metric-based, model-based, and optimization-based methods, while transfer learning is divided into hybrid and non-hybrid methods. Additionally, vision-language foundation model adaptation approaches encompasses few-shot parameter tuning, dynamic or unsupervised tuning, and training-free adaptation methods. This review also examines specialized FSIC methodologies in FSFGIC, CDFSIC, and FSCIL. Furthermore, an overview of benchmark datasets used in general classification, fine-grained classification, and cross-domain classification is provided. Direct comparisons among FSIC approaches are challenged by inconsistencies in dataset utilization, including variations in train–test split ratios, class allocation, and dataset availability. Existing FSIC evaluation protocols primarily assess average accuracy on novel tasks under 5-way 1-shot and 5-way 5-shot settings. However, discrepancies in the number of novel tasks lead to inconsistencies in reported results for the same methods. To ensure

Table 20
The 5-way K-shot accuracy of reviewed few-shot approaches on Stanford Dogs and Stanford Cars datasets.

Category	Method	Backbone	T	Stanford Dogs [237]		Stanford Cars [139]	
				1-shot	5-shot	1-shot	5-shot
Metric	CovaMNet [16]	ConvNet	600	49.10 ± 0.76	63.04 ± 0.65	56.65 ± 0.86	71.33 ± 0.62
	DN4 [177]	ConvNet	600 × 5	45.73 ± 0.76	66.33 ± 0.66	61.51 ± 0.85	89.60 ± 0.44
	LSANet [179]	ConvNet	1000	55.85	71.78	68.65	87.23
	MlSo [44]	ConvNet	600	55.62 ± 0.58	71.98 ± 0.71	67.83 ± 0.63	84.98 ± 0.48
	LRPABN$_{cpt}$ [14]	ConvNet	600	45.72 ± 0.75	60.94 ± 0.66	60.28 ± 0.76	73.29 ± 0.58
	TOAN [180]	ConvNet	1000	49.30 ± 0.77	67.16 ± 0.49	65.90 ± 0.72	84.24 ± 0.48
	OLSA [181]	ConvNet	2000	55.53 ± 0.45	71.68 ± 0.36	70.13 ± 0.48	84.29 ± 0.31
	HelixFormer [183]	ConvNet	2000	59.81 ± 0.50	73.40 ± 0.36	75.46 ± 0.37	89.68 ± 0.25
	AGPF [189]	ConvNet	600	60.89 ± 0.98	78.14 ± 0.62	78.14 ± 0.84	87.42 ± 0.57
	Bi-FRN [185]	ConvNet	10,000	64.74 ± 0.22	81.29 ± 0.14	75.74 ± 0.20	91.58 ± 0.09
	Bi-FRN+Snapshot [186]	ConvNet	10,000	67.74 ± 0.22	83.23 ± 0.14	78.21 ± 0.19	92.43 ± 0.09
	C2-Net [191]	ConvNet	2000	66.42 ± 0.50	81.23 ± 0.34	81.29 ± 0.45	91.08 ± 0.26
	BTG-Net [192]	ConvNet	2000	67.33 ± 0.52	81.08 ± 0.33	78.72 ± 0.46	90.78 ± 0.26
	RSaD [196]	ConvNet	600	59.42 ± 0.95	75.30 ± 0.69	65.43 ± 1.29	81.75 ± 0.88
	TOAN [180]	ConvNet	1000	51.83 ± 0.80	69.83 ± 0.66	76.62 ± 0.70	89.57 ± 0.40
	OLSA [181]	ResNet	2000	64.15 ± 0.49	78.28 ± 0.32	77.03 ± 0.46	88.85 ± 0.46
	HelixFormer [183]	ResNet	2000	65.92 ± 0.49	80.65 ± 0.36	79.40 ± 0.43	92.26 ± 0.15
	BSFA [190]	ResNet	2000	69.58 ± 0.50	82.59 ± 0.33	88.93 ± 0.38	95.20 ± 0.20
	Bi-FRN [185]	ResNet	10,000	76.89 ± 0.21	88.27 ± 0.12	90.44 ± 0.15	97.49 ± 0.05
	Bi-FRN+Snapshot [186]	ResNet	10,000	79.70 ± 0.20	89.92 ± 0.11	91.81 ± 0.13	97.88 ± 0.05
	C2-Net [191]	ResNet	2000	75.50 ± 0.49	87.65 ± 0.28	88.96 ± 0.37	95.16 ± 0.20
	BTG-Net [192]	ResNet	2000	76.06 ± 0.50	88.48 ± 0.29	90.28 ± 0.34	96.78 ± 0.15
	RSaD [196]	ResNet	600	73.75 ± 0.93	86.65 ± 0.54	87.27 ± 0.70	95.01 ± 0.49
Optimization	MattML [176]	ConvNet	2000	54.84 ± 0.53	71.34 ± 0.38	66.11 ± 0.54	82.80 ± 0.28
Augmentation	FOT+Baseline++ [193]	ConvNet	–	49.32 ± 0.74	68.18 ± 0.69	54.55 ± 0.73	73.69 ± 0.65
	Ortho-shot [116]	ConvNet	–	57.06 ± 0.63	73.15 ± 0.22	75.34 ± 0.41	94.38 ± 0.25

Table 21
The 5-way K-shot accuracy of reviewed metric-based few-shot approaches on Aircraft and NABirds datasets.

Method	T	ConvNet				ResNet			
		Aircraft [137]		NABirds [238]		Aircraft [137]		NABirds [238]	
		1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
LRPABN$_{cpt}$ [14]	10,000	–	–	67.73 ± 0.81	81.62 ± 0.58	–	–	–	–
BSNet (R&C) [182]	600	64.83 ± 1.00	80.25 ± 0.67	–	–	–	–	–	–
TOAN [180]	1000	–	–	70.02 ± 0.80	85.52 ± 0.50	–	–	76.14 ± 0.75	90.21 ± 0.40
OLSA [181]	2000	–	–	75.60 ± 0.49	87.21 ± 0.29	–	–	83.76 ± 0.44	92.61 ± 0.23
FRN [184]	10,000	53.20	71.17	–	–	70.17	83.81	–	–
FRN+TDM [188]	10,000	54.21	71.37	–	–	70.89	84.54	–	–
HelixFormer [183]	2000	70.37 ± 0.57	79.80 ± 0.42	78.63 ± 0.48	90.06 ± 0.26	74.01 ± 0.54	83.11 ± 0.41	84.51 ± 0.41	93.11 ± 0.19
LCCRN [187]	10,000	76.81 ± 0.21	88.21 ± 0.11	–	–	88.48 ± 0.17	94.61 ± 0.08	–	–

Table 22
The 5-way K-shot accuracy of reviewed few-shot approaches with ConNet backbone on iNat2017-based datasets.

Method	T	meta-iNat [239]		tiered meta-iNat [239]	
		1-shot	5-shot	1-shot	5-shot
FRN [184]	10,000	62.42	80.45	43.91	63.36
FRN+TDM [188]	10,000	63.97	81.60	44.05	62.91
C2-Net [191]	2000	71.47	85.47	49.04	67.25
BTG-Net [192]	2000	71.36	84.59	50.62	69.11

fair comparisons, standardized experimental settings with a consistent number of novel tasks should be adopted. FSIC has demonstrated broad applicability across multiple domains, including medical diagnosis, fault detection, and smart agriculture, by enabling high-performance models to function effectively with limited labeled data. Emerging research directions, such as Transformer-based architectures, cross-domain adaptation, foundation model adaptation, and self-supervised learning, continue to advance FSIC, enhancing model robustness and generalization in real-world scenarios.

CRediT authorship contribution statement

Jit Yan Lim: Writing – original draft, Visualization, Software, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Kian Ming Lim:** Writing – review & editing, Supervision, Resources, Methodology. **Chin Poo Lee:** Writing – review & editing, Validation. **Yong Xuan Tan:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Fundamental Research Grant Scheme of the Ministry of Higher Education (FRGS/1/2024/ICT02/MMU/03/1) which is awarded to the Multimedia University.

Table 23
The 5-way K-shot accuracy of reviewed few-shot approaches on the FWT benchmark [197] and the BSCD-FSL benchmark [199]. "FT" denotes the result after fine-tuning.

Method	Backbone	T	FT	FWT [197]								BSCD-FSL [199]							
				CUB-2011 [236]		Stanford Cars[a] [139]		Places [242]		Plantae [241]		CropDiseases [243]		EuroSAT [244]		ISIC [245,246]		ChestX [247]	
				1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
GNN+FWT [197]	ResNet	1000	–	47.47	66.98	31.61	44.90	55.77	73.94	35.95	53.85	–	–	–	–	–	–	–	
Transductive FT[b] [199]	ResNet	600	✓	–	–	–	–	–	–	–	–	–	90.64	–	81.76	–	49.68	–	26.09
STARTUP [200]	ResNet	600	–	–	–	–	–	–	–	–	–	75.93	93.02	63.88	82.29	32.66	47.22	23.09	26.94
CAN+LRP[b] [201]	ResNet	2000	–	48.35	66.57	32.35	42.57	61.60	76.90	38.48	51.63	–	–	–	–	–	–	–	
TPN+ATA[b] [202]	ResNet	2000	✓	51.89	70.14	38.07	55.23	57.26	73.87	40.75	59.02	82.47	93.56	70.84	85.47	35.55	49.83	22.45	24.74
NSAE[b] [204]	ResNet	600	✓	–	76.00	–	61.11	–	73.40	–	65.66	–	96.09	–	87.53	–	56.85	–	28.73
Meta-FDMixup [205]	ResNet	1000	–	63.24	79.46	51.31	66.52	58.22	78.92	51.03	69.22	–	–	–	–	–	–	–	–
GMeta-FDMixup [206]	ResNet	5000	–	63.85	80.48	53.10	71.80	59.39	78.80	51.28	69.45	70.21	87.32	91.07	95.87	70.90	84.07	53.07	55.37
TPN+AFA[b] [207]	ResNet	2000	–	50.85	65.86	38.43	47.89	60.29	72.81	40.27	55.67	72.44	85.69	66.17	80.12	34.25	46.29	21.69	23.47
P>M>F [208]	ResNet	600	✓	–	–	–	–	–	–	–	–	–	95.06	–	89.18	–	43.78	–	27.13
RDC[b] [209]	ResNet	2000	✓	51.20	67.77	39.13	53.75	61.50	74.65	44.33	60.63	86.33	93.55	71.57	84.67	35.84	49.06	22.27	25.48
StyleAdv [210]	ResNet	1000	✓	48.49	70.90	35.09	56.44	58.58	79.35	41.13	64.10	80.69	96.51	72.92	91.64	35.76	53.05	22.64	26.24
LDP-net[b] [211]	ResNet	600	✓	55.94	73.34	37.44	53.06	62.21	75.47	41.04	59.64	81.24	91.89	73.25	84.05	33.44	48.44	22.21	26.88
FLoR[b] [212]	ResNet	–	✓	55.94	74.06	40.01	57.98	61.27	74.25	41.70	61.70	86.30	93.60	71.38	83.76	41.67	57.54	23.12	26.89
P>M>F [208]	ViT	600	✓	–	–	–	–	–	–	–	–	–	92.96	–	85.98	–	50.12	–	27.27
StyleAdv [210]	ViT	1000	✓	84.01	95.82	40.48	66.02	72.64	88.33	55.52	78.01	84.11	95.99	74.93	90.12	33.99	51.23	22.92	26.97
FLoR [212]	ViT	–	✓	85.40	96.53	43.42	68.44	74.69	89.48	52.29	76.22	83.55	96.47	73.09	90.75	35.49	53.06	23.26	27.02
AttnTemp[b] [213]	ViT	–	✓	–	–	–	–	–	–	–	–	87.58	96.74	77.40	91.34	40.13	55.22	23.96	28.41
ADAPTER[b] [214]	CCT	600	✓	–	–	–	–	–	–	–	–	85.23	96.61	73.20	92.04	40.66	55.67	25.28	31.00

[a] Using 98/49/49 split ratio.
[b] Transductive result.

Table 24
The performance of reviewed few-shot approaches on Meta-Dataset [10]. Only datasets under "In-domain" are used for training.

Method	Backbone	T	In-domain	Out-domain								
			ImageNet	Omniglot	Aircraft	CUB-2011	DTD	Quick Draw	Fungi	VGG-Flower	Traffic signs	MSCOCO
Proto-MAML [10]	ResNet	600	49.53	63.37	55.95	68.66	66.49	51.52	39.96	87.15	48.83	43.74
RFS [94]	ResNet	1000	61.43	64.31	62.32	79.47	79.28	60.83	48.53	91.00	76.33	59.28
Meta-baseline [93]	ResNet	1000	59.2	69.1	54.1	77.3	76.0	57.3	45.4	89.6	66.2	55.7
P>M>F [208]	ResNet	600	67.08	75.33	75.39	72.08	86.42	66.79	50.53	94.14	86.54	58.2
P>M>F [208]	ViT	600	74.69	80.68	76.78	85.04	86.63	71.25	54.78	94.57	88.33	62.57
Method	Backbone	T	In-domain								Out-domain	
			ImageNet	Omniglot	Aircraft	CUB-2011	DTD	Quick Draw	Fungi	VGG-Flower	Traffic signs	MSCOCO
Proto-MAML [10]	ResNet	600	46.52	82.69	75.23	69.88	68.25	66.84	41.99	88.72	52.42	41.74
Classifier-baseline [93]	ResNet	1000	55.0	76.9	69.8	78.3	71.4	62.7	55.4	90.6	69.3	53.1
Meta-baseline [93]	ResNet	1000	48.0	89.4	81.7	77.3	64.5	74.5	60.2	83.8	59.5	43.6
P>M>F [208]	ResNet	600	67.51	85.91	80.3	81.67	87.08	72.84	60.03	94.69	87.17	58.92
P>M>F [208]	ViT	600	74.59	91.79	88.33	91.02	86.61	79.23	74.2	94.12	88.85	62.59

Table 25
The 5-way K-shot accuracy of existing approaches with different number of T.

Methods	T	*mini*ImageNet, 5-way		*tiered*ImageNet, 5-way		CIFAR-FS, 5-way		FC100, 5-way	
		1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
ProtoNet [13]	600	50.17 ± 0.77	67.96 ± 0.64	44.71 ± 0.80	65.81 ± 0.76	46.11 ± 0.87	68.70 ± 0.80	36.56 ± 0.75	49.66 ± 0.72
	2000	49.91 ± 0.43	67.79 ± 0.36	44.41 ± 0.46	66.52 ± 0.41	46.29 ± 0.47	68.30 ± 0.42	36.46 ± 0.40	49.68 ± 0.39
	10,000	49.88 ± 0.20	67.83 ± 0.16	43.80 ± 0.20	65.97 ± 0.19	46.14 ± 0.21	68.40 ± 0.19	36.79 ± 0.18	49.88 ± 0.17
RFS-Gen 0 [94]	600	52.76 ± 0.75	70.32 ± 0.61	50.04 ± 0.87	67.00 ± 0.77	62.38 ± 0.87	78.69 ± 0.70	34.73 ± 0.70	45.81 ± 0.73
	2000	52.97 ± 0.75	70.22 ± 0.34	50.31 ± 0.47	67.07 ± 0.41	62.22 ± 0.49	78.52 ± 0.38	34.58 ± 0.38	45.97 ± 0.39
	10,000	53.15 ± 0.19	70.20 ± 0.15	50.23 ± 0.21	67.24 ± 0.18	62.33 ± 0.22	78.58 ± 0.17	34.70 ± 0.17	45.75 ± 0.18
FEAT [32]	600	66.99 ± 0.83	82.21 ± 0.57	69.74 ± 0.94	85.10 ± 0.63	–	–	–	–
	2000	66.88 ± 0.46	82.11 ± 0.31	70.01 ± 0.51	84.89 ± 0.35	–	–	–	–
	10,000	66.70 ± 0.20	82.07 ± 0.14	70.40 ± 0.23	84.83 ± 0.16	–	–	–	–
SCL [104]	600	66.90 ± 0.80	83.12 ± 0.54	69.93 ± 0.89	85.04 ± 0.66	75.76 ± 0.82	88.45 ± 0.58	46.22 ± 0.76	62.62 ± 0.75
	2000	66.99 ± 0.42	83.39 ± 0.29	70.35 ± 0.50	85.43 ± 0.34	75.45 ± 0.46	88.10 ± 0.32	46.14 ± 0.43	62.59 ± 0.42
	10,000	66.96 ± 0.19	83.18 ± 0.13	70.99 ± 0.22	85.52 ± 0.15	75.42 ± 0.21	87.96 ± 0.15	45.91 ± 0.19	62.58 ± 0.19
SSL-ProtoNet [60]	600	52.17 ± 0.78	70.72 ± 0.63	54.68 ± 0.90	73.81 ± 0.71	59.86 ± 0.92	75.74 ± 0.71	–	–
	2000	52.22 ± 0.43	70.75 ± 0.35	54.94 ± 0.50	73.92 ± 0.39	59.83 ± 0.50	75.55 ± 0.39	–	–
	10,000	51.97 ± 0.20	70.58 ± 0.16	54.31 ± 0.22	73.35 ± 0.18	59.82 ± 0.23	75.74 ± 0.17	–	–

Data availability

The authors do not have permission to share data.

References

[1] W. Zhao, R. Chellappa, P.J. Phillips, A. Rosenfeld, Face recognition: A literature survey, ACM Comput. Surv. 35 (4) (2003) 399–458.
[2] J. Hu, J. Lu, Y.-P. Tan, Discriminative deep metric learning for face verification in the wild, in: 2014 IEEE Conference on Computer Vision and Pattern Recognition, 2014, pp. 1875–1882.
[3] J. Guo, X. Zhu, C. Zhao, D. Cao, Z. Lei, S.Z. Li, Learning meta face recognition in unseen domains, in: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020, pp. 6162–6171.
[4] J.-M. Perez-Rua, X. Zhu, T.M. Hospedales, T. Xiang, Incremental few-shot object detection, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020.
[5] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, 2016 IEEE Conf. Comput. Vis. Pattern Recognition (CVPR) (2016) 770–778.
[6] S. Zagoruyko, N. Komodakis, Wide residual networks, in: E.R.H. Richard C. Wilson, W.A.P. Smith (Eds.), Proceedings of the British Machine Vision Conference (BMVC), 2016, pp. 87.1–87.12.
[7] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, L. Chen, MobileNetV2: Inverted residuals and linear bottlenecks, in: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2018, pp. 4510–4520.
[8] M. Tan, Q. Le, EfficientNet: Rethinking model scaling for convolutional neural networks, in: International Conference on Machine Learning, PMLR, 2019, pp. 6105–6114.
[9] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, F.-F. Li, ImageNet: A large-scale hierarchical image database, 2009 IEEE Conf. Comput. Vis. Pattern Recognit. (2009) 248–255.
[10] E. Triantafillou, T. Zhu, V. Dumoulin, P. Lamblin, U. Evci, K. Xu, R. Goroshin, C. Gelada, K. Swersky, P. Manzagol, H. Larochelle, Meta-dataset: A dataset of datasets for learning to learn from few examples, in: 8th International Conference on Learning Representations, ICLR 2020, Addis Ababa, Ethiopia, April 26-30, 2020, 2020.
[11] L. Fei-Fei, R. Fergus, P. Perona, One-shot learning of object categories, IEEE Trans. Pattern Anal. Mach. Intell. 28 (4) (2006) 594–611.
[12] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, N. Houlsby, An image is worth 16x16 words: Transformers for image recognition at scale, Int. Conf. Learn. Represent. (2021).
[13] J. Snell, K. Swersky, R. Zemel, Prototypical networks for few-shot learning, in: Advances in Neural Information Processing Systems, 2017, pp. 4077–4087.
[14] H. Huang, J. Zhang, J. Zhang, J. Xu, Q. Wu, Low-rank pairwise alignment bilinear network for few-shot fine-grained image classification, IEEE Trans. Multimed. 23 (2020) 1666–1680.
[15] J. Zhang, C. Zhao, B. Ni, M. Xu, X. Yang, Variational few-shot learning, 2019 IEEE/ CVF Int. Conf. Comput. Vis. (ICCV) (2019) 1685–1694.
[16] W. Li, J. Xu, J. Huo, L. Wang, Y. Gao, J. Luo, Distribution consistency based covariance metric networks for few-shot learning, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 33, 2019, pp. 8642–8649, 01.
[17] Y. Song, T. Wang, P. Cai, S.K. Mondal, J.P. Sahoo, A comprehensive survey of few-shot learning: Evolution, applications, challenges, and opportunities, ACM Comput. Surv. 55 (13s) (2023) http://dx.doi.org/10.1145/3582688.
[18] J. Lu, P. Gong, J. Ye, J. Zhang, C. Zhang, A survey on machine learning from few samples, Pattern Recognit. 139 (2023) 109480.
[19] J. Wang, K. Liu, Y. Zhang, B. Leng, J. Lu, Recent advances of few-shot learning methods and applications, Sci. China Technol. Sci. 66 (4) (2023) 920–944.
[20] Y. Liu, H. Zhang, W. Zhang, G. Lu, Q. Tian, N. Ling, Few-shot image classification: Current status and research trends, Electronics 11 (11) (2022).
[21] X. Li, X. Yang, Z. Ma, J.-H. Xue, Deep metric learning for few-shot image classification: A review of recent developments, Pattern Recognit. 138 (2023) 109381.
[22] K. He, N. Pu, M. Lao, M.S. Lew, Few-shot and meta-learning methods for image understanding: A survey, Int. J. Multimed. Inf. Retr. 12 (2) (2023) 14.
[23] W. Zeng, Z.-y. Xiao, Few-shot learning based on deep learning: A survey, Math. Biosci. Eng. 21 (1) (2024) 679–711.
[24] J. Ren, C. Li, Y. An, W. Zhang, C. Sun, Few-shot fine-grained image classification: A comprehensive review, AI 5 (1) (2024) 405–425.
[25] G. Koch, R. Zemel, R. Salakhutdinov, Siamese neural networks for one-shot image recognition, in: ICML Deep Learning Workshop, vol. 2, Lille, 2015.
[26] O. Vinyals, C. Blundell, T. Lillicrap, D. Wierstra, et al., Matching networks for one shot learning, Adv. Neural Inf. Process. Syst. 29 (2016).
[27] F. Sung, Y. Yang, L. Zhang, T. Xiang, P.H.S. Torr, T.M. Hospedales, Learning to compare: Relation network for few-shot learning, 2018 IEEE/ CVF Conf. Comput. Vis. Pattern Recognit. (2018) 1199–1208.
[28] M. Ren, S. Ravi, E. Triantafillou, J. Snell, K. Swersky, J.B. Tenenbaum, H. Larochelle, R.S. Zemel, Meta-learning for semi-supervised few-shot classification, in: International Conference on Learning Representations, 2018.
[29] B. Oreshkin, P. Rodríguez López, A. Lacoste, Tadam: Task dependent adaptive metric for improved few-shot learning, Adv. Neural Inf. Process. Syst. 31 (2018) 721–731.
[30] K. Allen, E. Shelhamer, H. Shin, J. Tenenbaum, Infinite mixture prototypes for few-shot learning, in: International Conference on Machine Learning, PMLR, 2019, pp. 232–241.
[31] J. Liu, L. Song, Y. Qin, Prototype rectification for few-shot learning, in: European Conference on Computer Vision, Springer, 2020, pp. 741–756.
[32] H.-J. Ye, H. Hu, D.-C. Zhan, F. Sha, Few-shot learning via embedding adaptation with set-to-set functions, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 8808–8817.
[33] S. Lu, H.-J. Ye, D.-C. Zhan, Tailoring embedding function to heterogeneous few-shot tasks by global and local feature adaptors, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2021, pp. 8776–8783.
[34] Y. Liu, J. Lee, M. Park, S. Kim, E. Yang, S.J. Hwang, Y. Yang, Learning to propagate labels: Transductive propagation network for few-shot learning, in: International Conference on Learning Representations, 2019.
[35] R. Hou, H. Chang, M. Bingpeng, S. Shan, X. Chen, Cross attention network for few-shot classification, in: Advances in Neural Information Processing Systems, 2019, pp. 4003–4014.
[36] X. Li, L. Yu, C.-W. Fu, M. Fang, P.-A. Heng, Revisiting metric learning for few-shot image classification, Neurocomputing 406 (2020) 49–58.
[37] Z. Wu, Y. Li, L. Guo, K. Jia, PARN: Position-aware relation networks for few-shot learning, 2019 IEEE/ CVF Int. Conf. Comput. Vis. (ICCV) (2019) 6658–6666.
[38] J. He, R. Hong, X. Liu, M. Xu, Z. Zha, M.-C. Wang, Memory-augmented relation network for few-shot learning, Proc. the 28th ACM Int. Conf. Multimed. (2020).
[39] C. Zhang, Y. Cai, G. Lin, C. Shen, Deepemd: Few-shot image classification with differentiable earth mover's distance and structured classifiers, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 12203–12213.
[40] W.-Y. Chen, Y.-C. Liu, Z. Kira, Y.-C. Wang, J.-B. Huang, A closer look at few-shot classification, in: International Conference on Learning Representations, 2019.
[41] N. Fei, Z. Lu, T. Xiang, S. Huang, MELR: Meta-learning via modeling episode-level relationships for few-shot learning, in: International Conference on Learning Representations, 2021, URL https://openreview.net/forum?id=D3PcGLdMx0.
[42] J. He, R. Hong, X. Liu, M. Xu, Q. Sun, Revisiting local descriptor for improved few-shot classification, ACM Trans. Multimed. Comput. Commun. Appl. (TOMM) 18 (2s) (2022) 1–23.
[43] F. Gao, X. Luo, Z. Yang, Q. Zhang, Label smoothing and task-adaptive loss function based on prototype network for few-shot learning, Neural Netw. 156 (2022) 39–48.
[44] H. Zhang, H. Li, P. Koniusz, Multi-level second-order few-shot learning, IEEE Trans. Multimed. 25 (2023) 2111–2126, http://dx.doi.org/10.1109/TMM.2022.3142955.
[45] Z. Cui, N. Lu, W. Wang, G. Guo, Dual global-aware propagation for few-shot learning, Image Vis. Comput. 128 (2022) 104574.
[46] D. Zhou, O. Bousquet, T. Lal, J. Weston, B. Schölkopf, Learning with local and global consistency, Adv. Neural Inf. Process. Syst. 16 (2003).
[47] H. Zhu, P. Koniusz, Transductive few-shot learning with prototype-based label propagation by iterative graph refinement, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR, 2023, pp. 23996–24006.
[48] Y. Hu, V. Gripon, S. Pateux, Leveraging the feature distribution in transfer-based few-shot learning, in: International Conference on Artificial Neural Networks, Springer, 2021, pp. 487–499.
[49] O. Veilleux, M. Boudiaf, P. Piantanida, I. Ben Ayed, Realistic evaluation of transductive few-shot learning, Adv. Neural Inf. Process. Syst. 34 (2021) 9290–9302.
[50] M. Lazarou, T. Stathaki, Y. Avrithis, Iterative label cleaning for transductive and semi-supervised few-shot learning, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 8751–8760.
[51] Y. Hu, S. Pateux, V. Gripon, Adaptive dimension reduction and variational inference for transductive few-shot classification, in: International Conference on Artificial Intelligence and Statistics, PMLR, 2023, pp. 5899–5917.
[52] L. Tian, J. Feng, X. Chai, W. Chen, L. Wang, X. Liu, B. Chen, Prototypes-oriented transductive few-shot learning with conditional transport, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 16317–16326.
[53] C. Doersch, A. Gupta, A. Zisserman, Crosstransformers: Spatially-aware few-shot transfer, Adv. Neural Inf. Process. Syst. 33 (2020) 21981–21993.
[54] M. Hiller, R. Ma, M. Harandi, T. Drummond, Rethinking generalization in few-shot classification, Adv. Neural Inf. Process. Syst. 35 (2022) 3582–3595.
[55] F. Hao, F. He, L. Liu, F. Wu, D. Tao, J. Cheng, Class-aware patch embedding adaptation for few-shot image classification, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 18905–18915.

[56] H. Bao, L. Dong, S. Piao, F. Wei, BEit: BERT pre-training of image transformers, in: International Conference on Learning Representations, 2022, URL https://openreview.net/forum?id=p-BhZSz59o4.

[57] Z. Li, Z. Chen, F. Yang, W. Li, Y. Zhu, C. Zhao, R. Deng, L. Wu, R. Zhao, M. Tang, et al., Mst: Masked self-supervised transformer for visual representation, Adv. Neural Inf. Process. Syst. 34 (2021) 13165–13176.

[58] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, R. Girshick, Masked autoencoders are scalable vision learners, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 16000–16009.

[59] J. Zhou, C. Wei, H. Wang, W. Shen, C. Xie, A. Yuille, T. Kong, Image BERT pre-training with online tokenizer, in: International Conference on Learning Representations, 2022, URL https://openreview.net/forum?id=ydopy-e6Dg.

[60] J.Y. Lim, K.M. Lim, C.P. Lee, Y.X. Tan, SSL-ProtoNet: Self-supervised learning prototypical networks for few-shot learning, Expert Syst. Appl. 238 (2024) 122173.

[61] W.H. Liu, K.M. Lim, T.S. Ong, C.P. Lee, NegCosIC: Negative cosine similarity-invariance- covariance regularization for few-shot learning, IEEE Access 12 (2024) 52867–52877, http://dx.doi.org/10.1109/ACCESS.2024.3386808.

[62] W.H. Liu, K.M. Lim, T.S. Ong, C.P. Lee, DBDC-SSL: Deep Brownian distance covariance with self-supervised learning for few-shot image classification, IEEE Access 12 (2024) 58586–58596, http://dx.doi.org/10.1109/ACCESS.2024.3390034.

[63] J.Y. Lim, K.M. Lim, C.P. Lee, Y.X. Tan, SFT: Few-shot learning via self-supervised feature fusion with transformer, IEEE Access 12 (2024) 86690–86703, http://dx.doi.org/10.1109/ACCESS.2024.3416327.

[64] A. Santoro, S. Bartunov, M. Botvinick, D. Wierstra, T. Lillicrap, Meta-learning with memory-augmented neural networks, in: Proceedings of the 33rd International Conference on International Conference on Machine Learning - Volume 48, ICML '16, 2016, pp. 1842–1850.

[65] T. Munkhdalai, H. Yu, Meta networks, Proc. Mach. Learn. Res. 70 (2017) 2554–2563.

[66] N. Mishra, M. Rohaninejad, X. Chen, P. Abbeel, A simple neural attentive meta-learner, in: International Conference on Learning Representations, 2018.

[67] Q. Cai, Y. Pan, T. Yao, C.C. Yan, T. Mei, Memory matching networks for one-shot image recognition, 2018 IEEE/ CVF Conf. Comput. Vis. Pattern Recognit. (2018) 4080–4088.

[68] S. Qiao, C. Liu, W. Shen, A.L. Yuille, Few-shot image recognition by predicting parameters from activations, in: Conference on Computer Vision and Pattern Recognition, 2018.

[69] H. Qi, M. Brown, D.G. Lowe, Low-shot learning with imprinted weights, in: 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2018, pp. 5822–5830, http://dx.doi.org/10.1109/CVPR.2018.00610.

[70] S. Gidaris, N. Komodakis, Dynamic few-shot visual learning without forgetting, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 4367–4375.

[71] T. Hospedales, A. Antoniou, P. Micaelli, A. Storkey, Meta-learning in neural networks: A survey, IEEE Trans. Pattern Anal. Mach. Intell. 44 (09) (2022) 5149–5169.

[72] C. Finn, S. Levine, Meta-learning and universality: Deep representations and gradient descent can approximate any learning algorithm, in: International Conference on Learning Representations, 2018.

[73] M. Andrychowicz, M. Denil, S. Gomez, M.W. Hoffman, D. Pfau, T. Schaul, B. Shillingford, N. De Freitas, Learning to learn by gradient descent by gradient descent, in: Advances in Neural Information Processing Systems, 2016, pp. 3981–3989.

[74] S. Ravi, H. Larochelle, Optimization as a model for few-shot learning, in: International Conference on Learning Representations, 2017.

[75] C. Finn, P. Abbeel, S. Levine, in: D. Precup, Y.W. Teh (Eds.), Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks, in: Proceedings of Machine Learning Research, vol. 70, PMLR, International Convention Centre, Sydney, Australia, 2017, pp. 1126–1135.

[76] Z. Li, F. Zhou, F. Chen, H. Li, Meta-SGD: Learning to learn quickly for few shot learning, 2017, CoRR abs/1707.09835. arXiv:1707.09835.

[77] A. Nichol, J. Achiam, J. Schulman, On first-order meta-learning algorithms, 2018, ArXiv abs/1803.02999.

[78] E. Grant, C. Finn, S. Levine, T. Darrell, T. Griffiths, Recasting gradient-based meta-learning as hierarchical Bayes, in: International Conference on Learning Representations, 2018.

[79] C. Finn, K. Xu, S. Levine, Probabilistic model-agnostic meta-learning, Adv. Neural Inf. Process. Syst. 31 (2018).

[80] J. Yoon, T. Kim, O. Dia, S. Kim, Y. Bengio, S. Ahn, Bayesian model-agnostic meta-learning, in: Advances in Neural Information Processing Systems, 2018, pp. 7332–7342.

[81] Y. Lee, S. Choi, Gradient-based meta-learning with learned layerwise metric and subspace, in: International Conference on Machine Learning, PMLR, 2018, pp. 2927–2936.

[82] R. Vuorio, S.-H. Sun, H. Hu, J.J. Lim, Toward multimodal model-agnostic meta-learning, 2018, ArXiv arXiv:1812.07172.

[83] A. Antoniou, H. Edwards, A. Storkey, How to train your MAML, in: International Conference on Learning Representations, 2019.

[84] A. Rajeswaran, C. Finn, S.M. Kakade, S. Levine, Meta-learning with implicit gradients, Adv. Neural Inf. Process. Syst. 32 (2019).

[85] A.A. Rusu, D. Rao, J. Sygnowski, O. Vinyals, R. Pascanu, S. Osindero, R. Hadsell, Meta-learning with latent embedding optimization, in: International Conference on Learning Representations, 2019.

[86] L. Bertinetto, J.F. Henriques, P. Torr, A. Vedaldi, Meta-learning with differentiable closed-form solvers, in: International Conference on Learning Representations, 2018.

[87] K. Lee, S. Maji, A. Ravichandran, S. Soatto, Meta-learning with differentiable convex optimization, 2019 IEEE/ CVF Conf. Comput. Vis. Pattern Recognition (CVPR) (2019) 10649–10657.

[88] H. Yao, Y. Wei, J. Huang, Z. Li, Hierarchically structured meta-learning, in: International Conference on Machine Learning, PMLR, 2019, pp. 7045–7054.

[89] A. Raghu, M. Raghu, S. Bengio, O. Vinyals, Rapid learning or feature reuse? Towards understanding the effectiveness of MAML, Int. Conf. Learn. Represent. (ICLR) (2020).

[90] S. Lee, S. Lee, B.C. Song, Contextual gradient scaling for few-shot learning, in: 2022 IEEE/CVF Winter Conference on Applications of Computer Vision, WACV, 2022, pp. 3503–3512, http://dx.doi.org/10.1109/WACV51458.2022.00356.

[91] J. Jia, X. Feng, H. Yu, Few-shot classification via efficient meta-learning with hybrid optimization, Eng. Appl. Artif. Intell. 127 (2024) 107296, http://dx.doi.org/10.1016/j.engappai.2023.107296.

[92] Y. Wang, W. Chao, K.Q. Weinberger, L. van der Maaten, SimpleShot: Revisiting nearest-neighbor classification for few-shot learning, 2019, CoRR abs/1911.04623. arXiv:1911.04623.

[93] Y. Chen, Z. Liu, H. Xu, T. Darrell, X. Wang, Meta-baseline: Exploring simple meta-learning for few-shot learning, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 9062–9071.

[94] Y. Tian, Y. Wang, D. Krishnan, J.B. Tenenbaum, P. Isola, Rethinking few-shot image classification: A good embedding is all you need? in: European Conference on Computer Vision, Springer, 2020, pp. 266–282.

[95] J.Y. Lim, K.M. Lim, S.Y. Ooi, C.P. Lee, Efficient-PrototypicalNet with self knowledge distillation for few-shot learning, Neurocomputing 459 (2021) 327–337.

[96] H. Huang, J. Zhang, J. Zhang, Q. Wu, C. Xu, PTN: A Poisson transfer network for semi-supervised few-shot learning, in: AAAI Conference on Artificial Intelligence, 2020.

[97] J. Calder, B. Cook, M. Thorpe, D. Slepčev, Poisson learning: Graph based semi-supervised learning at very low label rates, in: International Conference on Machine Learning, 2020.

[98] B. Zhang, X. Li, Y. Ye, S. Feng, R. Ye, MetaNODE: Prototype optimization as a neural ODE for few-shot learning, in: AAAI Conference on Artificial Intelligence, 2022.

[99] S. Gidaris, A. Bursuc, N. Komodakis, P. Pérez, M. Cord, Boosting few-shot visual learning with self-supervision, 2019 IEEE/ CVF Int. Conf. Comput. Vis. (ICCV) (2019) 8058–8067.

[100] Q. Sun, Y. Liu, T.-S. Chua, B. Schiele, Meta-transfer learning for few-shot learning, in: 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR, 2019, pp. 403–412, http://dx.doi.org/10.1109/CVPR.2019.00049.

[101] Q. Sun, Y. Liu, Z. Chen, T.-S. Chua, B. Schiele, Meta-transfer learning through hard tasks, IEEE Trans. Pattern Anal. Mach. Intell. 44 (3) (2022) 1443–1456.

[102] J. Rajasegaran, S.H. Khan, M. Hayat, F.S. Khan, M. Shah, Self-supervised knowledge distillation for few-shot learning, in: Proceedings of the British Machine Vision Conference (BMVC), 2021.

[103] P. Singh, P. Mazumder, Dual class representation learning for few-shot image classification, Knowl.-Based Syst. 238 (2022) 107840.

[104] J.Y. Lim, K.M. Lim, C.P. Lee, Y.X. Tan, SCL: Self-supervised contrastive learning for few-shot image classification, Neural Netw. 165 (2023) 19–30.

[105] S. Gidaris, P. Singh, N. Komodakis, Unsupervised representation learning by predicting image rotations, in: International Conference on Learning Representations, 2018.

[106] B. Hariharan, R. Girshick, Low-shot visual recognition by shrinking and hallucinating features, in: Proceedings of the IEEE International Conference on Computer Vision, 2017, pp. 3018–3027.

[107] Y.-X. Wang, R. Girshick, M. Hebert, B. Hariharan, Low-shot learning from imaginary data, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 7278–7286.

[108] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, Y. Bengio, Generative adversarial nets, Adv. Neural Inf. Process. Syst. 27 (2014).

[109] A. Mehrotra, A. Dukkipati, Generative adversarial residual pairwise networks for one shot learning, 2017, arXiv preprint arXiv:1703.08033.

[110] R. Zhang, T. Che, Z. Ghahramani, Y. Bengio, Y. Song, Metagan: An adversarial approach to few-shot learning, Adv. Neural Inf. Process. Syst. 31 (2018).

[111] K. Li, Y. Zhang, K. Li, Y. Fu, Adversarial feature hallucination networks for few-shot learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 13470–13479.

[112] Z. Chen, Y. Fu, Y.-X. Wang, L. Ma, W. Liu, M. Hebert, Image deformation meta-networks for one-shot learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 8680–8689.

[113] H. Zhang, J. Zhang, P. Koniusz, Few-shot learning via saliency-guided hallucination of samples, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 2770–2779.
[114] J. Zhang, T. Zhang, Y. Dai, M. Harandi, R. Hartley, Deep unsupervised saliency detection: A multiple noisy labeling perspective, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 9029–9038.
[115] H. Tang, Z. Li, Z. Peng, J. Tang, Blockmix: Meta regularization and self-calibrated inference for metric-based meta-learning, in: Proceedings of the 28th ACM International Conference on Multimedia, 2020, pp. 610–618.
[116] U. Osahor, N.M. Nasrabadi, Ortho-shot: Low displacement rank regularization with data augmentation for few-shot learning, in: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2022, pp. 2200–2209.
[117] Y. Zhang, M. Gong, J. Li, K. Feng, M. Zhang, Few-shot learning with enhancements to data augmentation and feature extraction, IEEE Trans. Neural Netw. Learn. Syst. (2024).
[118] E. Schwartz, L. Karlinsky, J. Shtok, S. Harary, M. Marder, A. Kumar, R. Feris, R. Giryes, A. Bronstein, Delta-encoder: An effective sample synthesis method for few-shot object recognition, Adv. Neural Inf. Process. Syst. 31 (2018).
[119] Z. Chen, Y. Fu, Y. Zhang, Y.-G. Jiang, X. Xue, L. Sigal, Multi-level semantic feature augmentation for one-shot learning, IEEE Trans. Image Process. 28 (9) (2019) 4594–4605.
[120] Y. Xian, S. Sharma, B. Schiele, Z. Akata, f-vaegan-d2: A feature generating framework for any-shot learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 10275–10284.
[121] Y. Zhu, W. Min, S. Jiang, Attribute-guided feature learning for few-shot image recognition, IEEE Trans. Multimed. 23 (2020) 1200–1209.
[122] L. Zhang, S. Wang, X. Chang, J. Liu, Z. Ge, Q. Zheng, Auto-FSL: Searching the attribute consistent network for few-shot learning, IEEE Trans. Circuits Syst. Video Technol. 32 (3) (2021) 1213–1223.
[123] C. Xing, N. Rostamzadeh, B. Oreshkin, P.O. O Pinheiro, Adaptive cross-modal few-shot learning, Adv. Neural Inf. Process. Syst. 32 (2019).
[124] P. Tokmakov, Y.-X. Wang, M. Hebert, Learning compositional representations for few-shot recognition, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2019, pp. 6372–6381.
[125] S. Huang, M. Zhang, Y. Kang, D. Wang, Attributes-guided and pure-visual attention alignment for few-shot recognition, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 35, 2021, pp. 7840–7847, 9.
[126] C. Chen, X. Yang, M. Yan, C. Xu, Attribute-guided dynamic routing graph network for transductive few-shot learning, in: Proceedings of the 30th ACM International Conference on Multimedia, 2022, pp. 6259–6268.
[127] W. Xu, Y. Xian, J. Wang, B. Schiele, Z. Akata, Attribute prototype network for any-shot learning, Int. J. Comput. Vis. 130 (7) (2022) 1735–1753.
[128] Q. Lyu, W. Wang, Compositional prototypical networks for few-shot classification, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 37, 2023, pp. 9011–9019, 7.
[129] W. Xu, Y. Xian, J. Wang, B. Schiele, Z. Akata, Attribute prototype network for zero-shot learning, Adv. Neural Inf. Process. Syst. 33 (2020) 21969–21980.
[130] A. Cai, L. Chen, Y. Chen, Z. He, S. Tao, C. Zhou, Adaptive attribute distribution similarity for few-shot learning, Image Vis. Comput. (2024) 105118.
[131] Z. Wang, H. Li, Z. Zhang, F. Chen, J. Zhai, Attribute-and attention-guided few-shot classification, Multimedia Syst. 30 (1) (2024) 60.
[132] A. Radford, J.W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, et al., Learning transferable visual models from natural language supervision, in: International Conference on Machine Learning, PMLR, 2021, pp. 8748–8763.
[133] C. Jia, Y. Yang, Y. Xia, Y.-T. Chen, Z. Parekh, H. Pham, Q. Le, Y.-H. Sung, Z. Li, T. Duerig, Scaling up visual and vision-language representation learning with noisy text supervision, in: International Conference on Machine Learning, PMLR, 2021, pp. 4904–4916.
[134] J.-B. Alayrac, J. Donahue, P. Luc, A. Miech, I. Barr, Y. Hasson, K. Lenc, A. Mensch, K. Millican, M. Reynolds, et al., Flamingo: A visual language model for few-shot learning, Adv. Neural Inf. Process. Syst. 35 (2022) 23716–23736.
[135] J. Li, D. Li, C. Xiong, S. Hoi, Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation, in: International Conference on Machine Learning, PMLR, 2022, pp. 12888–12900.
[136] L. Fei-Fei, R. Fergus, P. Perona, Learning generative visual models from few training examples: An incremental bayesian approach tested on 101 object categories, in: 2004 Conference on Computer Vision and Pattern Recognition Workshop, IEEE, 2004, 178–178.
[137] S. Maji, E. Rahtu, J. Kannala, M. Blaschko, A. Vedaldi, Fine-grained visual classification of aircraft, 2013, arXiv preprint arXiv:1306.5151.
[138] O.M. Parkhi, A. Vedaldi, A. Zisserman, C. Jawahar, Cats and dogs, in: 2012 IEEE Conference on Computer Vision and Pattern Recognition, IEEE, 2012, pp. 3498–3505.
[139] J. Krause, M. Stark, J. Deng, L. Fei-Fei, 3D object representations for fine-grained categorization, in: 2013 IEEE International Conference on Computer Vision Workshops, 2013, pp. 554–561, http://dx.doi.org/10.1109/ICCVW.2013.77.
[140] M.-E. Nilsback, A. Zisserman, Automated flower classification over a large number of classes, in: 2008 Sixth Indian Conference on Computer Vision, Graphics & Image Processing, IEEE, 2008, pp. 722–729.
[141] L. Bossard, M. Guillaumin, L. Van Gool, Food-101–mining discriminative components with random forests, in: Computer Vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Proceedings, Part VI 13, Springer, 2014, pp. 446–461.
[142] K. Soomro, A.R. Zamir, M. Shah, UCF101: A dataset of 101 human actions classes from videos in the wild, 2012, URL https://arxiv.org/abs/1212.0402.
[143] B. Lester, R. Al-Rfou, N. Constant, The power of scale for parameter-efficient prompt tuning, in: Conference on Empirical Methods in Natural Language Processing, 2021, URL https://api.semanticscholar.org/CorpusID:233296808.
[144] T. Shin, Y. Razeghi, R.L.L. IV, E. Wallace, S. Singh, AutoPrompt: Eliciting knowledge from language models with automatically generated prompts, in: Empirical Methods in Natural Language Processing, EMNLP, 2020.
[145] X.L. Li, P. Liang, Prefix-tuning: Optimizing continuous prompts for generation, in: C. Zong, F. Xia, W. Li, R. Navigli (Eds.), Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers), Association for Computational Linguistics, Online, 2021, pp. 4582–4597, http://dx.doi.org/10.18653/v1/2021.acl-long.353, URL https://aclanthology.org/2021.acl-long.353/.
[146] L. Reynolds, K. McDonell, Prompt programming for large language models: Beyond the few-shot paradigm, in: Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems, 2021, pp. 1–7.
[147] X. Liu, Y. Zheng, Z. Du, M. Ding, Y. Qian, Z. Yang, J. Tang, GPT understands, too, AI Open 5 (2024) 208–215.
[148] N. Houlsby, A. Giurgiu, S. Jastrzebski, B. Morrone, Q. De Laroussilhe, A. Gesmundo, M. Attariyan, S. Gelly, Parameter-efficient transfer learning for NLP, in: International Conference on Machine Learning, PMLR, 2019, pp. 2790–2799.
[149] N. Ding, Y. Qin, G. Yang, F. Wei, Z. Yang, Y. Su, S. Hu, Y. Chen, C.-M. Chan, W. Chen, et al., Delta tuning: A comprehensive study of parameter efficient methods for pre-trained language models, 2022, arXiv preprint arXiv:2203.06904.
[150] S. Jie, Z.-H. Deng, S. Chen, Z. Jin, Convolutional bypasses are better vision transformer adapters, in: ECAI 2024, IOS Press, 2024, pp. 202–209.
[151] S. Jie, Z.-H. Deng, Fact: Factor-tuning for lightweight adaptation on vision transformer, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 37, 2023, pp. 1060–1068, 1.
[152] K. Zhou, J. Yang, C.C. Loy, Z. Liu, Learning to prompt for vision-language models, Int. J. Comput. Vis. 130 (9) (2022) 2337–2348.
[153] K. Zhou, J. Yang, C.C. Loy, Z. Liu, Conditional prompt learning for vision-language models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 16816–16825.
[154] T. Huang, J. Chu, F. Wei, Unsupervised prompt learning for vision-language models, 2022, arXiv preprint arXiv:2204.03649.
[155] G. Chen, W. Yao, X. Song, X. Li, Y. Rao, K. Zhang, PLOT: Prompt learning with optimal transport for vision-language models, in: The Eleventh International Conference on Learning Representations, 2023, URL https://openreview.net/forum?id=zqwryBoXYnh.
[156] C. Villani, et al., Optimal Transport: Old and New, vol. 338, Springer, 2009.
[157] Y. Lu, J. Liu, Y. Zhang, Y. Liu, X. Tian, Prompt distribution learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 5206–5215.
[158] K. Ding, Y. Wang, P. Liu, Q. Yu, H. Zhang, S. Xiang, C. Pan, Multi-task prompt tuning with soft context sharing for vision–language models, Neurocomputing 603 (2024) 128290.
[159] A. Bulat, G. Tzimiropoulos, Lasp: Text-to-text optimization for language-aware soft prompting of vision & language models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 23232–23241.
[160] P. Gao, S. Geng, R. Zhang, T. Ma, R. Fang, Y. Zhang, H. Li, Y. Qiao, Clip-adapter: Better vision-language models with feature adapters, Int. J. Comput. Vis. 132 (2) (2024) 581–595.
[161] M. Shu, W. Nie, D.-A. Huang, Z. Yu, T. Goldstein, A. Anandkumar, C. Xiao, Test-time prompt tuning for zero-shot generalization in vision-language models, Adv. Neural Inf. Process. Syst. 35 (2022) 14274–14289.
[162] D. Hendrycks, N. Mu, E.D. Cubuk, B. Zoph, J. Gilmer, B. Lakshminarayanan, AugMix: A simple data processing method to improve robustness and uncertainty, Proc. Int. Conf. Learn. Represent. (ICLR) (2020).
[163] R. Zhang, W. Zhang, R. Fang, P. Gao, K. Li, J. Dai, Y. Qiao, H. Li, Tip-adapter: Training-free adaption of clip for few-shot classification, in: European Conference on Computer Vision, Springer, 2022, pp. 493–510.
[164] O. Pantazis, G.J. Brostow, K.E. Jones, O.M. Aodha, SVL-adapter: Self-supervised adapter for vision-language pretrained models, in: BMVC, 2022, p. 580, URL https://bmvc2022.mpi-inf.mpg.de/580/.
[165] Z. Guo, R. Zhang, L. Qiu, X. Ma, X. Miao, X. He, B. Cui, Calip: Zero-shot enhancement of clip with parameter-free attention, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 37, 2023, pp. 746–754, 1.
[166] S. Pratt, I. Covert, R. Liu, A. Farhadi, What does a platypus look like? generating customized prompts for zero-shot image classification, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 15691–15701.

[167] T. Brown, B. Mann, N. Ryder, M. Subbiah, J.D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, et al., Language models are few-shot learners, Adv. Neural Inf. Process. Syst. 33 (2020) 1877–1901.

[168] S. Menon, C. Vondrick, Visual classification via description from large language models, in: The Eleventh International Conference on Learning Representations, 2023, URL https://openreview.net/forum?id=jlAjNL8z5cs.

[169] V. Udandarao, A. Gupta, S. Albanie, Sus-x: Training-free name-only transfer of vision-language models, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 2725–2736.

[170] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, B. Ommer, High-resolution image synthesis with latent diffusion models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 10684–10695.

[171] C. Schuhmann, R. Beaumont, R. Vencu, C. Gordon, R. Wightman, M. Cherti, T. Coombes, A. Katta, C. Mullis, M. Wortsman, et al., Laion-5b: An open large-scale dataset for training next generation image-text models, Adv. Neural Inf. Process. Syst. 35 (2022) 25278–25294.

[172] R. Zhang, X. Hu, B. Li, S. Huang, H. Deng, Y. Qiao, P. Gao, H. Li, Prompt, generate, then cache: Cascade of foundation models makes strong few-shot learners, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 15211–15222.

[173] A. Ramesh, M. Pavlov, G. Goh, S. Gray, C. Voss, A. Radford, M. Chen, I. Sutskever, Zero-shot text-to-image generation, in: International Conference on Machine Learning, Pmlr, 2021, pp. 8821–8831.

[174] M. Caron, H. Touvron, I. Misra, H. Jégou, J. Mairal, P. Bojanowski, A. Joulin, Emerging properties in self-supervised vision transformers, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 9650–9660.

[175] X.-S. Wei, P. Wang, L. Liu, C. Shen, J. Wu, Piecewise classifier mappings: Learning fine-grained learners for novel categories with few examples, IEEE Trans. Image Process. 28 (12) (2019) 6116–6125.

[176] Y. Zhu, C. Liu, S. Jiang, et al., Multi-attention meta learning for few-shot fine-grained image recognition., in: IJCAI, Beijing, 2020, pp. 1090–1096.

[177] W. Li, L. Wang, J. Xu, J. Huo, Y. Gao, J. Luo, Revisiting local descriptor based image-to-class measure for few-shot learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 7260–7268.

[178] O. Boiman, E. Shechtman, M. Irani, In defense of nearest-neighbor based image classification, in: 2008 IEEE Conference on Computer Vision and Pattern Recognition, 2008, pp. 1–8, http://dx.doi.org/10.1109/CVPR.2008.4587598.

[179] Y. Yu, D. Zhang, S. Wang, Z. Ji, Z. Zhang, Local spatial alignment network for few-shot learning, Neurocomputing 497 (2022) 182–190.

[180] H. Huang, J. Zhang, L. Yu, J. Zhang, Q. Wu, C. Xu, TOAN: Target-oriented alignment network for fine-grained image categorization with few labeled samples, IEEE Trans. Circuits Syst. Video Technol. 32 (2) (2021) 853–866.

[181] Y. Wu, B. Zhang, G. Yu, W. Zhang, B. Wang, T. Chen, J. Fan, Object-aware long-short-range spatial alignment for few-shot fine-grained image classification, in: Proceedings of the 29th ACM International Conference on Multimedia, 2021, pp. 107–115.

[182] X. Li, J. Wu, Z. Sun, Z. Ma, J. Cao, J.-H. Xue, BSNet: Bi-similarity network for few-shot fine-grained image classification, IEEE Trans. Image Process. 30 (2020) 1318–1331.

[183] B. Zhang, J. Yuan, B. Li, T. Chen, J. Fan, B. Shi, Learning cross-image object semantic relation in transformer for few-shot fine-grained image classification, in: Proceedings of the 30th ACM International Conference on Multimedia, 2022, pp. 2135–2144.

[184] D. Wertheimer, L. Tang, B. Hariharan, Few-shot classification with feature map reconstruction networks, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 8012–8021.

[185] J. Wu, D. Chang, A. Sain, X. Li, Z. Ma, J. Cao, J. Guo, Y.-Z. Song, Bi-directional feature reconstruction network for fine-grained few-shot image classification, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 37, 2023, pp. 2821–2829, 3.

[186] J. Wu, D. Chang, A. Sain, X. Li, Z. Ma, J. Cao, J. Guo, Y.-Z. Song, Bi-directional ensemble feature reconstruction network for few-shot fine-grained classification, IEEE Trans. Pattern Anal. Mach. Intell. (2024).

[187] X. Li, Q. Song, J. Wu, R. Zhu, Z. Ma, J.-H. Xue, Locally-enriched cross-reconstruction for few-shot fine-grained image classification, IEEE Trans. Circuits Syst. Video Technol. 33 (12) (2023) 7530–7540.

[188] S. Lee, W. Moon, J.-P. Heo, Task discrepancy maximization for fine-grained few-shot classification, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 5331–5340.

[189] H. Tang, C. Yuan, Z. Li, J. Tang, Learning attention-guided pyramidal features for few-shot fine-grained recognition, Pattern Recognit. 130 (2022) 108792.

[190] Z. Zha, H. Tang, Y. Sun, J. Tang, Boosting few-shot fine-grained recognition with background suppression and foreground alignment, IEEE Trans. Circuits Syst. Video Technol. 33 (8) (2023) 3947–3961.

[191] Z.-X. Ma, Z.-D. Chen, L.-J. Zhao, Z.-C. Zhang, X. Luo, X.-S. Xu, Cross-layer and cross-sample feature optimization network for few-shot fine-grained image classification, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 38, 2024, pp. 4136–4144, 5.

[192] Z.-X. Ma, Z.-D. Chen, L.-J. Zhao, Z.-C. Zhang, T. Zheng, X. Luo, X.-S. Xu, Bi-directional task-guided network for few-shot fine-grained image classification, in: Proceedings of the 32nd ACM International Conference on Multimedia, 2024, pp. 8277–8286.

[193] C. Wang, S. Song, Q. Yang, X. Li, G. Huang, Fine-grained few shot learning with foreground object transformation, Neurocomputing 466 (2021) 16–26.

[194] L. Zhao, G. Liu, D. Guo, W. Li, X. Fang, Boosting few-shot visual recognition via saliency-guided complementary attention, Neurocomputing 507 (2022) 412–427.

[195] L. Zhao, G. Liu, D. Guo, W. Li, X. Fang, Saliency-guided complementary attention for improved few-shot learning, in: 2021 IEEE International Conference on Multimedia and Expo, ICME, IEEE, 2021, pp. 1–6.

[196] H. Liu, C.P. Chen, X. Gong, T. Zhang, Robust saliency-aware distillation for few-shot fine-grained visual recognition, IEEE Trans. Multimed. (2024).

[197] H.-Y. Tseng, H.-Y. Lee, J.-B. Huang, M.-H. Yang, Cross-domain few-shot classification via learned feature-wise transformation, in: International Conference on Learning Representations, 2020, URL https://openreview.net/forum?id=SJl5Np4tPr.

[198] G. Cohen, S. Afshar, J. Tapson, A. Van Schaik, EMNIST: Extending MNIST to handwritten letters, in: 2017 International Joint Conference on Neural Networks, IJCNN, IEEE, 2017, pp. 2921–2926.

[199] Y. Guo, N.C. Codella, L. Karlinsky, J.V. Codella, J.R. Smith, K. Saenko, T. Rosing, R. Feris, A broader study of cross-domain few-shot learning, in: Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXVII 16, Springer, 2020, pp. 124–141.

[200] C.P. Phoo, B. Hariharan, Self-training for few-shot transfer across extreme task differences, in: International Conference on Learning Representations, 2021, URL https://openreview.net/forum?id=O3Y56aqpChA.

[201] J. Sun, S. Lapuschkin, W. Samek, Y. Zhao, N.-M. Cheung, A. Binder, Explanation-guided training for cross-domain few-shot classification, in: 2020 25th International Conference on Pattern Recognition, ICPR, IEEE, 2021, pp. 7609–7616.

[202] H. Wang, Z.-H. Deng, Cross-domain few-shot classification via adversarial task augmentation, in: Z.-H. Zhou (Ed.), Proceedings of the Thirtieth International Joint Conference on Artificial Intelligence, IJCAI-21, International Joint Conferences on Artificial Intelligence Organization, 2021, pp. 1075–1081, http://dx.doi.org/10.24963/ijcai.2021/149, Main Track.

[203] H. Wang, H. Mai, Y. Gong, Z.-H. Deng, Towards well-generalizing meta-learning via adversarial task augmentation, Artificial Intelligence (2023) 103875.

[204] H. Liang, Q. Zhang, P. Dai, J. Lu, Boosting the generalization capability in cross-domain few-shot learning via noise-enhanced supervised autoencoder, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 9424–9434.

[205] Y. Fu, Y. Fu, Y.-G. Jiang, Meta-fdmixup: Cross-domain few-shot learning guided by labeled target data, in: Proceedings of the 29th ACM International Conference on Multimedia, 2021, pp. 5326–5334.

[206] Y. Fu, Y. Fu, J. Chen, Y.-G. Jiang, Generalized meta-fdmixup: Cross-domain few-shot learning guided by labeled target data, IEEE Trans. Image Process. 31 (2022) 7078–7090.

[207] Y. Hu, A.J. Ma, Adversarial feature augmentation for cross-domain few-shot classification, in: European Conference on Computer Vision, Springer, 2022, pp. 20–37.

[208] S.X. Hu, D. Li, J. Stühmer, M. Kim, T.M. Hospedales, Pushing the limits of simple pipelines for few-shot learning: External data and fine-tuning make a difference, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 9068–9077.

[209] P. Li, S. Gong, C. Wang, Y. Fu, Ranking distance calibration for cross-domain few-shot learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 9099–9108.

[210] Y. Fu, Y. Xie, Y. Fu, Y.-G. Jiang, Styleadv: Meta style adversarial training for cross-domain few-shot learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 24575–24584.

[211] F. Zhou, P. Wang, L. Zhang, W. Wei, Y. Zhang, Revisiting prototypical network for cross domain few-shot learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 20061–20070.

[212] Y. Zou, Y. Liu, Y. Hu, Y. Li, R. Li, Flatten long-range loss landscapes for cross-domain few-shot learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 23575–23584.

[213] Y. Zou, R. Ma, Y. Li, R. Li, Attention temperature matters in vit-based cross-domain few-shot learning, in: The Thirty-Eighth Annual Conference on Neural Information Processing Systems, 2024, URL https://openreview.net/forum?id=o8m4RM5mBk.

[214] N. Paeedeh, M. Pratama, M.A. Ma'sum, W. Mayer, Z. Cao, R. Kowlczyk, Cross-domain few-shot learning via adaptive transformer networks, Knowl.-Based Syst. 288 (2024) 111458.

[215] X. Tao, X. Hong, X. Chang, S. Dong, X. Wei, Y. Gong, Few-shot class-incremental learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 12183–12192.

[216] H. Zhao, Y. Fu, M. Kang, Q. Tian, F. Wu, X. Li, Mgsvf: Multi-grained slow versus fast framework for few-shot class-incremental learning, IEEE Trans. Pattern Anal. Mach. Intell. 46 (3) (2021) 1576–1588.

[217] S. Dong, X. Hong, X. Tao, X. Chang, X. Wei, Y. Gong, Few-shot class-incremental learning via relation knowledge distillation, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 35, 2021, pp. 1255–1263, 2.

[218] L. Zhao, J. Lu, Y. Xu, Z. Cheng, D. Guo, Y. Niu, X. Fang, Few-shot class-incremental learning via class-aware bilateral distillation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 11838–11847.

[219] Z. Song, Y. Zhao, Y. Shi, P. Peng, L. Yuan, Y. Tian, Learning with fantasy: Semantic-aware virtual contrastive constraint for few-shot class-incremental learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 24183–24192.

[220] X. Wang, Z. Ji, Y. Pang, Y. Yu, A cognition-driven framework for few-shot class-incremental learning, Neurocomputing 600 (2024) 128118.

[221] J. Liu, Z. Ji, Y. Pang, Y. Yu, NTK-guided few-shot class incremental learning, IEEE Trans. Image Process. 33 (2024) 6029–6044, http://dx.doi.org/10.1109/TIP.2024.3478854.

[222] Z. Ji, Z. Hou, X. Liu, Y. Pang, X. Li, Memorizing complementation network for few-shot class-incremental learning, IEEE Trans. Image Process. 32 (2023) 937–948.

[223] H. Kang, J. Yoon, S.R.H. Madjid, S.J. Hwang, C.D. Yoo, On the soft-subnetwork for few-shot class incremental learning, in: The Eleventh International Conference on Learning Representations, 2023, URL https://openreview.net/forum?id=z57WK5lGeHd.

[224] K. Zhu, Y. Cao, W. Zhai, J. Cheng, Z.-J. Zha, Self-promoted prototype refinement for few-shot class-incremental learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 6801–6810.

[225] C. Zhang, N. Song, G. Lin, Y. Zheng, P. Pan, Y. Xu, Few-shot incremental learning with continually evolved classifiers, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 12455–12464.

[226] G. Shi, J. Chen, W. Zhang, L.-M. Zhan, X.-M. Wu, Overcoming catastrophic forgetting in incremental few-shot learning by finding flat minima, Adv. Neural Inf. Process. Syst. 34 (2021) 6747–6761.

[227] D.-W. Zhou, H.-J. Ye, L. Ma, D. Xie, S. Pu, D.-C. Zhan, Few-shot class-incremental learning by sampling multi-phase tasks, IEEE Trans. Pattern Anal. Mach. Intell. 45 (11) (2022) 12816–12831.

[228] D.-W. Zhou, F.-Y. Wang, H.-J. Ye, L. Ma, S. Pu, D.-C. Zhan, Forward compatible few-shot class-incremental learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 9046–9056.

[229] M. Hersche, G. Karunaratne, G. Cherubini, L. Benini, A. Sebastian, A. Rahimi, Constrained few-shot class-incremental learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 9057–9067.

[230] Y. Yang, H. Yuan, X. Li, Z. Lin, P. Torr, D. Tao, Neural collapse inspired feature-classifier alignment for few-shot class-incremental learning, in: The Eleventh International Conference on Learning Representations, 2023, URL https://openreview.net/forum?id=y5W8tpojhtJ.

[231] X. Wang, Z. Ji, Y. Yu, Y. Pang, J. Han, Model attention expansion for few-shot class-incremental learning, IEEE Trans. Image Process. (2024).

[232] K.-H. Park, K. Song, G.-M. Park, Pre-trained vision and language transformers are few-shot incremental learners, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 23881–23890.

[233] B.M. Lake, R. Salakhutdinov, J. Gross, J.B. Tenenbaum, One shot learning of simple visual concepts, Cogn. Sci. 33 (2011).

[234] A. Krizhevsky, Learning multiple layers of features from tiny images, 2009.

[235] P. Welinder, S. Branson, T. Mita, C. Wah, F. Schroff, S. Belongie, P. Perona, Caltech-UCSD Birds 200, Technical Report CNS-TR-2010-001, California Institute of Technology, 2010.

[236] C. Wah, S. Branson, P. Welinder, P. Perona, S. Belongie, The Caltech-Ucsd Birds-200–2011 Dataset, Technical Report CNS-TR-2011-001, California Institute of Technology, 2011.

[237] A. Khosla, N. Jayadevaprakash, B. Yao, F.-F. Li, Novel dataset for fine-grained image categorization: Stanford dogs, in: Proc. CVPR Workshop on Fine-Grained Visual Categorization, FGVC, vol. 2, Citeseer, 2011, 1.

[238] G. Van Horn, S. Branson, R. Farrell, S. Haber, J. Barry, P. Ipeirotis, P. Perona, S. Belongie, Building a bird recognition app and large scale dataset with citizen scientists: The fine print in fine-grained dataset collection, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2015, pp. 595–604.

[239] D. Wertheimer, B. Hariharan, Few-shot learning with localization in realistic settings, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 6558–6567.

[240] N. Hilliard, L. Phillips, S. Howland, A. Yankov, C.D. Corley, N.O. Hodas, Few-shot learning with metric-agnostic conditional embeddings, 2018, arXiv preprint arXiv:1802.04376.

[241] G. Van Horn, O. Mac Aodha, Y. Song, Y. Cui, C. Sun, A. Shepard, H. Adam, P. Perona, S. Belongie, The inaturalist species classification and detection dataset, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 8769–8778.

[242] B. Zhou, A. Lapedriza, A. Khosla, A. Oliva, A. Torralba, Places: A 10 million image database for scene recognition, IEEE Trans. Pattern Anal. Mach. Intell. 40 (6) (2017) 1452–1464.

[243] S.P. Mohanty, D.P. Hughes, M. Salathé, Using deep learning for image-based plant disease detection, Front. Plant Sci. 7 (2016) 1419.

[244] P. Helber, B. Bischke, A. Dengel, D. Borth, Eurosat: A novel dataset and deep learning benchmark for land use and land cover classification, IEEE J. Sel. Top. Appl. Earth Obs. Remote. Sens. 12 (7) (2019) 2217–2226.

[245] P. Tschandl, C. Rosendahl, H. Kittler, The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions, Sci. Data 5 (1) (2018) 1–9.

[246] N. Codella, V. Rotemberg, P. Tschandl, M.E. Celebi, S. Dusza, D. Gutman, B. Helba, A. Kalloo, K. Liopyris, M. Marchetti, et al., Skin lesion analysis toward melanoma detection 2018: A challenge hosted by the international skin imaging collaboration (isic), 2019, arXiv preprint arXiv:1902.03368.

[247] X. Wang, Y. Peng, L. Lu, Z. Lu, M. Bagheri, R.M. Summers, Chestx-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 2097–2106.

[248] D. Hughes, M. Salathé, et al., An open access repository of images on plant health to enable the development of mobile disease diagnostics, 2015, arXiv preprint arXiv:1511.08060.

[249] M. Cimpoi, S. Maji, I. Kokkinos, S. Mohamed, A. Vedaldi, Describing textures in the wild, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2014, pp. 3606–3613.

[250] J. Jongejan, H. Rowley, T. Kawashima, J. Kim, N. Fox-Gieg, The quick, draw!-ai experiment.(2016), 2016, URL http://quickdraw.withgoogle.com. 4.

[251] B. Schroeder, Y. Cui, Fgvcx fungi classification challenge 2018, 2018, Available Online: Github.Com/Visipedia/Fgvcx_fungi_comp (Accessed 14 July 2021).

[252] S. Houben, J. Stallkamp, J. Salmen, M. Schlipsing, C. Igel, Detection of traffic signs in real-world images: The german traffic sign detection benchmark, in: The 2013 International Joint Conference on Neural Networks, IJCNN, Ieee, 2013, pp. 1–8.

[253] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, C.L. Zitnick, Microsoft coco: Common objects in context, in: Computer Vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Proceedings, Part V 13, Springer, 2014, pp. 740–755.

[254] A. Cai, W. Hu, J. Zheng, Few-shot learning for medical image classification, in: Artificial Neural Networks and Machine Learning–ICANN 2020: 29th International Conference on Artificial Neural Networks, Bratislava, Slovakia, September 15–18, 2020, Proceedings, Part I 29, Springer, 2020, pp. 441–452.

[255] X. Chen, L. Yao, T. Zhou, J. Dong, Y. Zhang, Momentum contrastive learning for few-shot COVID-19 diagnosis from chest CT images, Pattern Recognit. 113 (2021) 107826.

[256] S. Jadon, COVID-19 detection from scarce chest x-ray image data using few-shot deep learning approach, in: Medical Imaging 2021: Imaging Informatics for Healthcare, Research, and Applications, vol. 11601, SPIE, 2021, pp. 161–170.

[257] Z. Dai, J. Yi, L. Yan, Q. Xu, L. Hu, Q. Zhang, J. Li, G. Wang, Pfemed: Few-shot medical image classification using prior guided feature enhancement, Pattern Recognit. 134 (2023) 109108.

[258] A. Ouahab, O.B. Ahmed, ProtoMed: Prototypical networks with auxiliary regularization for few-shot medical image classification, Image Vis. Comput. 154 (2025) 105337.

[259] A. Krenzer, S. Heil, D. Fitting, S. Matti, W.G. Zoller, A. Hann, F. Puppe, Automated classification of polyps using deep learning architectures and few-shot learning, BMC Med. Imaging 23 (1) (2023) 59.

[260] J. Xu, J. Ma, Auto parts defect detection based on few-shot learning, in: 2022 3rd International Conference on Computer Vision, Image and Deep Learning & International Conference on Computer Engineering and Applications (CVIDL & ICCEA), 2022, pp. 943–946, http://dx.doi.org/10.1109/CVIDLICCEA56201.2022.9823993.

[261] G. Tong, C. Chen, Q. Peng, H. Shen, L. Huang, X. Hu, Research for defect detection based on few-shot learning of siamese network, in: 2022 IEEE 6th Advanced Information Technology, Electronic and Automation Control Conference, IAEAC, 2022, pp. 835–838, http://dx.doi.org/10.1109/IAEAC54830.2022.9929914.

[262] T. Tang, C. Qiu, T. Yang, J. Wang, J. Zhao, M. Chen, J. Wu, L. Wang, A novel lightweight relation network for cross-domain few-shot fault diagnosis, Measurement 213 (2023) 112697.

[263] J. Hu, W. Li, A. Wu, Z. Tian, Novel joint transfer fine-grained metric network for cross-domain few-shot fault diagnosis, Knowl.-Based Syst. 279 (2023) 110958.

[264] J. Hu, W. Li, Y. Zhang, Z. Tian, Cross-domain few-shot fault diagnosis based on meta-learning and domain adversarial graph convolutional network, Eng. Appl. Artif. Intell. 136 (2024) 108970.

[265] H. Shao, X. Zhou, J. Lin, B. Liu, Few-shot cross-domain fault diagnosis of bearing driven by task-supervised ANIL, IEEE Internet Things J. (2024).

[266] X. Zhang, W. Huang, R. Wang, J. Wang, C. Shen, Dual prototypical contrastive network: a novel self-supervised method for cross-domain few-shot fault diagnosis, J. Intell. Manuf. 36 (1) (2025) 475–490.

[267] X. Liang, Few-shot cotton leaf spots disease classification based on metric learning, Plant Methods 17 (2021) 1–11.

[268] D. Argüeso, A. Picon, U. Irusta, A. Medela, M.G. San-Emeterio, A. Bereciartua, A. Alvarez-Gila, Few-shot learning approach for plant disease classification using images taken in the field, Comput. Electron. Agric. 175 (2020) 105542.

[269] C. Wang, J. Zhou, C. Zhao, J. Li, G. Teng, H. Wu, Few-shot vegetable disease recognition model based on image text collaborative representation learning, Comput. Electron. Agric. 184 (2021) 106098.

[270] F. Zhong, Z. Chen, Y. Zhang, F. Xia, Zero-and few-shot learning for diseases recognition of citrus aurantium l. using conditional adversarial autoencoders, Comput. Electron. Agric. 179 (2020) 105828.

[271] M. Rezaei, D. Diepeveen, H. Laga, M.G. Jones, F. Sohel, Plant disease recognition in a low data scenario using few-shot learning, Comput. Electron. Agric. 219 (2024) 108812.

[272] Y. Li, J. Yang, Few-shot cotton pest recognition and terminal realization, Comput. Electron. Agric. 169 (2020) 105240.

[273] G. Jiang-sheng, F. Jing-yi, F. Xia-ping, Hyperspectral imaging for detection of leguminivora glycinivorella based on 3D few-shot meta-learning model, Spectrosc. Spectr. Anal. 41 (7) (2021) 2171–2174.

[274] D. Zhang, F. Pan, Q. Diao, X. Feng, W. Li, J. Wang, Seeding crop detection framework using prototypical network method in UAV images, Agriculture 12 (1) (2021) 26.

[275] W.-S. Kim, D.-H. Lee, T. Kim, G. Kim, H. Kim, T. Sim, Y.-J. Kim, One-shot classification-based tilled soil region segmentation for boundary guidance in autonomous tillage, Comput. Electron. Agric. 189 (2021) 106371.

[276] L. Li, J. Li, C. Lv, Y. Yuan, B. Zhao, Maize residue segmentation using siamese domain transfer network, Comput. Electron. Agric. 187 (2021) 106261.

[277] Z. Wang, Q. Gao, D. Li, J. Liu, H. Wang, X. Yu, Y. Wang, Insulator anomaly detection method based on few-shot learning, IEEE Access 9 (2021) 94970–94980, http://dx.doi.org/10.1109/ACCESS.2021.3071305.

[278] E. Patsiouras, A. Tefas, I. Pitas, Few-shot image recognition for UAV sports cinematography, in: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, CVPRW, 2020, pp. 965–969, http://dx.doi.org/10.1109/CVPRW50498.2020.00127.

[279] Z. Ji, X. Liu, Y. Pang, W. Ouyang, X. Li, Few-shot human-object interaction recognition with semantic-guided attentive prototypes network, IEEE Trans. Image Process. 30 (2020) 1648–1661.

[280] Z. Ji, P. An, X. Liu, C. Gao, Y. Pang, L. Shao, Semantic-aware dynamic generation networks for few-shot human–object interaction recognition, IEEE Trans. Neural Networks Learn. Syst. (2023).

[281] Y. Xu, H. Bi, H. Yu, W. Lu, P. Li, X. Li, X. Sun, Attention-based contrastive learning for few-shot remote sensing image classification, IEEE Trans. Geosci. Remote Sens. (2024).

[282] J. Li, M. Gong, H. Liu, Y. Zhang, M. Zhang, Y. Wu, Multiform ensemble self-supervised learning for few-shot remote sensing scene classification, IEEE Trans. Geosci. Remote Sens. 61 (2023) 1–16, http://dx.doi.org/10.1109/TGRS.2023.3234252.

[283] L. Hou, Z. Ji, X. Wang, Y. Yu, Y. Pang, Diversity-infused network for unsupervised few-shot remote sensing scene classification, IEEE Geosci. Remote. Sens. Lett. (2024).

[284] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, H. Jegou, Training data-efficient image transformers & distillation through attention, in: M. Meila, T. Zhang (Eds.), Proceedings of the 38th International Conference on Machine Learning, in: Proceedings of Machine Learning Research, vol. 139, PMLR, 2021, pp. 10347–10357.

[285] H. Lin, G. Han, J. Ma, S. Huang, X. Lin, S.-F. Chang, Supervised masked knowledge distillation for few-shot transformers, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR, 2023, pp. 19649–19659.

[286] Y. Chen, Y. Zheng, Z. Xu, T. Tang, Z. Tang, J. Chen, Y. Liu, Cross-domain few-shot classification based on lightweight Res2Net and flexible GNN, Knowl.-Based Syst. 247 (2022) 108623.

[287] T. Ma, Y. Sun, Z. Yang, Y. Yang, Prod: Prompting-to-disentangle domain knowledge for cross-domain few-shot image classification, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR, 2023, pp. 19754–19763.

[288] P. Li, F. Liu, L. Jiao, S. Li, L. Li, X. Liu, X. Huang, Knowledge transduction for cross-domain few-shot learning, Pattern Recognit. 141 (2023) 109652.

[289] Y. Zhao, T. Zhang, J. Li, Y. Tian, Dual adaptive representation alignment for cross-domain few-shot learning, IEEE Trans. Pattern Anal. Mach. Intell. 45 (10) (2023) 11720–11732.

[290] R. Xu, L. Xing, B. Liu, D. Tao, W. Cao, W. Liu, Cross-domain few-shot classification via class-shared and class-specific dictionaries, Pattern Recognit. 144 (2023) 109811.

[291] Y. Liu, Y. Zou, R. Li, Y. Li, Spectral decomposition and transformation for cross-domain few-shot learning, Neural Netw. 179 (2024) 106536.

[292] S. Rao, J. Huang, Z. Tang, Rdprotofusion: Refined discriminative prototype-based multi-task fusion for cross-domain few-shot learning, Neurocomputing 599 (2024) 128117.

[293] L. Zhuo, Y. Fu, J. Chen, Y. Cao, Y.-G. Jiang, Unified view empirical study for large pretrained model on cross-domain few-shot learning, ACM Trans. Multimed. Comput. Commun. Appl. 20 (9) (2024) 1–18.

[294] X. Li, Z. He, L. Zhang, S. Guo, B. Hu, K. Guo, CDCNet: Cross-domain few-shot learning with adaptive representation enhancement, Pattern Recognit. (2025) 111382.

[295] L. Yue, L. Feng, Q. Shuai, Z. Li, L. Xu, Multi-level adaptive feature representation based on task augmentation for cross-domain few-shot learning, Appl. Intell. 55 (4) (2025) 1–12.

[296] Y. Bi, H. Zhu, J. Shi, B. Song, TsCANet: Three-stream contrastive adaptive network for cross-domain few-shot learning, J. Supercomput. 81 (1) (2025) 139.

[297] S. Hui, S. Zhou, Y. Deng, P. Li, J. Wang, Meta channel masking for cross-domain few-shot image classification, Neurocomputing 620 (2025) 128956.

[298] W.-H. Li, X. Liu, H. Bilen, Universal representation learning from multiple domains for few-shot classification, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 9526–9535.

[299] W.-H. Li, X. Liu, H. Bilen, Cross-domain few-shot learning with task-specific adapters, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2022, pp. 7161–7170.

[300] Y. Guo, R. Du, Y. Dong, T. Hospedales, Y.-Z. Song, Z. Ma, Task-aware adaptive learning for cross-domain few-shot learning, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 1590–1599.

[301] R. Perera, S. Halgamuge, Discriminative sample-guided and parameter-efficient feature space adaptation for cross-domain few-shot learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 23794–23804.

[302] F. Peng, X. Yang, L. Xiao, Y. Wang, C. Xu, Sgva-clip: Semantic-guided visual adapting of vision-language models for few-shot image classification, IEEE Trans. Multimed. (2023).

[303] L. Song, R. Xue, H. Wang, H. Sun, Y. Ge, Y. Shan, et al., Meta-adapter: An online few-shot learner for vision-language model, Adv. Neural Inf. Process. Syst. 36 (2023) 55361–55374.

[304] Y. Ouali, A. Bulat, B. Matinez, G. Tzimiropoulos, Black box few-shot adaptation for vision-language models, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 15534–15546.

[305] J. Silva-Rodriguez, S. Hajimiri, I. Ben Ayed, J. Dolz, A closer look at the few-shot adaptation of large vision-language models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 23681–23690.

[306] B. Chen, Y.S. Koh, G. Dobbie, SSAT-adapter: Enhancing vision-language model few-shot learning with auxiliary tasks, in: Proceedings of the 32nd ACM International Conference on Multimedia, 2024, pp. 1004–1013.

[307] K. Palanisamy, Y.-W. Chao, X. Du, Y. Xiang, et al., Proto-clip: Vision-language prototypical network for few-shot learning, in: 2024 IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS, IEEE, 2024, pp. 2594–2601.

[308] Z. Xu, Z. Shi, J. Wei, F. Mu, Y. Li, Y. Liang, Towards few-shot adaptation of foundation models via multitask finetuning, in: The Twelfth International Conference on Learning Representations, 2024, URL https://openreview.net/forum?id=1jbh2e0b2K.

[309] M. Zanella, I. Ben Ayed, Low-rank few-shot adaptation of vision-language models, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024, pp. 1593–1603.

[310] F. Shakeri, Y. Huang, J. Silva-Rodríguez, H. Bahig, A. Tang, J. Dolz, I. Ben Ayed, Few-shot adaptation of medical vision-language models, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2024, pp. 553–563.

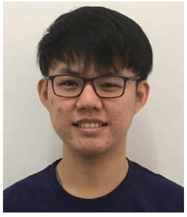
[311] Y. Zhou, H. Yan, K. Ding, T. Cai, Y. Zhang, Few-shot image classification of crop diseases based on vision–language models, Sensors 24 (18) (2024) 6109.

[312] T. Chang, S. Chen, G. Fan, Z. Feng, A vision-language model based on prompt learner for few-shot medical images diagnosis, in: 2024 27th International Conference on Computer Supported Cooperative Work in Design, CSCWD, 2024, pp. 1455–1460, http://dx.doi.org/10.1109/CSCWD61410.2024.10580842.

[313] P. Mangla, N. Kumari, A. Sinha, M. Singh, B. Krishnamurthy, V.N. Balasubramanian, Charting the right manifold: Manifold mixup for few-shot learning, in: The IEEE Winter Conference on Applications of Computer Vision, 2020, pp. 2218–2227.

[314] J.-C. Su, S. Maji, B. Hariharan, When does self-supervision improve few-shot learning? in: European Conference on Computer Vision, Springer, 2020, pp. 645–666.

[315] D. Chen, Y. Chen, Y. Li, F. Mao, Y. He, H. Xue, Self-supervised learning for few-shot image classification, in: ICASSP 2021 - 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2021, pp. 1745–1749.



Jit Yan Lim received B.IT (Hons) in Artificial Intelligence and Ph.D. (I.T.) from Multimedia University. He is currently a Lecturer with the School of Information Technology, Monash University Malaysia. His research interests are few-shot learning, computer vision and deep learning.

Kian Ming Lim received B.IT (Hons) in Information Systems Engineering, Master of Engineering Science (MEngSc) and Ph.D. (I.T.) degrees from Multimedia University. He is currently an Associate Professor with the School of Computer Science, University of Nottingham Ningbo China. His research and teaching interests includes machine learning, deep learning, and computer vision and pattern recognition.

Chin Poo Lee is an Assistant Professor in the School of Computer Science, University of Nottingham Ningbo China. She obtained her Master of Science and Ph.D. in Information Technology, with a specialization in video-based abnormal behavior detection and gait recognition. Her research interests include computer vision, natural language processing, and deep learning.

Yong Xuan Tan received B.IT (Hons) in Artificial Intelligence and Master of Science (Information Technology) from Multimedia University. He is currently an Assistant Lecturer with the School of Information Technology, Monash University Malaysia. His research interests are pattern recognition and computer vision.